

H2-SHIFT Round 2 model: hydrogen retrofit of factory-CNG passenger cars versus continuing on CNG

Team Wookies, NIT Warangal. SRDI Igniters Innovation Challenge 2026, Engineering Track, Option 2. Research cut-off 14 September 2026.

Purpose: one consistent set of inputs feeding the presentation, annexure and pitch. All outputs are formulas; change inputs only on the Inputs sheet.

Scenario selector: Inputs!C3 (1 = Base, 2 = Downside for the retrofit case, 3 = Upside for the retrofit case). 'Downside' means the retrofit looks worse.

Colour code: blue text on yellow = input; black = formula; green header = key output.

Evidence classes on the Inputs sheet: A = verified external evidence; B = our calculated estimate; C = explicit assumption or sensitivity range; D =

Options compared for an owner deciding in the decision year: A = continue operating the CNG donor; B = retrofit the donor to hydrogen (H2-ICE, passenger car); C = compact fleet BEV.

Fairness rules: the donor's resale value is a cash inflow only in option C (sold at t_0); options A and B receive the donor's residual value at the end of the horizon in addition to capital cost. Battery replacement is not assumed within the horizon (warranty evidence on Inputs). Charging during idle hours carries no time cost in all three options.

Levelised cost per km = NPV of costs divided by discounted kilometres. Break-even hydrogen price = the year-1 delivered price at which option B has the same price trajectory shape.

Hydrogen price is the delivered price at the dispenser and is built up on the H2_Supply sheet (production, transport, compression electricity, station capital, margin and taxes). Station capital is recovered through the fuel price only, never allocated twice.

Sheets: Inputs; Donor_Tech (donor and technology comparison); BOM (conversion cost, pilot versus commercial volume); Energy_Range (energy balance, NPVs, break-even); H2_Supply (fleet demand, station sizing, delivered price, utilisation); Market (bottom-up sizing); Pilot_Budget (phases, budget, milestones); Sensitivities (grids and one-at-a-time); Sources (register).

Recalculated with LibreOffice Calc (headless) on the build date; open in Excel and press F9 if values do not appear.

Limitations: many external documents were inaccessible through the build environment's network policy; such inputs are flagged in Sources with 'e' against originals before publication.

Inputs (single source of truth)

Change values in the Base / Downside / Upside columns only; Active column follows the scenario selector.

Scenario selector (1 Base, 2 Downside, 3 Upside):

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Key	Parameter	Base	Downside	Upside	Active	Unit	Class	Source IDs	Notes
General frame									
decision_year	Decision year for the owner's choice (first commercial kits after certification)	2,028	2,028	2,028		2,028 year	C		Pilot vehicles run from 2028; owner decision modelled at commercial availability
horizon	Analysis horizon (fixed at 5 years in the cash-flow table)	5	5	5		5 years	C		Remaining useful life of a 3-year-old fleet sedan
disc	Discount rate (nominal)	12.00%	14.00%	10.00%		0.12 fraction	A	W4-S29	NBFC fleet loan rates 10-16 percent
days	Operating days per year	330	330	330		330 days	C	R1	Fleet taxi benchmark (Round 1)
km_day	Kilometres per operating day	167	212	150		167 km/day	A	W4-S30	55,000 km/yr base; NITI Aayog Feb 2026 plans 70,000 km/yr for 4W taxis (downside); Round 1 used 150
fuel_esc	Annual escalation of petrol and CNG prices	3.00%	2.00%	4.00%		0.03 fraction/yr	C		Nominal
elec_esc	Annual escalation of electricity tariffs	2.00%	1.00%	3.00%		0.02 fraction/yr	C		Nominal
vot	Value of in-shift time lost to refuelling or charging	150	150	150		150 Rs/hour	C		Approximate contribution margin per hour of a fleet taxi
Donor vehicle (Maruti Suzuki Dzire Tour S CNG, about 3 years old at decision)									
donor_value_now	Resale value of donor at decision date (taxi-registered)	350,000	400,000	300,000		350,000 Rs	C	W4-S28	New Tour S CNG ex-showroom about Rs 7.1 lakh; taxi retention 45-55 percent at 3-4 years; higher value favours selling for a BEV
donor_value_end	Residual value of donor at end of horizon (about 8 years old, high km)	100,000	70,000	130,000		100,000 Rs	C	W4-S28	
cng_price	CNG retail price, Delhi (IGL), September 2026	87	78	95		87 Rs/kg	A	W4-S16	Rs 86.98/kg on 14 Sep 2026 (aggregator); Ahmedabad Rs 94; lower CNG price is downside for the retrofit
cng_kmkg	CNG consumption in fleet use (real world, AC, load)	25	27	23		25 km/kg	C	W1-S3b;W1-S8a	Declared 32.1 km/kg (Tour S); owner reports 24-30; fleet use assumed lower
petrol_price	Petrol retail price, Delhi	102	95	111		102 Rs/L	A	W4-S19	Rs 102.12/L verified 25 May 2026; aggregators show Rs 111 in Sep 2026
petrol_kml	Petrol consumption in fleet use (real world, AC)	15	16	14		15 km/L	C	W1-S3a	Declared 26.1 km/L (Tour S); fleet use assumed lower
petrol_share_cng	Share of km run on petrol by the CNG car (CNG unavailable, queues)	5.00%	5.00%	5.00%		0.05 fraction	C		
maint_cng	Maintenance and tyres, CNG car	0.900	0.700	1.2		0.9 Rs/km	C	W4-S22	CEEW 2025 Table 5 to be transcribed; Round 1 value retained
ins_cng	Insurance per year, CNG car	18,000	18,000	18,000		18,000 Rs/yr	C		
cng_retest	CNG cylinder hydrostatic test every 3 years	2,000	2,000	2,000		2,000 Rs per test	A	W2-S24	Gas Cylinder Rules: 3-yearly test
time_cng_h	In-shift time lost refuelling CNG per day (queues)	0.500	0.400	0.600		0.5 hours/day	C		
Hydrogen retrofit (H2-ICE, port injection, 350 bar composite cylinder, petrol retained as automatic backup)									
kit_cost	Installed conversion cost at commercial volume, incl. taxes (from BOM sheet)	322,123	402654.35	273804.958		322,123.48 Rs	B	BOM	Linked to BOM volume total
kit_cost_pilot	Installed conversion cost per pilot vehicle (from BOM sheet)	774,375	774,375	774,375		774,375 Rs	B	BOM	
salvage_cng	Credit for removed CNG cylinder and kit	10,000	0	20,000		10,000 Rs	C		Requalified cylinder resale
downtime_days	Conversion downtime	3	4	2		3 days	C		
lost_margin_day	Lost contribution per day of downtime	1,500	1,500	1,500		1,500 Rs/day	C		
h2_cons	Hydrogen consumption, H2-ICE mode, fleet use (real world, AC)	1.85	2.3	1.6		1.85 kg/100 km	B	W1-S12a;W1-S12c;W1-S12d	Energy balance versus CNG (Energy_Range); Ford Focus H2RV 1.4, Mazda RX-8 RE 2.4, HiAce van 3.1 kg/100 km
h2_share	Share of km driven on hydrogen (rest on petrol backup)	65.00%	50.00%	85.00%		0.65 fraction	C		Two depot fills per day; check Energy_Range max share
h2_price_y1	Delivered hydrogen price at dispenser, year 1 (2028), unsubsidised, shared hub station	800	1,000	550		800 Rs/kg	B	W4-S22;W4-S46	CEEW: small-network refuelling today above USD 10/kg; H2_Supply build-up gives the pilot-station price
h2_change	Annual change in delivered hydrogen price	-10.00%	-3.00%	-15.00%		-0.1 fraction/yr	C	W4-S22	CEEW path to USD 4.2/kg (about Rs 360) delivered in 2030
maint_h2	Maintenance and tyres, retrofitted car	1	1.2	0.850		1 Rs/km	C	W4-S22	Adds injector, sensor, regulator service
ins_h2	Insurance per year, retrofitted car	21,000	24,000	19,000		21,000 Rs/yr	C	W2-S53	No IMT endorsement exists for hydrogen kits; assumes insurer accepts endorsed retrofit
h2_inspect_yr	Hydrogen system inspection and 3-yearly cylinder requalification, annualised	4,000	6,000	3,000		4,000 Rs/yr	C	W2-S24	
time_h2_h	In-shift time lost for hydrogen refuelling and detour per day	0.400	0.600	0.300		0.4 hours/day	C		Two short fills at a depot dispenser
mod_resid_factor	Residual value factor for a modified vehicle	90.00%	80.00%	1		0.9 fraction	C		
kit_resid	Residual value of the kit at end of horizon	0	0	20,000		0 Rs	C		
BEV alternative (sell donor, buy compact fleet BEV)									
bev_price	BEV ex-showroom price incl. GST (compact fleet BEV, 24-30 kWh)	1,050,000	900,000	1,300,000		1,050,000 Rs	A	W3-S27;W3-S25	Tata Tiago EV Rs 6.99-10.09 lakh, Punch EV Rs 9.69-12.59 lakh (2026), Xpres-T fleet EV from Rs 13.04 lakh
bev_onroad	Registration and first-year extras beyond ex-showroom	30,000	20,000	40,000		30,000 Rs	A	W4-S25	Delhi EV Policy 2026 waives road tax and registration fee to March 2030
bev_kwh100	BEV consumption measured at the meter (incl. charging losses)	15	13	17		15 kWh/100 km	A	W3-S30;W4-S23	Real-world tests 12.8-15.3 kWh/100 km at the battery (Tiago EV, Punch EV, Nexon EV) plus 5-15 percent charging losses
depot_tariff	Depot or home charging tariff, landed	6	5	8		6 Rs/kWh	A	W4-S20;W3-S32	Delhi DERC EV-charging tariff Rs 4.50/kWh (LT) plus surcharges and taxes
pub_tariff	Public DC fast-charging price	20	16	25		20 Rs/kWh	A	W4-S21;W3-S33	2026 network prices Rs 16-25/kWh incl. GST
pub_share	Share of energy from public fast charging	30.00%	20.00%	50.00%		0.3 fraction	C		
maint_bev	Maintenance and tyres, BEV	0.500	0.400	0.800		0.5 Rs/km	C	W4-S22	
ins_bev	Insurance per year, BEV	30,000	28,000	34,000		30,000 Rs/yr	C		
bev_resid	BEV residual value at end of horizon	420,000	480,000	340,000		420,000 Rs	C	W4-S22	40 percent of price at 5 years (no Indian fleet-EV residual data found); higher residual is downside for the retrofit
time_bev_h	In-shift time lost for charging per day	0.500	0.300	1		0.5 hours/day	C		Overnight depot charging is idle time
Energy and storage physics									
cng_lhv	Lower heating value of CNG	48	48	48		48 MJ/kg	A	W4-S62	IPCC natural gas 48.0 TJ/Gg
petrol_lhv	Lower heating value of petrol	32.2	32.2	32.2		32.2 MJ/L	A	W4-S62	44.3 MJ/kg x 0.74 kg/L
h2_lhv	Lower heating value of hydrogen	120	120	120		120 MJ/kg	A	W1-S13a	
cyl_vol	Hydrogen cylinder internal (water) volume	60	60	60		60 L	A	W1-S2	Luxfer G-Stor Pro V060H35 (Type III, 350 bar): 60 L, OD 400 mm, length 767 mm, 36 kg, 1.45 kg H2
h2_density_350	Hydrogen density at 350 bar, 15 C	24	24	24		24 kg/m3	A	W1-S13a	Full-fill density per SAE J2601 definition
h2_density_700	Hydrogen density at 700 bar, 15 C	40.2	40.2	40.2		40.2 kg/m3	A	W1-S13a	
fill_soc	State of charge reached in a non-cooled 350 bar fill (hot ambient)	85.00%	80.00%	95.00%		0.85 fraction	C	W1-S13d	SAE J2601: 95-100 percent needs pre-cooled dispensing; small tanks may fall outside light-duty tables
resid_density	Residual gas density at minimum usable pressure	1.7	2	1.5		1.7 kg/m3	C	W1-S13c	About 20 bar residual
fills_day	Depot fills per day available to the driver	2	1.5	2.5		2 fills/day	C		
cyl_mass	Composite cylinder mass (60 L, 350 bar)	36	42	32		36 kg	A	W1-S2	Luxfer Type III 36 kg; Type IV similar

kit_other_mass	Valve, frame, regulator, lines, ECU mass	14	16	12	14 kg	C		
cng_cyl_mass	Removed CNG cylinder (55 L Type I) mass	55	55	55	55 kg	A	W1-S14a;W1-S14b	50 L steel cylinder 48 kg; 60 L 61-69 kg
cng_kit_mass	Removed CNG regulator, rail, lines mass	8	8	8	8 kg	C		
petrol_ps	Donor engine power on petrol (Tour S 1.2 L)	81.6	81.6	81.6	81.6 PS	A	W1-S3a	60 kW at 5,700 rpm
cng_ps	Donor engine power on CNG	70	70	70	70 PS	C	W1-S6	Third-party 69 bhp; official CNG rating not captured
h2_power_factor	H2 lean-burn peak power relative to petrol	52.00%	45.00%	60.00%	0.52 fraction	B	W1-S9a;W1-S9b	About 30 percent air displacement (20 percent loss at stoichiometric) plus
petrol_tank	Petrol tank capacity	37	37	37	37 L	A	W1-S3a	35-40 percent loss at lambda 2; strategy uses lambda 1.6-2.2
Hydrogen supply and station								
pilot_cars	Pilot fleet size	30	30	30	30 cars	D		Statistical spread across three fleets, one dispensing point
h2_prod_cost	Green hydrogen production cost at plant gate (2026-28)	400	560	300	400 Rs/kg	A	W4-S33;W4-S34;W4-S35	CEEW USD 3.5-5/kg; industry Rs 397-560; SIGHT ammonia auctions imply
h2_prod_cost_2030	Green hydrogen production cost at plant gate (2030)	240	320	180	240 Rs/kg	A	W4-S22;W4-S57;W4-S58	up to Rs 302 for the hydrogen molecule
transport_cost	Tube-trailer transport to station	45	60	35	45 Rs/kg	B	W4-S44	CEEW USD 2-2.5; IEA about USD 3; NITI-RMI best case USD 1.6
station_capex_pilot	Dispensing point capex, pilot (compressor, cascade, one dispenser, civil, safety, PESO)	100,000,000	150,000,000	80,000,000	100,000,000 Rs	B	W4-S56;W4-S43	About 150 km, 300 kg payload
station_cap_pilot	Dispensing point design capacity, pilot	150	150	150	150 kg/day	C		Indian trade source Rs 14-20 crore for 200-500 kg/day; delivered
station_util_pilot	Station utilisation, pilot	50.00%	35.00%	65.00%	0.5 fraction	C	W4-S22	hydrogen, no electrolyser
station_capex_2030	Station capex, 2030 hub scale (500 kg/day)	120,000,000	200,000,000	90,000,000	120,000,000 Rs	B	W4-S56	Shared with bus/truck pilot users on the Sahibabad-Faridabad-Delhi route
station_cap_2030	Station design capacity, 2030 hub	500	500	500	500 kg/day	C		
station_util_2030	Station utilisation, 2030 hub	70.00%	50.00%	85.00%	0.7 fraction	C	W4-S22;W4-S46	CEEW assumes 30 percent in 2030 rising to 80 percent by 2050; ICCT 50-
station_life	Station capital recovery period	10	10	10	10 years	C		70 percent
station_om	Fixed operation and maintenance, share of capex per year	4.00%	5.00%	3.00%	0.04 fraction/yr	A	W4-S22	
comp_kwh	Compression and dispensing electricity	3	3.5	2.5	3 kWh/kg	C		CEEW 5 percent
ind_tariff	Industrial electricity tariff	9	10	8	9 Rs/kWh	C		Delivered gas at 200 bar to 450 bar cascade
h2_loss	Venting, purge and metering losses	3.00%	5.00%	2.00%	0.03 fraction	C		
h2_margin_tax	Retailer margin and taxes on delivered hydrogen	10.00%	15.00%	5.00%	0.1 fraction	C		GST treatment of hydrogen as automotive fuel to be confirmed
trailer_kg	Tube-trailer payload	300	250	350	300 kg	C		Typical 200 bar tube trailer
Emissions factors								
ef_cng	Tailpipe CO2 per kg CNG	2.69	2.69	2.75	2.69 kg CO2/kg	A	W4-S62	IPCC 2006 default 56.1 t/TJ at 48 TJ/Gg
wt_cng	Well-to-tank uplift for CNG in India	25.00%	15.00%	35.00%	0.25 fraction	C	W4-S63;W4-S17	Imported spot LNG share raises upstream emissions; India-specific factor
ef_petrol	Tailpipe CO2 per litre petrol	2.27	2.27	2.31	2.27 kg CO2/L	A	W4-S62	not found
wt_petrol	Well-to-tank uplift for petrol	20.00%	15.00%	25.00%	0.2 fraction	C	W4-S63	IPCC 2006 default
ci_h2_green	Carbon intensity, green hydrogen incl. compression and delivery	2	3.5	1.2	2 kg CO2e/kg H2	A	W4-S63;W4-S64;W4-S65	India green hydrogen standard: 2 kg CO2e/kg well-to-gate
ci_h2_grid	Carbon intensity, grid-electrolysis hydrogen (India grid 2025)	39	43	35	39 kg CO2e/kg H2	B	W4-S59	0.71 t/MWh x 50-60 kWh/kg
ci_h2_grey	Carbon intensity, grey (SMR) hydrogen	12	14	10	12 kg CO2e/kg H2	A	W4-S64	
grid_now	Grid emission factor at the meter, 2025	0.860	0.860	0.860	0.86 kg CO2/kWh	A	W4-S59	CEA v21 weighted average 0.710 t/MWh (FY2024-25) adjusted for about 17
grid_2030	Grid emission factor at the meter, 2030	0.580	0.520	0.660	0.58 kg CO2/kWh	A	W4-S60	percent T&D losses
bev_mfg	Manufacturing emissions, compact BEV incl. 27 kWh battery	7	6	8.5	7 t CO2e	B	W4-S63;W4-S66	CEA NEP path 0.548 (2026-27) to 0.430 (2031-32) net, adjusted for T&D
bev_life_km	Lifetime km over which BEV manufacturing is amortised	150,000	180,000	120,000	150,000 km	C		losses; lower grid factor is downside for the retrofit
kit_mfg	Manufacturing emissions of the retrofit kit	0.500	0.700	0.400	0.5 t CO2e	B		Glider 4-6 t plus 27 kWh x 70 kg CO2e/kWh
nox_bs6	BS-VI NOx limit, M1 positive ignition	60	60	60	60 mg/km	A	W4-S71	Carbon-fibre cylinder dominant
Market sizing								
cng_stock	CNG vehicles on road in India, all categories (March 2025)	8,195,000	7,500,000	8,800,000	8,195,000 vehicles	A	W3-S2	81.95 lakh (PNGRB citing Vahan, Aug 2025); 72.10 lakh in Jul 2024 (Lok
cng_pv_share	Share of passenger cars in the CNG stock	55.00%	45.00%	60.00%	0.55 fraction	B	W3-S10;W3-S12;W3-S13	Sabha reply); Round 1 said 7.5 million
eligible_share	Share of CNG cars that are factory-fitted, boot-cylinder layout, under 6 years old at 2028	50.00%	45.00%	55.00%	0.5 fraction	B	W3-S10;W3-S12	No official split; estimated from cumulative factory-CNG car sales (Maruti
fleet_share	Share of eligible cars in commercial or institutional fleets	20.00%	15.00%	25.00%	0.2 fraction	C	W3-S18;W3-S25	about 2.6 million FY20-FY26 plus others) and aftermarket conversions;
hub_share	Share of fleet cars in cities with a hydrogen hub by 2030	40.00%	30.00%	50.00%	0.4 fraction	C	W3-S36;W3-S37	Round 1 said 65 percent
adopter_share	Plausible early-adopter share (return-to-base fleets)	5.00%	3.00%	8.00%	0.05 fraction	C		Factory CNG sales FY23-FY26 versus total CNG cars
hub_stations_2030	Fleet-hub stations serving cars by 2030	8	4	15	8 stations	D	W3-S37	Delhi alone registers about 60,000-85,000 taxis; no national fleet split

Retrofit bill of materials and conversion cost

Pilot batch (about 30 kits) versus commercial volume (about 10,000 kits per year with localised cylinders). Unit costs are estimates unless a source is given.

#	Component / activity	Qty	Unit cost pilot (Rs)	Unit cost volume (Rs)	Pilot total (Rs)	Volume total (Rs)	Class	Basis / source
Storage and mounting								
	Composite hydrogen cylinder, 60 L water volume, 350 bar (Type III or IV), 15-year life							Scaled from DOE/Strategic Analysis Type IV system cost records (S-CYLCOST): about \$33/kWh at low volume and \$21/kWh at 10k/yr for a 5.6 kg 700 bar system, applied to a 1.45 kg (48 kWh) cylinder with a small-size penalty; example product Luxfer G-Stor Pro V060H35 (S-CYL)
1	On-tank valve with solenoid, excess-flow, manual shut-off and TPRD (UN R134.2 class)	1	250,000	95,000	250,000	95,000	B	Production parts exist (S-OTV); no public price; estimate
	Mounting frame, straps, boot bulkhead shield, TPRD vent duct, boot ventilation	1	55,000	18,000	55,000	18,000	C	Fabricated steel frame per UN R110-type 20 g / 8 g loads; estimate
		1	30,000	11,000	30,000	11,000	C	
Fuel delivery								
4	H35 receptacle, check valve, high-pressure filter	1	25,000	8,000	25,000	8,000	C	ISO 17268 H35 receptacle; estimate
5	Stainless (316L) high-pressure lines and fittings	1	18,000	7,000	18,000	7,000	C	Estimate
	Two-stage pressure regulator 350 bar to injection pressure, with pressure and 6 temperature sensors	1	55,000	15,000	55,000	15,000	C	Production FCEV regulators exist (S-REG); estimate for ICE outlet pressure
7	Hydrogen port-fuel injectors (4) and rail	1	40,000	12,000	40,000	12,000	C	Bosch / PHINIA H2 PFI injectors announced 2024-25 (S-INJ); estimate
8	Low-pressure hose, fuel shut-off, pressure relief	1	8,000	3,500	8,000	3,500	C	Estimate
Controls and safety								
	Hydrogen ECU (dual-map, CAN, automatic petrol fallback, diagnostics) with 9 harness	1	45,000	15,000	45,000	15,000	C	Analogous to sequential CNG ECU at volume; pilot uses low-volume controller
	Hydrogen leak sensors (boot and engine bay), controller, cabin warning, vent 10 fan	1	22,000	7,000	22,000	7,000	C	Automotive H2 sensors exist (S-SENS); estimate
Engine and emissions								
	Cold-grade spark plugs, PCV/oil-separator upgrade, coolant and knock sensor 11 check	1	8,000	4,000	8,000	4,000	C	Backfire and pre-ignition mitigation items (S-H2ICE)
	Lean-NOx aftertreatment provision (lean NOx trap) and lambda sensor	1	35,000	15,000	35,000	15,000	C	Needed if lean calibration alone does not meet BS-VI NOx over the cycle (S-NOX); estimate
Installation and validation								
13	Removal of CNG kit and cylinder, installation labour (hours)	40	600	600	24,000	8,400	C	Pilot 40 h; volume 14 h at Rs 600/h (see qty note)
14	Per-vehicle leak test, commissioning, road test and documentation	1	10,000	2,500	10,000	2,500	C	Estimate
15	Certification and engineering amortisation per kit	1	0	8,000	0	8,000	B	Volume: one-time development and type-approval cost (Pilot_Budget) spread over 50,000 kits; pilot carries it in the programme budget
Subtotal parts, labour and validation					625,000	229,400		
	Warranty provision (share of subtotal)		5%	4%	31,250	9,176		
	Installer and kit-maker margin (share of subtotal)		0%	15%	0	34,410		
	GST on kit and installation (rate assumed; CNG kits attract 28 percent, hydrogen kit cl		18%	18%	118,125	49,137		

Installed conversion cost per vehicle (incl. taxes)	Pilot (G) / Volume (H):	774,375	774,375	322,123
Share of cylinder and valve in volume kit cost		49%		
Volume kit cost in Rs lakh		3.22		
Pilot kit cost in Rs lakh		7.74		
Reference: Round 1 assumed Rs 1.8 to 2.2 lakh installed. Reference: sequential CNG retrofit kit incl. cylinder and fitting in India about Rs 0.6 to 0.9 lakh (S-CNGKIT).				

Energy balance, storage, range, mass and power

Column vectors of formulas referencing Inputs. Hydrogen consumption is an input cross-checked against the CNG energy balance.

Quantity	Value	Unit	Formula / note
Energy use per km on CNG (fleet real-world)		1.92 MJ/km	LHV / km per kg
Energy use per km on petrol (fleet real-world)		2.15 MJ/km	
Hydrogen consumption assumed (input)		1.85 kg/100 km	Input h2_cons
Energy use per km on hydrogen		2.22 MJ/km	
Implied H2-ICE efficiency relative to CNG operation (>1 = better)		0.86 ratio	Cross-check: lean PFI hydrogen at part load can match or exceed CNG efficiency; retrofit calibration usually loses some of that
Hydrogen economy		54.05 km/kg	
Cylinder internal volume		60.00 L	
Nominal hydrogen at 350 bar, 15 C		1.44 kg	Density 24.0 kg/m3 (S-DENS)
Nominal hydrogen if 700 bar cylinder of same volume		2.41 kg	Design option; needs 700 bar dispensing
Hydrogen after a non-cooled 350 bar fill		1.22 kg	Fill SOC input; hot ambient, no pre-cooling
Residual hydrogen at minimum usable pressure		0.10 kg	About 20 bar residual
Usable hydrogen per fill		1.12 kg	Key output
Range on hydrogen per fill		60.65 km	Key output
Range on hydrogen per fill, 700 bar option		105.31 km	
Petrol backup range		555.00 km	Existing tank retained
Daily hydrogen need at full hydrogen operation		3.09 kg/day	
Fills per day needed for 100 percent hydrogen operation		2.75 fills/day	
Maximum share of km on hydrogen with the available depot fills		0.73 fraction	Compare with input h2_share
Share of km on hydrogen used in the model (input)		0.65 fraction	
Hydrogen used per operating day in the model		2.01 kg/day	
Mass added: cylinder + valve/frame/regulator/lines/ECU		50.00 kg	
Mass removed: CNG cylinder + CNG kit		63.00 kg	
Net kerb mass change		-13.00 kg	Negative = lighter than CNG configuration
Peak power on petrol		81.60 PS	Donor spec
Peak power on CNG (factory)		70.00 PS	Third-party figure; official CNG rating not captured
Estimated peak power on hydrogen (lean PFI)		42.43 PS	About 30 percent air displacement plus lean operation (S-H2PWR)
Power loss versus petrol		0.48 fraction	
Power loss versus CNG		0.39 fraction	
Power-to-mass, hydrogen mode, 4 occupants (kerb 1045 kg + 300 kg)		31.86 PS per tonne	Urban duty reference: a loaded 800 cc hatchback is about 40 PS per tonne
HCNG 18 vol% blend: hydrogen share of energy		0.06 fraction	Volumetric LHVs 10.8 and 35.8 MJ/Nm3 (S-HCNG); no vehicle modification

Ownership cost comparison for one vehicle: A continue on CNG, B hydrogen retrofit, C sell and buy BEV

Nominal cash costs (Rs). Negative values are inflows (residual values). Years in columns; t0 = decision date.

Year index	0	1	2	3	4	5
Calendar year	2028	2029	2030	2031	2032	2033
Discount factor	1.000	0.893	0.797	0.712	0.636	0.567
Annual km	0	55,110	55,110	55,110	55,110	55,110
Petrol price (Rs/L)		102.0	105.1	108.2	111.5	114.8
CNG price (Rs/kg)		87.0	89.6	92.3	95.1	97.9
Delivered hydrogen price (Rs/kg)		800	720	648	583	525
Hydrogen price trajectory multiplier (relative to year 1)		1.000	0.900	0.810	0.729	0.656
Depot tariff (Rs/kWh)		6.00	6.12	6.24	6.37	6.49
Public fast-charge price (Rs/kWh)		20.00	20.40	20.81	21.22	21.65
Hydrogen consumed (kg)		663	663	663	663	663

Option A: continue operating the CNG donor

Capital / transaction at t0	0	0	0	0	0	0
CNG fuel	0	182,194	187,659	193,289	199,088	205,061
Petrol fuel (share of km)	0	18,737	19,300	19,879	20,475	21,089
Maintenance and tyres	0	49,599	49,599	49,599	49,599	49,599
Insurance	0	18,000	18,000	18,000	18,000	18,000
Cylinder periodic test (annualised)	0	667	667	667	667	667
In-shift refuelling time cost	0	24,750	24,750	24,750	24,750	24,750
Residual value at end of horizon (inflow)	0	0	0	0	0	-100,000
Total cash cost	0	293,947	299,975	306,183	312,578	219,165
Present value	0	262,452	239,138	217,935	198,649	124,360
NPV of costs (Rs)		1,042,535				

Option B: retrofit the donor to hydrogen (H2-ICE, petrol backup retained)

Capital / transaction at t0	316,623	0	0	0	0	0
Hydrogen fuel	0	530,158	477,142	429,428	386,485	347,837
Petrol fuel (backup share of km)	0	131,162	135,097	139,150	143,324	147,624
Maintenance and tyres	0	55,110	55,110	55,110	55,110	55,110
Insurance	0	21,000	21,000	21,000	21,000	21,000
Hydrogen system inspection and cylinder requalification	0	4,000	4,000	4,000	4,000	4,000
In-shift refuelling time and detour cost	0	19,800	19,800	19,800	19,800	19,800
Residual value at end of horizon (inflow)	0	0	0	0	0	-90,000
Total cash cost	316,623	761,230	712,149	668,488	629,719	505,371
Present value	316,623	679,670	567,721	475,816	400,198	286,761
NPV of costs (Rs)		2,726,789				
NPV of option B excluding hydrogen fuel (Rs)		1,124,410				

Option C: sell the donor and buy a compact fleet BEV

Capital / transaction at t0	730,000	0	0	0	0	0
Electricity (depot and public mix)	0	84,318	86,005	87,725	89,479	91,269
Maintenance and tyres	0	27,555	27,555	27,555	27,555	27,555
Insurance	0	30,000	30,000	30,000	30,000	30,000
In-shift charging time cost	0	24,750	24,750	24,750	24,750	24,750
Residual value at end of horizon (inflow)	0	0	0	0	0	-420,000
Total cash cost	730,000	166,623	168,310	170,030	171,784	-246,426
Present value	730,000	148,771	134,175	121,024	109,172	-139,829
NPV of costs (Rs)		1,103,313				

SUMMARY (active scenario)

Discounted km over horizon	198,659	
NPV A: continue CNG (Rs)	1,042,535	
NPV B: hydrogen retrofit (Rs)	2,726,789	
NPV C: sell and buy BEV (Rs)	1,103,313	
Levelised cost per km, A (Rs/km)	5.25	
Levelised cost per km, B (Rs/km)	13.73	
Levelised cost per km, C (Rs/km)	5.55	
Extra cost of B versus A over horizon (Rs)	1,684,254	
Extra cost of B versus C over horizon (Rs)	1,623,476	
Extra cost of C versus A over horizon (Rs)	60,778	
Year-1 fuel cost per km, CNG mode (Rs/km)	3.48	
Year-1 fuel cost per km, hydrogen mode (Rs/km)	14.80	
Year-1 fuel cost per km, petrol mode (Rs/km)	6.80	
Year-1 energy cost per km, BEV (Rs/km)	1.53	
Year-1 blended fuel cost per km, option B (Rs/km)	12.00	
Year-1 blended fuel cost per km, option A (Rs/km)	3.65	
Year-1 non-fuel operating cost per km, A (Rs/km)	1.69	
Year-1 non-fuel operating cost per km, B (Rs/km)	1.81	
Year-1 non-fuel operating cost per km, C (Rs/km)	1.49	
Discounted hydrogen kg, trajectory-weighted	2,003	
Discounted hydrogen kg, flat	2,389	
Break-even year-1 hydrogen price versus A, same trajectory (Rs/kg)	-41	
Break-even flat hydrogen price versus A (Rs/kg)	-34	
Break-even year-1 hydrogen price versus C (BEV), same trajectory (Rs/kg)	-11	
Fuel-parity hydrogen price versus CNG (fuel cost per km equal) (Rs/kg)	188	
Fuel-parity hydrogen price versus petrol (Rs/kg)	368	
Maximum conversion cost for parity with A at the active hydrogen price (Rs)	0	
Maximum conversion cost for parity with C at the active hydrogen price (Rs)	0	
Year-1 operating saving of B versus A (Rs/yr; negative = B costs more)	-467,283	
Simple payback of conversion versus A (years)	none	
Annuity factor (sum of discount factors, years 1-5)	3.605	
Minimum annual km for payback within horizon (year-1 prices)	n/a	
Coefficient a (B-A): fixed and capital terms (Rs)	327,284	
Coefficient b (B-A): per-km non-hydrogen terms (Rs per km)	-4.453	
Coefficient c: hydrogen term (Rs per km per Rs/kg of year-1 price)	0.03635	
Check: $a + b \cdot km + c \cdot km \cdot P$ reproduces B-A (should be ~0)	0.00	
Coefficient a (B-C): fixed and capital terms (Rs)	-261,993	
Coefficient b (B-C): per-km non-hydrogen terms (Rs per km)	5.137	
Check: $a + b \cdot km + c \cdot km \cdot P$ reproduces B-C (should be ~0)	0.00	

Hydrogen supply: fleet demand, station sizing and delivered price build-up

Delivered price = (production + transport + compression electricity + station capital recovery + O&M) x (1 + losses) x (1 + margin and taxes). Capital is recovered only here.

Quantity	Pilot (2028)	Hub (2030)	Unit	Note
Cars served	30.00	37.35 cars		Hub: cars per station at design utilisation (row 27)
Hydrogen per car per operating day	2.01	2.01 kg/car/day		From Energy_Range
Fleet hydrogen demand per operating day	60.25	75.00 kg/day		
Station design capacity	150.00	500.00 kg/day		
Station utilisation assumed for pricing	50%	70% fraction		Pilot station shared with bus/truck pilot users
Hydrogen dispensed per year at that utilisation	27,375	127,750 kg/yr		
Fills per day (cars x fills)	60	75 fills/day		
Peak window length	3.00	3.00 hours		Start-of-shift window
Share of fills in the peak window	50%	40% fraction		Assumption
Fills per hour in the peak window	10.0	10.0 fills/h		
Dispenser positions needed (6 fills per hour each, 1.2 kg fills at 350 bar)	2	2 dispensers		Assumption 6 fills/h per H35 dispenser incl. connection
Compressor throughput needed (16 operating hours)	9.38	31.25 kg/h		Sized on design capacity
On-site storage buffer (1.5 days of demand)	90.37	112.50 kg		Cascade at 450 bar
Tube-trailer deliveries per week	1.4	1.8 deliveries/week		
Production cost at plant gate	400	240 Rs/kg		
Transport (tube trailer)	45	45 Rs/kg		
Compression and dispensing electricity	27	27 Rs/kg		
Station capex	100,000,000	120,000,000 Rs		
Capital recovery factor	0.177	0.177 fraction		
Station capital recovery per kg	647	166 Rs/kg		At the assumed utilisation
Station fixed O&M per kg	146	38 Rs/kg		
Delivered price at dispenser	1433	584 Rs/kg		Key output; compare with Inputs h2_price_y1
Cars a station can serve at design utilisation	37	174 cars		
Utilisation implied by the car fleet alone	40%	15% fraction		

Delivered price versus station utilisation (Rs/kg)

Utilisation	Pilot station (2028)	Hub station (2030)
20%	2780	1162
35%	1818	815
50%	1433	677
65%	1226	602
80%	1096	556
95%	1007	524

Utilisation needed to reach a target delivered price

Target	Target price (Rs/kg)	Pilot station utilisation needed	Hub station utilisation needed
CNG fuel-parity price (from TCO)		188 not reachable at any	not reachable at any utilisation
Full-TCO break-even versus continuing on CNG (from TCO)		-41 not reachable at any	not reachable at any utilisation
Petrol fuel-parity price (from TCO)		368 not reachable at any	999%
Full-TCO break-even versus BEV (from TCO)		-11 not reachable at any	not reachable at any utilisation

Note: values above 100 percent mean the target cannot be met even at full utilisation; 'not reachable' means production plus transport plus electricity alone exceed the target.

Bottom-up addressable market and supply-constrained deployment

Funnel from national CNG stock to what stations can actually serve by 2030.

Step	Vehicles	Share applied	Note
CNG vehicles in India, all categories	8,195,000		Input, class A
Passenger cars among them	4,507,250	55%	Share input
Technically eligible: factory-fitted, boot-cylinder layout, under 6 years old at 2028	2,253,625	50%	Excludes older vehicles and layouts without a compatible cylinder bay
In commercial fleets (taxi, aggregator, corporate, institutional)	450,725	20%	Private owners excluded from the first segment
In cities with a hydrogen hub station by 2030	180,290	40%	Delhi-NCR, Gujarat corridor, Pune-Mumbai, Kochi and similar
Plausible early adopters (return-to-base fleets willing to convert)	9,015	5%	Hypothesis to be validated with fleet interviews
Supply-constrained capacity: stations x cars per station	1,394		From H2_Supply
Realistic 2030 converted fleet (minimum of demand and supply)	1,394		Key output
Round 1 claim for 2030	100,000		For comparison

Programme budget by phase and rollout quantities

Rs. Phases run October 2026 to September 2030. Contingency applied to all phases.

Phase	Line item	Quantity	Unit cost (Rs)	Total (Rs)	Class	Basis
Phase 0: Feasibility and validation (Oct 2026 - Mar 2027)						
	Regulator and test-agency consultations, route definition (AISC, MoRTH, ARAI/ICAT, PESO)	1.0	2,500,000	2,500,000	C	Engineering time, travel, legal review
	Fleet and fuel-partner validation: 30 operator interviews, duty-cycle telematics on 20 cars	1.0	2,500,000	2,500,000	C	Customer demand is a hypothesis; this buys the evidence
	Donor inspection, packaging study, CAD and mule vehicle (boot mock-up)	1.0	4,000,000	4,000,000	C	
	Phase subtotal			9,000,000		
Phase 1: Engineering and controlled validation (Apr 2027 - Mar 2028)						
	Hydrogen engine test-cell time and instrumentation (calibration, NOx, abnormal combustion, 500 h)	1.0	30,000,000	30,000,000	C	Hydrogen test cells exist at ARAI, iCAT and IOCL R&D (S-TEST)
	Prototype conversions (3 vehicles) at pilot kit cost	3.0	774,375	2,323,125	B	BOM pilot cost
	Prototype donor vehicles (used Tour S CNG)	3.0	500,000	1,500,000	C	
	Storage system qualification samples and tests (burst, cycling, bonfire, drop, permeation)	1.0	15,000,000	15,000,000	C	UN GTR 13 / R134 type tests via supplier and PESO
	Vehicle-level safety tests (leak, sled/crash structure, ventilation) at test agency	1.0	12,000,000	12,000,000	C	
	Engineering team (8 FTE x 12 months)	96.0	250,000	24,000,000	C	Rs 2.5 lakh per FTE-month fully loaded
	Phase subtotal			84,823,125		
Phase 2: Certification and pilot build (Apr 2028 - Mar 2029)						
	Kit type approval and emissions testing at ARAI/ICAT (analogous to CNG kit approval)	1.0	10,000,000	10,000,000	C	Industry reports Rs 40-50 lakh for CNG kit approval (S-TAFEE); hydrogen scope larger
	Durability demonstration (2 vehicles x 80,000 km accelerated)	1.0	8,000,000	8,000,000	C	
	Pilot conversions (30 vehicles) at pilot kit cost	30.0	774,375	23,231,250	B	BOM pilot cost
	Instrumentation and telematics (30 cars + 20 comparison cars)	50.0	60,000	3,000,000	C	CAN logger, GPS, hydrogen sensor logging
	Hydrogen dispensing point at partner site (compressor, cascade, one dispenser, safety, PESO)	1.0	100,000,000	100,000,000	B	Input station_capex_pilot; delivered hydrogen, no on-site electrolyser
	Workshop upgrade (ventilation, detection, training) at 2 service points	2.0	4,000,000	8,000,000	C	NFPA 2 type ventilation and detection
	Phase subtotal			152,231,250		
Phase 3: Pilot operation and evaluation (Apr 2029 - Sep 2030)						
	Hydrogen fuel for 30 cars, 18 months (kg at delivered pilot price)	29,821.4	1,433	42,729,079	B	Quantity from Energy_Range and H2_Supply
	Operations, safety officer, data analysis and independent evaluation (18 months)	18.0	1,500,000	27,000,000	C	Per month
	Quarterly emissions and consumption tests at test agency	6.0	800,000	4,800,000	C	
	Insurance, incident reserve and driver training	1.0	6,000,000	6,000,000	C	
	Phase subtotal			80,529,079		
	Subtotal all phases			326,583,454		
	Contingency (share of subtotal)	15%		48,987,518		
	Total programme budget to September 2030 (Rs)			375,570,972		
	Total in Rs crore			37.6		
	Proposed funding split (shares are proposals, not commitments)					
	Suzuki R&D Center India / Maruti Suzuki (engineering, prototypes, test cells)	40%		150,228,389		
	Public support sought (MNRE NGHM R&D or pilot scheme, state hydrogen policy)	35%		131,449,840		
	Fuel and station partner (OMC or gas utility): dispensing point	15%		56,335,646		
	Fleet partner (vehicles, drivers, operations)	10%		37,557,097		

Rollout quantities (conditional expansion after gates)					
Item	2028	2029	2030		Note
Converted cars in operation (cumulative)	3		30	1,394	2030 value from Market sheet (supply-constrained scenario, conditional on gates)
Fleet-hub stations serving cars	1		1	8	
Hydrogen demand from cars (t/yr)	2.0		19.9	924.0	

Emissions per km and at fleet scale, same service boundary for all options

Tailpipe, well-to-tank and amortised manufacturing. Hydrogen combustion emits NOx (not CO2); see NOx rows.

Quantity	Value	Unit	Note
A: CNG tailpipe CO2 per km (CNG share of km)		102.2 g/km	
A: petrol tailpipe CO2 per km (petrol share)		7.6 g/km	
A: well-to-tank CO2e per km		27.1 g/km	Uplift factors on tailpipe
A: total well-to-wheel CO2e per km		136.9 g/km	Key
B: hydrogen share of km		0.7 fraction	
B: hydrogen consumption		1.9 kg/100 km	
B: well-to-wheel CO2e per km, green hydrogen		87.6 g/km	Tailpipe CO2 on hydrogen km is zero; petrol backup km counted
B: well-to-wheel CO2e per km, grid-electrolysis hydrogen		532.5 g/km	Reversal case
B: well-to-wheel CO2e per km, grey (SMR) hydrogen		207.9 g/km	Reversal case
B: retrofit kit manufacturing amortised over horizon km		1.8 g/km	
B: total per km incl. kit manufacturing, green hydrogen		89.4 g/km	Key
C: BEV electricity per km at 2025 grid factor		129.0 g/km	Metered kWh already include charging losses
C: BEV electricity per km at 2030 grid factor		87.0 g/km	
C: BEV manufacturing amortised per km		46.7 g/km	New vehicle and battery; counterfactual is keeping the donor
C: total per km incl. manufacturing, 2025 grid		175.7 g/km	Key
C: total per km incl. manufacturing, 2030 grid		133.7 g/km	Key
Reduction of B (green) versus A		0.3 fraction	
Reduction of C (2030 grid) versus A		0.0 fraction	Negative = BEV replacement emits more than continuing on CNG within the horizon
Grid factor at which BEV equals green-hydrogen retrofit per km		0.3 kg CO2/kWh	Reversal threshold
Hydrogen carbon intensity at which retrofit equals continuing on CNG		5.9 kg CO2e/kg H2	Retrofit is worse than CNG above this hydrogen carbon intensity
Per-car CO2e avoided over the horizon, B (green) versus A		13.1 t CO2e	
Abatement cost of the retrofit versus continuing on CNG (active scenario)		128,869.4 Rs per t CO2e	Extra ownership cost divided by CO2e avoided
Pilot fleet (30 cars) CO2e avoided per year, green hydrogen		78.4 t CO2e/yr	
2030 converted fleet CO2e avoided per year, green hydrogen		3,644.6 t CO2e/yr	Supply-constrained fleet from Market
2030 converted fleet hydrogen demand		924.0 t/yr	
NOx: BS-VI limit for M1 petrol		60.0 mg/km	Applies to the converted vehicle by analogy; test procedure for H2-ICE retrofits does not exist yet
NOx: lean hydrogen combustion engine-out (lambda above 2) reported range	5 to 20	mg/km class	Literature indicates near-zero NOx above lambda about 2 (S-NOX); transients and high load exceed it, hence aftertreatment provision

When does the recommendation change? Grids use the closed-form decomposition from the TCO sheet (same trajectory shape, active scenario for all other inputs).

Values in Rs lakh over the horizon; negative = the hydrogen retrofit is cheaper than the comparator.

Grid 1: retrofit minus continue-on-CNG (Rs lakh) by year-1 delivered hydrogen price (rows) and annual km (columns)

Rs/kg \ km/yr	30,000	40,000	50,000	60,000	70,000
100	3.0	2.9	2.9	2.8	2.7
150	3.6	3.7	3.8	3.9	4.0
200	4.1	4.4	4.7	5.0	5.2
250	4.7	5.1	5.6	6.1	6.5
300	5.2	5.9	6.5	7.1	7.8
400	6.3	7.3	8.3	9.3	10.3
500	7.4	8.8	10.1	11.5	12.9
600	8.5	10.2	11.9	13.7	15.4
800	10.7	13.1	15.6	18.0	20.5

Grid 2: retrofit minus sell-and-buy-BEV (Rs lakh) by year-1 hydrogen price (rows) and BEV price in Rs lakh (columns)

Rs/kg \ BEV price	9	10	11	12	13	14
100	3.7	2.7	1.7	0.7	-0.3	-1.3
150	4.7	3.7	2.7	1.7	0.7	-0.3
200	5.7	4.7	3.7	2.7	1.7	0.7
250	6.7	5.7	4.7	3.7	2.7	1.7
300	7.7	6.7	5.7	4.7	3.7	2.7
400	9.7	8.7	7.7	6.7	5.7	4.7
500	11.7	10.7	9.7	8.7	7.7	6.7
600	13.7	12.7	11.7	10.7	9.7	8.7
800	17.7	16.7	15.7	14.7	13.7	12.7

Grid 3: retrofit minus continue-on-CNG (Rs lakh) by year-1 hydrogen price (rows) and installed kit cost in Rs lakh (columns)

Rs/kg \ kit cost	1.5	2.0	2.5	3.0	4.0	6.0
100	1.1	1.6	2.1	2.6	3.6	5.6
150	2.1	2.6	3.1	3.6	4.6	6.6
200	3.1	3.6	4.1	4.6	5.6	7.6
250	4.1	4.6	5.1	5.6	6.6	8.6
300	5.1	5.6	6.1	6.6	7.6	9.6
400	7.1	7.6	8.1	8.6	9.6	11.6
500	9.1	9.6	10.1	10.6	11.6	13.6
600	11.1	11.6	12.1	12.6	13.6	15.6
800	15.1	15.6	16.1	16.6	17.6	19.6

One-at-a-time sensitivities: extra cost of retrofit versus continuing on CNG (Rs lakh), other inputs at active values

Driver	Low value	High value	Extra cost at low	Extra cost at high Note
Year-1 delivered hydrogen price (Rs/kg)	550	1,000	11.8	20.8Upside and downside inputs
Installed kit cost (Rs)	273,804.96	402,654.35	16.4	17.6
Annual km	30,000	70,000	10.7	20.5 More km increases the fuel-cost penalty at current hydrogen prices
Hydrogen consumption (kg/100 km)	1.6	2.3	14.7	20.7 Scales the hydrogen term
CNG price (Rs/kg)	95	78	16.2	17.6Dearer CNG helps the retrofit
Share of km on hydrogen	0.85	0.5	18.9	15.3At current prices, more hydrogen km cost more than petrol km

Donor platform and technology-route comparison

Donor figures are manufacturer-declared unless marked third-party; hydrogen figures are model outputs.

Criterion	Dzire Tour S CNG (primary)	WagonR S-CNG / Tour H3	Ertiga S-CNG / Tour M	Eeco CNG	Source / note
Engine	1197 cc, 4-cyl K12N	998 cc, 3-cyl K10C	1462 cc, 4-cyl K15C	1197 cc, 4-cyl K12N	S-DONOR
Power petrol / CNG (PS)	81.6 / ~70 (CNG third-party)	~66 / ~57 (CNG third-party)	103 / 87.8	82 / n.a.	S-DONOR; official CNG ratings for 1.0 and 1.2 L cars not captured
Kerb mass CNG (kg)	~1045 (third-party)	~850-930	1250-1255	1085	S-DONOR
Gross vehicle weight (kg)	1375 (Dzire S-CNG)	1340	1820	n.a.	S-DONOR
CNG cylinder water volume (L) and location	55, boot, transverse	60, boot	60, boot	65, under rear floor area	S-DONOR
Boot volume without cylinder (L)	382	335	209 (3rd row up)	van	S-DONOR
Declared CNG economy (km/kg)	32.1	34.4	26.5	27.0	S-DONOR (MIDC declared); fleet real-world 24-30
Ground clearance (mm)	163	165	185	160	Under-floor hydrogen cylinders not credible; boot bay only
Fleet presence (taxi / aggregator)	Very high (app cabs, airport)	High (Delhi taxis, private)	Medium (outstation, 7 seats)	Medium (tier-2 taxi, school van)	Qualitative
Hydrogen cylinder that fits the bay (350 bar)	60 L composite, OD 400 x 767 mm	60 L composite	60-74 L composite	60-74 L composite	Luxfer G-Stor Pro V060H35 example: 36 kg, 1.45 kg H2 (S-CYL)
Power margin after 45 percent hydrogen derate	adequate for urban duty	marginal (3-cyl, 41 PS)	best (48 PS)	adequate	Model estimate
Verdict as first donor	PRIMARY: fleet workhorse, 4-cyl, sedan boot	Alternative: private/taxi hatch, weakest power	Alternative for larger storage (2 cylinders, 5-seat mode)	Alternative for tier-2 fleets	

Technology route comparison for the same donor and duty (matched criteria)

Criterion	H2-ICE retrofit (selected)	Fuel-cell conversion	New BEV (sell donor)	Continue on CNG	Basis
What changes	Fuel storage and delivery, injectors, ECU, sensors; engine retained	Entire powertrain: stack, battery, motor, inverter, cooling, tank, controls	New vehicle	Nothing	Engineering scope
Conversion or purchase cost (Rs lakh, model)		3.22 15 to 25 (pilot), 8 to 12 (volume)		10.50 0	BOM; fuel-cell figures are estimates from stack and battery cost literature (S-FCCOST)
Energy cost per km, year 1 (Rs/km)		12.00 about half of H2-ICE per km on hydrogen		1.53	3.65 TCO sheet
Range per fill or charge (km)		60.65 about 2x H2-ICE for same tank	187 to 260 (Tiago EV, Punch EV tests)	about 250 on CNG at 25 km/kg	Energy_Range; Autocar India real-world tests (W3-S30)
Refuel or recharge time	3 to 5 min (350 bar, non-cooled fill)	3 to 5 min	about 58 min 10-80 percent DC (Tiago EV 24 kWh); 6-8 h AC	3 to 5 min	SAE J2601; Tata specifications (W3-S27)
Peak power versus petrol		0.52 restored (electric drive)	typically higher	about 0.85	Model / spec
Tailpipe CO2 on hydrogen km	zero (NOx present)	zero (water only)	zero	about 105 g/km	Emissions sheet
Approval route today	None for retrofit; new-vehicle standard AIS-195 exists	None for retrofit; AIS-157 covers new FCEVs	Established	Established	WS2 findings
Supplier readiness	Injectors, valves, regulators production-intent; small cylinders scarce	Stacks imported; no Indian light-duty supply chain	Mature	Mature	WS1/WS3
Verdict	Selected: only route that is a retrofit at an affordable scope; economics hinge on hydrogen price	Not a retrofit in cost terms; equals a new vehicle	Benchmark to beat; lowest running cost	Baseline	

Source register

Access date 14 September 2026 unless noted. Status: observed / modelled / target / announced / spec / regulation. 'excerpt' or 'summary' in Limitations means the original could not be opened in the build environment.

ID	Claim / input supported	Publisher	Document title	URL	Publication / data date	Page / table / section	Geography	Unit	Status	Limitations
W1-S1	Hexagon Type IV cylinder dimensions/mass/H2 capacity (350 & 700 bar)	Hexagon Purus	Hydrogen Type 4 cylinder information	https://s3.eu-central-1.amazonaws.com/hexagonpurus-website/HPU_0222_12_HydrogenType4_Tabelle_2pages.pdf	REV. 08/22	p.2 table	Global (Canada/Germany/USA plants)	bar, mm, kg, L spec		"frequently built units; minimum quantities may apply"
W1-S2	Luxfer Type III 350 bar 60/74/205 L cylinders	Luxfer Gas Cylinders	G-Stor Pro H2 Type 3 (EU) spec table	https://www.luxfercylinders.com/product/g-stor-h2-hydrogen-cylinders/	2023-25 (page)	EU table	EU	bar, mm, kg, L spec		Firecrawl excerpt; Type III not Type IV
W1-S3a	Tour S petrol power/torque, FE, tanks, GC	Maruti Suzuki	Dzire Tour S page / Maruti Suzuki Commercial page	https://www.marutisuzuki.com/tour/dzire ; https://www.marutisuzukicommercial.com/dzire-tour-s	2025-26	spec block	India	kW, Nm, km/l, spec L		excerpt; CNG power not in excerpt
W1-S3b	Tour S C-SNG 32.12 km/kg	Maruti Suzuki	Media release: Maruti Suzuki launches the All-new Tour S	https://www.marutisuzuki.com/content/dam/arena-ed/s/corporate/press-release/assets/documents/2023/february/Media_Release_Maruti_Suzuki_launches_the_All-new_Tour_S.pdf	Feb 2023	body	India	km/kg spec (declared)		excerpt
W1-S3c	Dzire 4th gen kerb/GVW/FE/boot/tanks	Maruti Suzuki	Arena Dzire brochure; Dzire price page	https://www.marutisuzuki.com/content/dam/msil/arena/in/en/assets/cars/dzire/documents/Arena-Dzire-Brochure.pdf ; https://www.marutisuzuki.com/arena/dzire/price	2024-26	spec table	India	kg, L, km/l, km/kg	spec	excerpt
W1-S3d	WagonR kerb 850, GVW 1340, tank 32, boot 335; engines K10C/K12N	Maruti Suzuki	WagonR price page; WagonR page	https://www.marutisuzuki.com/arena/wagon-r/price ; https://www.marutisuzuki.com/arena/wagon-r	2025-26	spec table	India	kg, L	spec	excerpt; variant for kerb/GVW not stated
W1-S3d2	WagonR Tour H3 25.22 km/l, 34.41 km/kg	Maruti Suzuki	WagonR Tour H3 page	https://www.marutisuzuki.com/tour/wagonr	2025-26	FAQ	India	km/l, km/kg	spec	excerpt
W1-S3e	Swift 1197 cc 60 kW @5700	Maruti Suzuki	Swift page	https://www.marutisuzuki.com/arena/swift	2025-26	spec	India	kW	spec	excerpt
W1-S3f	Ertiga 1462 cc, 75.8 kW @6000, 139 Nm @4300	Maruti Suzuki	Ertiga page	https://www.marutisuzuki.com/arena/ertiga	2025-26	spec	India	kW, Nm	spec	excerpt
W1-S3g	1.5 L K-Series S-CNG 64.6 kW @5500, 121.5 Nm	Maruti Suzuki	Press release: All New Brezza S-CNG	https://www.marutisuzuki.com/corporate/media/press-releases/2023/march/maruti-suzuki-introduces-all-new-brezza-s-cng	Mar 2023	body	India	kW, Nm	spec	same engine as Ertiga S-CNG; excerpt
W1-S3h	Ertiga CNG 121.5 Nm @4200	Maruti Suzuki	Arena Ertiga brochure	https://www.marutisuzuki.com/content/dam/msil/arena/in/en/assets/cars/ertiga/document/Arena-Ertiga-Brochure.pdf	2024	spec	India	Nm	spec	excerpt
W1-S3i	Ertiga kerb 1150-1205/1250-1255, GVW 1760/1820, tanks 45/60	Maruti Suzuki	Ertiga price page; 6-page brochure	https://www.marutisuzuki.com/arena/ertiga/price ; https://www.marutisuzuki.com/content/dam/msil/arena/in/en/assets/cars/ertiga/document/ErtigaBrochure_6_Pgs_High_Res.pdf	2024-26	spec	India	kg, L	spec	excerpt
W1-S3j	Ertiga 20.51 km/l, 26.11 km/kg	Maruti Suzuki	Press release Next-Gen Ertiga	https://www.marutisuzuki.com/corporate/media/press-releases/2022/april/inspire-togetherness-again-maruti-suzuki-next-gen-ertiga-arrives-at-arena-showrooms	Apr 2022	body	India	km/l, km/kg	spec	excerpt
W1-S3k	Ertiga Tour M 21.10 km/l, 26.54 km/kg, 45 L	Maruti Suzuki	Ertiga Tour M page	https://www.marutisuzuki.com/tour/ertiga	2025-26	FAQ	India	km/l, km/kg	spec	excerpt
W1-S3l	Eeco Tour V 60.5 kW @6000, 105.5 Nm @3000, 20.20 km/l	Maruti Suzuki	Eeco Tour V page	https://www.marutisuzuki.com/tour/eeco	2025-26	spec	India	kW, Nm, km/l	spec	excerpt
W1-S3m	Eeco S-CNG 27.05 km/kg, 104.4 Nm @3000	Maruti Suzuki	Press release New Eeco	https://www.marutisuzuki.com/corporate/media/press-releases/2022/november/maruti-suzuki-new-eeco-more-power-more-fuel-efficient-and-more-style	Nov 2022	body	India	km/kg, Nm	spec	excerpt
W1-S3n	Eeco kerb 970, 19.71 km/l, price from ₹5,20,900	Maruti Suzuki	Eeco price page	https://www.marutisuzuki.com/arena/eeco/price	2025-26	spec	India	kg, km/l, ₹	spec	excerpt
W1-S3o	Eeco CNG kerb 1085, 26.78 km/kg	Maruti Suzuki	Arena Eeco brochure	https://www.marutisuzuki.com/content/dam/msil/arena/in/en/assets/cars/eeco/document/Arena-Eeco-Brochure.pdf	2024-25	spec	India	kg, km/kg	spec	excerpt
W1-S3p	Eeco S-CNG 65 L tank	Maruti Suzuki	Arena Eeco page	https://www.marutisuzuki.com/arena/eeco	2025-26	body	India	L	spec	excerpt
W1-S3q	Swift 24.8/25.75 km/l	Maruti Suzuki	Swift page	https://www.marutisuzuki.com/arena/swift	2025-26	spec	India	km/l	spec	excerpt
W1-S3r	Swift S-CNG 32.85 km/kg	Maruti Suzuki	Press release Epic New Swift S-CNG; Swift S-CNG brochure	https://www.marutisuzuki.com/corporate/media/press-releases/2024/sepember/epic-new-swift-s-cng ; https://www.marutisuzuki.com/content/dam/msil/arena/in/en/assets/cars/swift/document/Swift_SCNG_VerticalBrochure.pdf	Sep 2024	body	India	km/kg	spec	excerpt; torque truncated
W1-S3s	Swift kerb 920/990, GVW 1355/1425, tanks 37/55	Maruti Suzuki	Swift price page	https://www.marutisuzuki.com/arena/swift/price	2025-26	spec	India	kg, L	spec	excerpt
W1-S4	Aura CNG 50.5 kW, 95.2 Nm, tanks 37/65, dims, price	Hyundai Motor India	Aura specification/performance/price pages	https://www.hyundai.com/in/en/find-a-car/aura/specification ; https://www.hyundai.com/in/en/find-a-car/aura/performance ; https://www.hyundai.com/in/en/find-a-car/aura/price	2026	spec table	India	kW, Nm, L, ₹	spec	excerpt; kerb weight absent
W1-S5	Tigor iCNG specs, ARAI mileage, tanks, price	CarDekho (third party)	Tata Tigor specifications	https://www.cardekho.com/tata/tigor/specs	2025-26	spec table	India	bhp, Nm, km/kg, L, ₹	spec (third party)	not OEM page
W1-S5b	Twin-cylinder frees 200-205 L boot	carandbike (third party)	Tata Tigor page	https://www.carandbike.com/tata-cars/tigor	2025-26	body	India	L	claimed	third party
W1-S6	Tour S CNG 69 bhp/101.8 Nm, 34.3 km/kg, kerb 1045, boot 382, GC 163	CarDekho / ZigWheels (third party)	Dzire Tour S CNG overview; Tour S specs	https://www.cardekho.com/overview/Maruti_Dzire_Tour_S/ ; https://www.zigwheels.com/maruti-suzuki-cars/dzire-tour-s/specifications/	2026	spec	India	bhp, Nm, km/kg, kg	spec (third party)	search-summary
W1-S7	2025-26 ex-showroom prices (Tour S CNG ₹7.10 L; WagonR CNG ₹5.79-7.62 L; Ertiga CNG –₹9.5 L; Eeco AC CNG ₹6.46 L; Swift VXI CNG ₹7.57 L)	CarDekho/CarWale/ZigWheels (aggregators)	various price pages	https://www.cardekho.com/overview/Maruti_Dzire_Tour_S/ ; https://www.maruti-suzuki-cars/wagon-r/cng/ ; https://www.carwale.com/maruti-suzuki-cars/ertiga/cng/ ; https://www.cardekho.com/overview/Maruti_Eeco/Maruti_Eeco_S_Seater_AC_CNG.htm ; https://www.cardekho.com/overview/Maruti_Swift/Maruti_Swift_VXI_CNG.htm	Sep 2026 snapshot	price block	India	₹	third-party	search-summary; unverified
W1-S7b	WagonR CNG 56 bhp/82.1 Nm, 60 L, 33.47 km/kg	CarWale (third party)	WagonR CNG page	https://www.carwale.com/maruti-suzuki-cars/wagon-r/cng/	2026	body	India	bhp, Nm, L, km/kg	spec (third party)	search-summary
W1-S8a	Dzire CNG real-world 28-30 km/kg city	Cars24 (blog)	Maruti Suzuki Dzire Real World Mileage	https://www.cars24.com/article/maruti-suzuki-dzire-real-world-mileage/	2025-26	body	India	km/kg	anecdotal	not a fleet study
W1-S8b	WagonR CNG real-world 24-28 km/kg; 3-yr hydrotest	VahanBazaar (blog)	CNG Mileage Real-World India	https://vahanbazaar.in/tips/cng-mileage-real-world-india-vahanbazaar	2026	body	India	km/kg	anecdotal	low confidence
W1-S9a	H2 ≈30 % of mixture volume; ~20 % power loss at stoich PFI	Int. J. Hydrogen Energy (Elsevier)	Hydrogen-fuelled ICEs: Direct Injection versus Port-Fuel Injection	https://www.sciencedirect.com/science/article/pii/S0360319924028106	2024	intro/results	lab	%, kg/m³	observed/ reviewed	excerpt + search-summary; full text blocked
W1-S9b	λ=2 torque ~34.9-40.7 %; BTE 40.4 % (DI e-turbo); NOx <10 ppm above λ 2.7; BTE ~38 % lean	multiple (ACS Omega 2024; IJHE 2025; Springer IJAT 2026)	see URLs	https://pubs.acs.org/doi/10.1021/acsomega.4c02222 ; https://www.sciencedirect.com/science/article/pii/S036031992501715X ; https://link.springer.com/article/10.1007/s12239-026-00469-z	2024-26	abstracts	lab	%, ppm	observed	search-summary; attribution to individual papers not verified
W1-S9c	Supercharging + EGR power recovery	Int. J. Hydrogen Energy	Increasing the power output of hydrogen ICEs by means of supercharging and EGR (Verhelst et al.)	https://www.sciencedirect.com/science/article/abs/pii/S036031990900442X	2009	title/abstract	lab	—	observed	content blocked
W1-S9d	">30 % of gasoline" power with supercharged stoich EGR	LAPSE/PSE Community (review preprint)	Progress of Performance, Emission, and Technical Measures of H2 ICEs	https://pseccommunity.org/wp-content/plugins/wporg/includes/file/2302/LAPSE-2023.9491-1v1.pdf	2023	excerpt	review	%	reviewed	wording ambiguous
W1-S9e	Backfire mitigation by injection control (review exists)	Int. J. Hydrogen Energy	Hydrogen port fuel injection: Review of fuel injection control strategies to mitigate backfire	https://www.sciencedirect.com/science/article/abs/pii/S036031992401348X	2024	title	review	—	reviewed	content blocked
W1-S9f	Timed manifold injection overcomes backfire (IIT Delhi)	Int. J. Hydrogen Energy	Performance characteristics of a hydrogen fuelled S.I. engine using timed manifold injection (Das) ; Hydrogen engine: R&D programmes in IIT Delhi	https://www.sciencedirect.com/science/article/abs/pii/S036031999190038K ; https://www.sciencedirect.com/science/article/abs/pii/S036031990101781	1991; 2002	abstracts	India (lab)	—	observed	search-summary

W1-S9g	Knock at spark >26° CA; smaller advance than gasoline	OSTI ETDE record (Indian study)	Implications of combustion parameters on the ...	https://www.osti.gov/etdweb/servlets/purl/21400931	n.d.	abstract	India (lab)	* CA	observed	search-summary; blocked
W1-S9h	NOx peak at λ=1.3, ~2.7× λ=1	KAUST repository (Onorati et al., Int. J. Engine Res.)	The potential of hydrogen ICEs for heavy-duty ... / role of H2 for future ICEs	https://repository.kaust.edu.sa/bitstreams/e16f5716-15c8-4c8f-9951-bab74c8db2c9/download	2022	excerpt	review	ratio	reviewed	excerpt only
W1-S10a	Bosch PFI for H2 engines product page	Bosch Mobility	Port fuel injection for hydrogen engines	https://www.bosch-mobility.com/en/solutions/powertrain/gas/port-fuel-injection-for-hydrogen-engines/	2024-25	page	Global	—	announced	blocked; title via search
W1-S10b	PHINIA low-pressure H2 PFI injectors; DI-CHG10/15 ratings	PHINIA; MarkLines	Hydrogen Fuel Injectors; IAA 2024 note	https://www.phinia.com/technologies/technologies/hydrogen-solutions/fuel-injectors ; https://www.marklines.com/en/news/314084	2024	page	Global	kW/cyl	announced	search-summary
W1-S10c	Bosch DI injector 17 g/s @40 bar, 2026; Liebherr H964/H966; ~60 bar limit	Power Progress; Mobility Engineering Tech	Fuel injection must follow the hydrogen; Liebherr Q&A	https://www.powerprogress.com/news/fuel-injection-must-follow-the-hydrogen/8036089.article ; https://www.mobilityengineeringtech.com/component/content/article/tohe/pub/regulars/q-a/47237	2024-25	body	Global	g/s, bar	announced	search-summary
W1-S10d	OMB Saleri HPR 350 two-stage regulator; Bosch-OMB agreement; H2 valve catalogue	OMB Saleri; Green Car Congress	Hydrogen valves; catalogue 2023; Bosch partners with OMB Saleri	https://www.omb-saleri.it/products/hydrogen/ ; https://www.omb-saleri.it/uploads/categories/catalogo2023h2.pdf ; https://www.greencarcongress.com/2021/11/202111118-bosch.html	2021-23	page/catalogue	EU	bar	production	search-summary
W1-S10e	Lexier 350 bar HPR and OTV	Lexier	Pressure Regulator 350 bar; On Tank Valve 350 bar	https://lexierweb.com/en/hydrogen/valves/pressure-regulators/hpr-350bar/ ; https://lexierweb.com/en/hydrogen/vehicles-components/on-tank-valve-350bar/	2025	page	EU	bar	production	search-summary
W1-S10f	Poppe + Potthoff TOPAQ OTV 2.0, R134/HGV 3.1 tests, TPRD variants	Poppe + Potthoff; Hydrogen Central	On Tank Valve; certification news	https://www.poppe-potthoff.com/hydrogen/on-tank-valve/ ; https://hydrogen-central.com/hydrogen-tank-valves-by-poppe-potthoff-successfully-complete-required-as-part-of-the-certification-according-to-r134/	Sep 2024	page	EU	—	production-intent	search-summary
W1-S10g	Sensirion STC42A automotive H2 sensor specs	Sensirion	STC42A product page / press	https://sensirion.com/products/catalog/STC42A ; https://sensirion.com/company/news/press-releases-and-news/article/sensirion-announces-global-availability-of-the-stc42a-automotive-h2-sensor-for-thermal-runaway-detection	2024-25	datasheet summary	Global	vol %	production	search-summary
W1-S10h	Bosch H2 leak-detection sensors	Bosch Mobility	Sensors for fuel cell-electric systems	https://www.bosch-mobility.com/en/solutions/powertrain/fuel-cell-electric/sensors-for-fuel-cell-electric-systems/	2025	page	Global	—	production	search-summary
W1-S10i	NTM SenseH2	NTM Sensors / AZoSensors	SenseH2 Hydrogen Sensor	https://www.azosensors.com/equipment-details.aspx?EquipID=339	n.d.	page	US	—	production	search-summary
W1-S11a	ISO 11439: ≤2 % H2 for steel UTS >950 MPa; 0.1 % wet gas	iso11439.com FAQ (secondary)	CNG Cylinder standard ISO 11439 – FAQ	http://www.iso11439.com/faq.php	n.d.	FAQ	Intl	vol %	standard text (secondary)	standard itself not opened
W1-S11b	MoRTH allowed 18 % H-CNG in CNG engines; CMVR amendment	Autocar Professional	MoRTH allows use of H-CNG in CNG engines	https://www.autocarpro.in/news-national/morth-allows-use-of-hcng-in-cng-engines-67299	Sep 2020	body	India	vol %	search-summary	
W1-S11c	ARAI: 18 % HCNG –70 % CO, –25 % HC, +4-5 % fuel economy (HD BS-IV)	IAS Gyan (secondary, quoting PIB)	Notification allowing use of H-CNG issued	https://www.iasgyan.in/daily-current-affairs/notification-allowing-use-of-h-cng-issued	2020	body	India	%	reported	secondary; PIB blocked
W1-S11d	IOCL reformer +50 DTC buses; Rajghat Depot-1, Oct 2020	IndianOil; pv magazine India	IndianOil press release 58744; Delhi HCNG trial	https://iocl.com/NewsDetails/58744 ; https://www.pv-magazine-india.com/2020/10/21/pollution-hit-delhi-starts-hydrogen-enhanced-cng-trial-for-buses/	Oct 2020	body	India	—	announced	IOCL page blocked
W1-S11e	SIAM-led trial: 18 % chosen; NMHC –58 %, THC –23 %, NOx –50 %	NPC / DOE (topic paper)	NPC Future Transportation Fuels Study – Topic paper 25 HCNG	https://www.npc.org/FTF_Topic_papers/25-HCNG.pdf ; https://www.energy.gov/sites/default/files/2022-10/25-HCNG.pdf	2012 (data ~2010)	body	US paper on Indian trial	%	reported	blocked; search-summary
W1-S12a	BMW Hydrogen 7: 191 kW, 8 kg LH2, ~201 km, 3.7 / 2.1 kg/100 km	Wikipedia summarising Wallner et al., IJHE 2008; BMW press	BMW Hydrogen 7; Fuel economy and emissions evaluation of BMW Hydrogen 7 Mono-Fuel demonstration vehicles	https://en.wikipedia.org/wiki/BMW_Hydrogen_7 ; https://www.sciencedirect.com/science/article/abs/pii/S0360319908011439 ; https://www.press.bmwgroup.com/global/article/detail/T0012215EN/	2006-08	body	Germany/US (ANL test)	kW, kg, km, kg/100 km	observed (lab cycle)	primary blocked; summary
W1-S12b	Mazda RX-8 Hydrogen RE: 110 L @350 bar, 2.4 kg, ~107 hp, ~100 km	Wikipedia; Green Car Journal	Mazda RX-8 Hydrogen RE	https://en.wikipedia.org/wiki/Mazda_RX-8_Hydrogen_RE ; https://greencarjournal.com/dont-miss/mazdas-hydrogen-rx-8-re/	2006	body	Japan	L, bar, kg, hp, km	claimed	secondary
W1-S12c	Ford Focus H2RV: 2.3 L SC H2 ICE, 110 hp + 33 hp, 45 mi/kg, 125 mi	EE Power; AutoIntell	Ford Unveils Hydrogen ICE; H2ICE Ford Focus	https://eepower.com/news/ford-unveils-hydrogen-internal-combustion-engine/ ; https://www.autointell.com/News-2003/September-2003/September-2003-4/September-24-03-p4.htm	2003	body	US	hp, mi/kg, mi	claimed	secondary
W1-S12c2	Ford E-450 H2ICE shuttle: 6.8 L V10 SC, 350 bar 26 GGE tank, ~150 mi	Green Car Congress; TRID	Ford H2ICE Shuttle Bus; Engineering the Ford H2 IC Engine Powered E-450	https://www.greencarcongress.com/2004/09/ford_hsub2subic.html ; https://trid.trb.org/view/1813766	2004-06	body	US	mi, bar	claimed	hp not confirmed
W1-S12d	Toyota Hydrogen HiAce: 3.4 L V6, 120 kW, 354 Nm, SCR/Euro VI, 7 vehicles, Nov 2023	Toyota Australia (via RACQ/CarExpert/WhichCar)	Toyota unveils Hydrogen HiAce; reviews	https://www.toyota.com.au/discover/news/toyota-unveils-hydrogen-powertrain-technology-for-pilot-program-in-australia ; https://www.racq.com.au/articles/drive/2023/11/toyota-tests-hydrogen-fuelled-v6-powered-hiace ; https://www.whichcar.com.au/news/hydrogen-powered-toyota-hiace-revealed-with-200km-range	Nov 2023	body	Australia	kW, Nm	announced	Toyota pages blocked
W1-S12e	HiAce ~6.2 kg, 141 L, <200 km, >70 L/100 km; CPB month-long pilot	Drive.com.au; CarExpert; CPB Contractors	2024 Toyota HiAce hydrogen prototype review; CPB pilot news	https://www.drive.com.au/reviews/2024-toyota-hiace-hydrogen-prototype-quick-drive/ ; https://www.carexpert.com.au/car-reviews/2024-toyota-hydrogen-hiace-prototype-review-quick-drive ; https://www.cpbcon.com.au/news-and-media/2024/cpb-contractors-completes-australian-first-toyota-hydrogen-fuelled-vehicle-pilot	2024	body	Australia	kg, L, km	claimed/derived	consumption derived by me
W1-S12f	GR Corolla H2 liquid hydrogen; range doubled; tank >220 L; 2026 superconducting pump	Toyota Global Newsroom / Toyota Europe	Super Taikyu releases	https://global.toyota/en/newsroom/corporate/43550115.html ; https://newsroom.toyota.eu/evolved-liquid-hydrogen-powered-gr-corolla-to-participate-in-super-taikyu-fuji-24-hours-race/	2023-26	body	Japan	L	announced	no kg/100 km
W1-S12g	HySE formation and member roles	Honda Global	Ministerial approval granted to form HySE	https://global.honda/en/newsroom/news/2023/c230517eng.html	17 May 2023	body	Japan	—	announced	—
W1-S12h	Ashok Leyland/RIL H2-ICE truck; >20 trucks ~250,000 km	Business Standard; Hydrogen Central	Indian CV majors drive hydrogen transition; AL & RIL unveil	https://www.business-standard.com/industry/auto/indian-cv-majors-drive-hydrogen-transition-with-h2-vehicles-125061600725_1.html ; https://hydrogen-central.com/ashok-leyland-ril-unveil-indias-first-hydrogen-powered-heavy-duty-truck/	2023-25	body	India	km	announced	press
W1-S12i	IIT Delhi H2 SI genset with KOEL and IOCL R&D	IIT Delhi	Technology for Hydrogen Utilization in SI Engine Generator	https://home.iitd.ac.in/show.php?id=16&in_sections=Press	n.d. (2023-24)	press	India	—	announced	—
W1-S12j	Delhi hydrogen 3-wheeler fleet (15 vehicles)	capacity4dev (EU)	World's first hydrogen-powered three-wheelers launched in New Delhi	https://capacity4dev.europa.eu/discussions/worlds-first-hydrogen-powered-three-wheelers-launched-new-delhi_en	2012	body	India	—	announced	old
W1-S13a	24.0 g/L (35 MPa) and 40.2 g/L (70 MPa) at 15 °C	MDPI Energies	Effect of Hydrogen Refueling Parameters on Final State of Charge	https://www.mdpi.com/1996-1073/12/4/645	2019	\$2	—	g/L	modelled (EOS)	excerpt
W1-S13b	39.75 kg/m³ at 700 barg, 20 °C	Demaco	The energy density of hydrogen	https://demaco-cryogenics.com/blog/energy-density-of-hydrogen/	n.d.	body	—	kg/m³	modelled	vendor blog
W1-S13c	Regulator to ≤20 bar; ~5 % unusable; 20 bar min	USPTO patents (Hyundai/Toyota-type)	e.g. US 8927171; US 11394041; US 10868315	https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/8927171	2010s-2020s	descriptions	—	bar, %	design practice	patents, not standards
W1-S13d	J2601 scope ≥49.7 L; 85 °C; T40 3-min 95-100 % SOC; T20/T30 slower	SAE / ANSI store; CEC	SAE J2601-2020 abstract; SAE J2601 status presentation	https://webstore.ansi.org/standards/sae/sae26012020 ; https://efiling.energy.ca.gov/GetDocument.aspx?tn=66608	2020; 2014	abstract	Intl	L, °C, min	standard	full tables not accessible

W1-S13e	Non-communication fills need tight pre-cooling tolerance; 85 °C liner limit	OSTI (NREL); Atawey	Impact of hydrogen SAE J2601 fueling methods; Understanding SAE J2601	https://www.osti.gov/servlets/purl/1389635 ; https://atawey.com/en/sae-j2601-understanding-the-international-standard-for-gaseous-hydrogen-refueling/	2017; 2025	body	—	—	reviewed	search-summary
W1-S13f	Time Technoplast PESO approvals (250 L Type IV H2; BV 150 L)	Business Standard; Machine Maker; Upstox	PESO approval news	https://www.business-standard.com/markets/capital-market-news/time-technoplast-gets-final-approval-from-peso-for-manufacturing-composite-cylinders-124062100121_1.html ; https://themachinemaker.com/news/time-technoplast-receives-peso-approval-to-make-hydrogen-cylinders/	Mar-Jun 2024; May 2025	body	India	L	announced	press
W1-S13g	Everest Kanto Type-4 composite cylinders	Everest Kanto (via search)	Type IV cylinders	https://ekcuae.com/3d-flip-book/type-4-composite-cylinder/	n.d.	page	India/UAE	—	announced	blocked
W1-S13h	Mirai 2021: 5.6 kg; 52/25.3/64.9 L = 142.2 L	Encyclopedia.pub (citing Toyota 2022 Mirai product guide)	Application of Hydrogen Storage Tanks	https://encyclopedia.pub/entry/47333	2023	§3	Japan/US	kg, L	spec (secondary)	Toyota guide not opened
W1-S13i	Nexo 156.6 L / 6.33 kg; new Nexo 162.2 L / 6.69 kg	Encyclopedia.pub; InsideEVs; Hyundai News	see URLs	https://encyclopedia.pub/entry/47333 ; https://insideevs.com/news/755466/2026-hyundai-nexo-hydrogen-fuel-cell-official-specs/ ; https://www.hyundainews.com/assets/documents/original/52348-2023NEXOProductSpecifications20220720.pdf	2018-2025	spec	Korea/US	L, kg	spec	secondary
W1-S13j	DOE Record 19008 costs/capacities	US DOE HFTO (Strategic Analysis)	Onboard Type IV Compressed Hydrogen Storage System – Cost and Performance Status 2019	https://www.hydrogen.energy.gov/docs/hydrogenprogramlibraries/pdfs/19008_onboard_storage_cost_performance_status.pdf	25 Nov 2019	p.1-2	US	\$/kWh, kWh/kg, kWh/L	modelled (DFMA) blocked;	search-summary
W1-S13k	DOE Record 24006 costs	US DOE HFTO (Strategic Analysis)	Onboard Type IV Compressed Hydrogen Storage System – Cost and Performance Status	https://www.hydrogen.energy.gov/docs/hydrogenprogramlibraries/pdfs/24006-onboard-storage-cost-performance-status.pdf	28 Apr 2025	p.1-2	US	\$/kWh	modelled (DFMA) blocked;	search-summary
W1-S13l	~\$33/kWh low-volume (2013 basis)	US DOE Record 13010 / SA 2013 final report	Onboard storage performance and cost	https://www.hydrogen.energy.gov/docs/hydrogenprogramlibraries/pdfs/13010_onboard_storage_performance_cost.pdf ; https://www.osti.gov/servlets/purl/1082754	2013	—	US	\$/kWh	modelled	low confidence
W1-S13m	15-yr life / 11,000 cycles (GTR 13, GB/T 42612)	ResearchGate (comparative analysis paper); UNECE GTR 13	A comparative analysis of the RCS for on-board high-pressure hydrogen storage cylinders; GTR 13 Amend.1	https://www.researchgate.net/publication/374204598 ; https://unece.org/sites/default/files/2023-07/ECE-TRANS-180-Add.13-Amend1e.pdf	2023	body	Intl	yr, cycles	standard	search-summary
W1-S13n	AIS-195A: GCR 2016; PESO may use IS 16735, ISO 19881, R134, GTR 13	ARAI (HMR)	AIS-195A Safety and Procedural Requirements for hydrogen vehicles	https://hmr.araiindia.com/api/AISFiles/AIS-195A_dc4d3241-c4b7-466f-8c37-6bd23851f0cc.pdf	2023-24	clause on containers	India	—	standard	blocked; search-summary
W1-S13o	PESO Gas Cylinders (Amendment) Rules 2024 (Hydrogen)	PESO	Gas Cylinders (Amendment) Rules, 2024 (Hydrogen)	https://peso.gov.in/web/en/gas-cylinders-amendment-rules-2024-hydrogen	2024	page	India	—	regulation	blocked
W1-S14a	50 L CNG cylinder: 48 kg, 835 mm, 316 mm	GAIL Gas (CGD)	CNG FAQ	https://gailcgd.gail.co.in/CGD/Entry/CNGfaq	n.d.	FAQ	India	kg, mm	spec (typical)	excerpt
W1-S14b	60 L Type 1: OD 325 mm, 61.4 kg, ISO 11439, 200 bar	Anhui Clean Energy (vendor)	Diameter 325 mm 60L CNG cylinder	https://www.gascylindersupplier.com/diameter-325-mm-60l-cng-cylinder	n.d.	page	China	mm, kg	spec	non-Indian vendor
W1-S14c	60 L Type 1: 69 kg	PWENT (vendor)	Steel 60 Liter CNG Gas Cylinder 200 Bar	https://pwent.eu/en/products/products/gas-cylinder-products-high-pressure-low-pressure/cng-gas-cylinder/steel-60-liter-cng-gas-cylinder-200-bar/	n.d.	page	EU	kg	spec	vendor
W1-S14d	R110: 20 g / 8 g for M1, N1 (10/5 M2,N2; 6.6/5 M3,N3)	UNECE	Regulation No. 110 Rev.3	https://www.unece.org/fileadmin/DAM/trans/main/wp29/wp29regs/2015/R110r3e.pdf	2015	container installation clause	Intl	g	regulation	excerpt (verbatim)
W1-S14e	AIS-024 / AIS-028 Ver.4 CNG code of practice	ARAI (HMR) / MoRTH	AIS-024 (Version 4) AIS-028 (Version 4)	https://hmr.araiindia.com/Control/AIS/825201593205AM4_AIS-024_and_AIS-028_CNG-Ver4_F.pdf	2015	—	India	—	standard	blocked
W2-S1	H-CNG added to CMVR Rule 115; IS 17314:2019	MoRTH (Gazette); PIB	G.S.R. 585(E) "hydrogen CNG"; PIB "Notification allowing use of H-CNG issued"	https://morth.nic.in/sites/default/files/notifications_document/GSR%20585(E)%2025%20September%202020%20hydrogen%20CNG.pdf ; https://www.pib.gov.in/PressReleasePage.aspx?PRID=1659706	25-09-2020	Rule 115 insertion	India	In force	Blocked; wording from search summary	
W2-S2	IS 17314:2019 H-CNG spec, 18 % H2	BIS (via PIB/press)	IS 17314:2019 Hydrogen enriched CNG for automotive purposes	(BIS catalogue; not accessed)	2019	—	India	In force	18 % figure from press, not standard text	
W2-S3	Rule 115 CNG kit provisions; s.52 14-day endorsement	Indian Kanoon (CMVR text)	Section 115 / 115B, CMVR 1989	https://indiankanoon.org/doc/10585089/ ; https://indiankanoon.org/doc/67316961/	—	Rule 115 (sub-rules not numbered here)	India	In force	Summary only	
W2-S4	ARAI retrofitment certification (Rule 115, Rule 126 agencies)	ARAI	Vehicle Retrofitment Certification in India	https://www.araiindia.com/certification/certification/retrofitment	—	—	India	Current	Blocked	
W2-S5	Retrofit framework (CNG/LPG/EV; AIS-123)	EVreporter (ARAI compiled)	Regulatory Framework for Retrofitting in India	https://evreporter.com/regulatory-framework-for-retrofitting-in-india/	—	—	India	—	Secondary	
W2-S6	BS-VI CNG/LPG retrofit permitted	MoRTH; PIB; Business Standard	G.S.R. 625(E) "retrofitment in BS VI"	https://morth.gov.in/sites/default/files/notifications_document/7GSR%20625(E)%2011th%20August%202022%20retrofitment%20in%20BS%20VI.pdf ; https://pib.gov.in/PressReleasePage.aspx?PRID=1853835	11-08-2022	—	India	In force	Blocked	
W2-S7	AIS-024/028 Version 4 exist	ARAI (HMR)	AIS-024 (Version 4) / AIS-028 (Version 4)	https://hmr.araiindia.com/Control/AIS/825201593205AM4_AIS-024_and_AIS-028_CNG-Ver4_F.pdf	—	—	India	Published	Blocked; title only	
W2-S8	TA cost Rs 40-50 lakh, 3-yr validity	Autocar Professional	Delays in MoRTH guidelines for BS VI CNG kits hurting retrofit business	https://www.autocarpro.in/news-national/delays-in-morth-guidelines-for-bs-vi-cng-kits-hurting-retrofit-business-92660	2022 (approx.)	—	India	—	Industry claim	
W2-S9	Rule 126 provisos (retrofit self-certification; imported CBUs for testing/research)	Commercial Law Publishers (CMVR e-book)	The Central Motor Vehicles Rules, 1989 — page 335 (Rule 126)	https://ebook.commercialawpublishers.com/fa/cmvr/files/basic-html/page335.html	ed. incl. GSR 221(E) 30-03-2021	Rule 126 provisos	India	In force (pre-2026 text)	Verbatim via Firecrawl; may pre-date 2026 amendment	
W2-S10	2026 amendment substituting Rule 126 provisos; self-certification	The Bar Bulletin; TeamLease RegTech	Centre Notifies Changes to Motor Vehicle Type Approval Norms...; Draft CMV (...Amendment) Rules 2025	https://thebarbulletin.com/centre-revises-motor-vehicle-type-approval-norms-self-certification-retrofitted-vehicles/ ; https://teamleaseregtech.com/updates/article/52441/draft-central-motor-vehicles-amendment-rules-2025/	2026	Rule 126	India	In force (per source)	GSR no./date NOT FOUND	
W2-S11	AIS 157:2020 notified for M/N FCEVs; ISO 14687	MoRTH via India Code; GKToday; Mercom	9th GSR 579(E) dated 23 September 2020 hydrogen fuel cell	https://www.indiacode.nic.in/ViewFileUploaded?path=AC_CEN_30_42_00009_198859_1517807326286/notificationindividualfile&file=9th_gsr_579%28e%29_dated_23_september_2020_hydrogen_fuel_cell.pdf ; https://indiacode.ecourtsindia.com/rules/7ec3e5bb/ ; https://gktoday.in/current-affairs/standards-for-safety-evaluation-of-vehicles-propelled-by-hydrogen-fuel-cells ; https://www.mercomindia.com/inclusion-hydrogen-fuel-central-motor-vehicles-act	23-09-2020	new rule (see S12)	India	In force	Blocked; snippet	
W2-S12	FCEV rule numbered 125MA; fuel IS 16061:2021; H2-ICE Rule 125M	Applus IDIADA	IRIS Highlights – July 2026	https://www.digitalolutions.applusidiada.com/blogs/news/iris-highlights-july-2026	07-2026	—	India	—	Secondary digest	
W2-S13	Rule 125M; AIS 195:2023; M & N	MoRTH	8th GSR 746(E) dated 16 October 2023 Hydrogen ICE	https://morth.nic.in/sites/default/files/notifications_document/8th%20GSR%20746(E)%20dated%2016th%20october%202023%20Hydrogen%20ICE.pdf	16-10-2023	Rule 125M	India	In force	Blocked; summary	
W2-S14	GSR 720(E) 04-10-2023 also cites AIS 195:2023	(search index; MoRTH)	—	—	04-10-2023	—	India	Unresolved	Not verified	
W2-S15	AIS-195 with Amd 1 exists; Annexure VIII LH2	ARAI (HMR); Scribd copy	AIS-195 with Amd 1	https://hmr.araiindia.com/api/AISFiles/AIS-195%20with%20Amd%201_eccd18bd-9eae-4e08-bbb2-7823b7964666.pdf ; https://www.scribd.com/document/856143651/AIS-195-cc6897f6-30be-4dc9-b188-b7c52017bebb	2023 / Amd 1 05-2025	1.0 Scope; Annex VIII	India	Published	Blocked; scope not captured	
W2-S16	AIS-195A CEV; TPRD definitions/tests	ARAI (HMR); Scribd	AIS-195A Safety and Procedural Requirements for ... Hydrogen Powered CEV	https://hmr.araiindia.com/api/AISFiles/AIS-195A_dc4d3241-c4b7-466f-8c37-6bd23851f0cc.pdf ; https://www.scribd.com/document/1057373433/AIS-195-A	2024-25	1.0 Scope; TPRD clauses	India	Published	Fragments only	

W2-S17	AIS-206 L-category scope; "AIS 157 and AIS 195 published for M & N"	Scribd (AIS-206 copy)	Hydrogen Vehicle Safety Standards AIS-206	https://www.scribd.com/document/902873977/AIS-206	2025	1.0 Scope	India	Published/draft	Fragment
W2-S18	AIS-157A/DF (FC CEV) draft	ARAI (HMR)	Draft AIS-157A/DF June 2024	https://hmr.aralindia.com/api/AISFiles/AIS157A_CE_V_DF_67e85335-2f04-40ee-ac66-ecb704d7bd46.pdf	06-2024	Scope	India	Draft	Blocked
W2-S19	ARAI AIS list updated 7 May 2026	ARAI (HMR)	List of published AIS (07-05-2026)	https://hmr.aralindia.com/storage/ais-details/1778147654_0_List_of_AIS_published-07_05_2026.pdf	07-05-2026	—	India	Current	Blocked
W2-S20	Hydrogen registration mark	MoRTH via Complinty/autopdp	CMV (Fourth Amendment) Rules 2025, G.S.R. 819(E)	https://complinty.com/legal-update/central-motor-vehicles-fourth-amendment-rules-2025-21166/	03-11-2025	—	India	In force	Secondary
W2-S21	7-year permit exemption incl. hydrogen	MoRTH via Autocar Pro/ChiniMandi	Electric, Ethanol, Hydrogen CVs Get 7 Year Permit Exemption	https://www.autocarpro.in/news/electric-ethanol-hydrogen-cvs-get-7-year-permit-exemption-133500	07-2026 (Gazette 6 s.66(1) MV Act July)	—	India	In force	GSR no. not captured
W2-S22	Draft +5 yr age limit (Rule 88)	MoRTH via Business Standard/FPJ	Govt proposes 5-yr age extension for EV, CNG, hydrogen CVs	https://www.business-standard.com/industry/auto/govt-proposes-5-yr-age-extension-for-ev-cng-hydrogen-commercial-vehicles-126081800231_1.html	18-08-2026	Rule 88	India	Draft	Secondary
W2-S23	Draft GCR 2024 GSR 157(E) hydrogen provisions	DPIIT/PESO via TaxGuru, Lexplosion, RenewableWatch	Draft Gas Cylinders (Amendment) Rules, 2024	https://taxguru.in/corporate-law/draft-gas-cylinders-amendment-rules-2024.html ; https://peso.gov.in/web/en/gas-cylinders-amendment-rules-2024-hydrogen	05-03-2024	defs; rule 4; Sch. I	India	Draft (superseded by 2025 rules)	Blocked
W2-S24	GCR 2025 GSR 225(E): H2 cylinder on vehicle; 3-yr test; 15-yr life; codes	DPIIT via Lexplosion, LegalitySimplified, TeamLease, Sparrow RMS	Gas Cylinders (Amendment) Rules, 2025	https://lexplosion.in/ministry-of-commerce-and-industry-notifies-the-gas-cylinders-amendment-rules-2025/ ; https://www.legalitysimplified.com/gas-cylinders-amendment-rules-2025/ ; https://sparrowrms.in/sparrow-rms/wp-content/uploads/2025/04/GCR-amendment-2025-1.pdf	11-04-2025	rule 2, 4, 35(?) , Sch. I/IV	India	In force	Blocked; clause numbers unverified
W2-S25	GSR 386(E), 501(E), 839(E) 2025 amendments	DPIIT via WorldtradesScanner, LegalitySimplified, TeamLease	GCR Third/Fourth Amendment Rules 2025	https://worldtradesScanner.com/DPIIT%20Notification%20G.S.R.16-06/28-07/12-11-2025 ; https://www.worldtradesScanner.com/DPIIT%20Notification%20G.S.R.%20839(E)%20dated%2012%20November%202025.htm	16-06 / 28-07 / 12-11-2025	rule 4	India	In force	Secondary
W2-S26	SMPV(U) 2025 amendment hydrogen	PESO; TaxGuru	SMPV(Unfired)(Amendment) Rules, 2025	https://peso.gov.in/web/en/smpvunfiredamendment-rules-2025 ; https://taxguru.in/corporate-law/static-mobile-pressure-vessels-unfired-amendment-rules-2025.html	eff. 05-06-2025	defs; Form AS-1	India	In force	Blocked
W2-S27	PESO Form-H online module for H2 dispensing stations	IndiaLaw; LegalitySimplified; VahanBazaar	PESO's New Online Module for Licensing Compressed Hydrogen Dispensing Stations	https://www.indialaw.in/blog/civil/pesos-module-hydrogen-stations-in-india/	14-05-2025	GCR Form H	India	Current	Secondary
W2-S28	PESO SOPs (SMPV 02-12-2024; GCR Jan 2025)	PESO	SOP for SMPV(U) Rules 2016; GCR SOP	https://peso.gov.in/web/sites/default/files/2024-12/SOP%20for%20SMPV(U)%20Rules,%202016-Final%20-%2002_12_2024.pdf ; https://www.peso.gov.in/web/sites/default/files/2025-01/GCR%20SOP.pdf	2024-25	—	India	Current	Blocked
W2-S29	PESO licence to IOCL Faridabad for H-CNG dispensing	Business Standard	PESO grants permission to Indian Oil to set up hydrogen plant	https://www.business-standard.com/content/b2b-chemicals/peso-grants-permission-to-indian-oil-to-set-up-hydrogen-plant-115051100485_1.html	05-2015	—	India	Historical	Secondary
W2-S30	Type IV H2 cylinder PESO approvals (Time Technoplast)	Business Standard; Autocar Pro; company PR	Time Technoplast gets final approval from PESO...; 250 L approval	https://www.business-standard.com/markets/capital-markets/news/time-technoplast-gets-final-approval-from-peso-for-manufacturing-composite-cylinders-124062100121_1.html ; https://www.timetechnoplast.com/pdf/investors-calendar/press-release/approval-for-manufacturing-of-type-iv-composite-cylinders-for-hydrogen.pdf	03-2024 / 06-2024 / 01-2026	—	India	Current	Company-reported
W2-S31	Indian hydrogen standards landscape (AIS 195A, AIS 206, IS 19036, IS 16061, IS 18538-8)	MNRE with CEEW (PIB-hosted)	Green Hydrogen Standards and Approval Systems in India	https://static.pib.gov.in/WriteReadData/specificdocs/documents/2024/may/doc2024510336201.pdf ; https://nghtm.mnre.gov.in/admin/uploads/resources/171557269282523Green%20Hydrogen%20Standards%20and%20Approval%20Systems%20in%20India.pdf	05-2024	tables	India	Current	Blocked
W2-S32	BIS hydrogen standards list (23)	BIS	List of Indian Standards related to Hydrogen	https://www.bis.gov.in/wp-content/uploads/2023/10/List_Standards_Hydrogen.pdf	10-2023	—	India	Current	Blocked
W2-S33	MNRE transport pilot guidelines	MNRE; JSA; Business Standard	Scheme Guidelines for pilot projects ... green hydrogen in the transport sector	https://mnre.gov.in/en/notice/scheme-guidelines-for-implementation-of-pilot-projects-for-use-of-hydrogen-in-the-transport-sector-under-the-national-green-hydrogen-mission-nghtm/	14-02-2024	objectives; steering committee	India	In force	Blocked
W2-S34	5 pilots, 37 vehicles, 9 HRS, 10 routes, Rs 208 cr	Lok/Rajya Sabha reply via DownToEarth, DD News, Elets	As told to Parliament (25 March 2025)	https://www.downtoearth.org.in/environment/as-told-to-parliament-march-25-2025-37-cross-country-vehicles-9-fuelling-stations-planned-under-green-hydrogen-mission ; https://ddnews.gov.in/en/india-rolls-out-pilot-projects-for-hydrogen-powered-buses-and-trucks/	25-03-2025	—	India	Current	Secondary
W2-S35	Mirai pilot with ICAT	Zee News; BusinessToday	Gadkari inaugurates pilot project ... Toyota Mirai	https://zeenews.india.com/electric-vehicles/nitin-gadkari-inaugurates-pilot-project-to-test-viability-of-hydrogen-vehicles-in-india-2445693.html	16-03-2022	—	India	Historical	Press
W2-S36	Tata H2 truck trials, 16 vehicles, 24 months	Tata Motors; News on AIR	Tata Motors Drives India's Green Future with Country's First Hydrogen Truck Trials	https://www.tatamotors.com/press-releases/tata-motors-drives-indias-green-future-with-countrys-first-hydrogen-truck-trials ; https://www.newsair.gov.in/first-trials-of-hydrogen-powered-heavy-duty-trucks-flagged-off	04-03-2025	—	India	Current	Company/press
W2-S37	ISO 14687 grades; 2025 edition	ISO	ISO 14687:2019 / ISO 14687:2025 Hydrogen fuel quality — Product specification	https://www.iso.org/standard/69539.html ; https://www.iso.org/standard/82660.html	2019 / 2025	Table 1 grades	Intl	2025 current	Blocked; abstract
W2-S38	ISO 17268 pressure classes	ISO	ISO 17268:2020; ISO 17268-1:2025	https://www.iso.org/standard/68442.html	2020 / 2025	scope	Intl	Current	Abstract
W2-S39	SAE J2601 scope, edition	H2Tools (PNNL)	SAE J2601 Fueling Protocols for Light Duty Gaseous Hydrogen Surface Vehicles	https://h2tools.org/fuel-cell-codes-and-standards/sae-j2601-fueling-protocols-light-duty-gaseous-hydrogen-surface	4th ed. 05-2020	—	Intl	Current (revision open)	Verbatim page text captured
W2-S40	ISO 19881:2025; ISO 19880-1/-2	ISO; iTeh; H2Tools	ISO 19881:2025; ISO/AWI 19880-1; ISO 19880-2:2025	https://www.iso.org/standard/19881.html ; https://www.iso.org/standard/93897.html ; https://www.iso.org/obp/ui/en/#iso:std:85839:en	2025	scope	Intl	Current	Abstracts
W2-S41	GTR 13 Amend. 1 (2023); R134 02 series	UNECE; InterRegs; GAR	ECE/TRANS/180/Add.13/Amend.1	https://unece.org/sites/default/files/2023-07/ECE-TRANS-180-Add.13-Amend1e.pdf ; https://www.interregs.com/articles/spotlight/297/updated-unece-regulations-relating-to-hydrogen-fuelled-vehicles-published	21-06-2023	—	Intl	In force	Blocked
W2-S42	GTR 13 numeric limits (4 %/8 %; leak; single failure)	Frontiers in Energy Research	Analysis of safety technical standards for hydrogen storage in fuel cell vehicles	https://www.frontiersin.org/journals/energy-research/articles/10.3389/fenrg.2024.1376739/full	2024	—	Intl	—	Review; "118 NL/h" vs NL/min discrepancy
W2-S43	H2 properties	H2Tools (PNNL)	Basic Hydrogen Properties Chart; Flammability; Auto-Ignition	https://h2tools.org/basic-hydrogen-properties-chart ; https://h2tools.org/faq-keywords/flammability	—	—	Intl	Current	Blocked; summary
W2-S44	H2 vs other fuels (relative)	US DOE EERE	Hydrogen: Similar but Different (fact sheet)	https://www1.eere.energy.gov/hydrogenandfuelcells/pdfs/h2_safety_fsheets.pdf	—	—	US	Current	Blocked
W2-S45	NFPA 2 repair-garage ventilation	NFPA via UpCodes; NFPA	NFPA 2 (2023) Hydrogen Technologies Code — repair garages	https://up.codes/s/repair-garages-for-vehicles-fueled-by-lighter-than-air-motor-fuels ; https://www.nfpa.org/product/nfpa-2-hydrogen-technologies-code/p0002code	2023 ed.	Ch. 6; repair garages	US	Current	Secondary
W2-S46	ISO/TR 15916:2015 → ISO/TS 15916:2026	ISO	Basic considerations for the safety of hydrogen systems	https://www.iso.org/standard/56546.html ; https://www.iso.org/standard/89423.html	2015 / 2026	—	Intl	TS current	Abstract

W2-S47	UK guidance (HSG261; hydrogen hub; withdrawn C&U note)	HSE; GOV.UK	HSG261; Hydrogen Safety; Construction and use of hydrogen-powered vehicles [withdrawn]	https://www.hse.gov.uk/pubns/books/hsg261.htm ; https://solutions.hse.gov.uk/safe-net-zero/hydrogen-safety ; https://www.gov.uk/government/publications/construction-and-use-of-hydrogen-powered-vehicles/construction-and-use-of-hydrogen-powered-vehicles	—	—	UK	HSG261 current; Blocked GOV.UK note withdrawn	
W2-S48	Kjørbo root cause	Nel ASA; gasworld; Gexcon	Status and Q&A regarding the Kjørbo incident; Root cause ... identified	https://nelhydrogen.com/status-and-qa-regarding-the-kjorbo-incident/ ; https://www.gasworld.com/story/root-cause-of-norway-hydrogen-station-explosion-identified/	06/07-2019	—	Norway	—	Company statements
W2-S49	Santa Clara causes	Air Products (via H2Tools); Gexcon	Report on the June 2019 Hydrogen Explosion and Fire Incident in Santa Clara	https://h2tools.org/sites/default/files/2021-06/AP_Santa_Clara_Incident_Review_Report_Rev1.pdf	2021	—	US	—	Blocked; summary
W2-S50	Gangneung 2019; Busan 2024; HIAD	ScienceDirect (IJHE); CleanTechnica	Evaluating hydrogen risk management policy PR: Lessons learned from three hydrogen accidents in South Korea	https://www.sciencedirect.com/science/article/abs/pii/S0360319923012132 ; https://cleantechnica.com/2025/01/16/korean-hydrogen-explosions-leaks-auction-failure-impeachment-may-cause-pivot/	2023 / 2025	—	Korea	—	Abstract/press
W2-S51	Euro NCAP Mirai 2021; Nexo 2018	Euro NCAP; Hyundai	Toyota Mirai — Euro NCAP 2021 results — 5 stars; Hyundai Nexo 5-star	https://www.euroncap.com/en/results/toyota/mirai/43501 ; https://www.hyundai.com/worldwide/en/newsroom/detail/all-new-hyundai-nexo-achieves-maximum-five-star-rating-in-euro-ncap-safety-assessment-0000001089	2021 / 2018	—	EU	—	Blocked; summary
W2-S52	DOE on NOx from hydrogen combustion	US DOE EERE	Does the use of hydrogen produce air pollutants such as nitrogen oxides?	https://www.energy.gov/eere/fuelcells/does-use-hydrogen-produce-air-pollutants-such-nitrogen-oxides	—	—	US	Current	Requested /cmei/fuels/ URL not found; content from summary
W2-S53	IMT-25; Rs 60 TP loading	India Motor Tariff via GoDigit; IRDAI policy wordings; Acko; HDFC Ergo	India Motor Tariff (IMT) Endorsements — IMT.25	https://www.godigit.com/content/dam/godigit/directportal/en/downloads/car/Motor-IMT%20Endorsements.pdf ; https://www.hdfcergo.com/blogs/car-insurance/insurance-guide-for-cng-lpg-converted-cars-2025	—	IMT-25	India	Current	Blocked; summary
W2-S54	Form 22D (Rule 47B(1)); RTO documents	MoRTH (Parivahan); Policybazaar; RTOIndiaHelp	FORM 22D — Endorsement of alteration/retrofitment	https://parivahan.gov.in/sites/default/files/DownloadForm/cmvr/FORM-22D.pdf ; https://www.policybazaar.com/rto/form-22d/	—	Rule 47B(1)	India	Current	Blocked
W2-S55	Maruti warranty exclusion for retrofitted CNG	Maruti owner's manual (via GoMechanic/RideNRepair)	Dzire Owner's Manual (warranty section)	https://marutistorage.new.blog.core.windows.net/msinitwebpdf/Dzire_Owners_Manual_1.pdf	—	warranty exclusions	India	Current	Not verified
W2-S56	Rule 41 trade-certificate purposes	Indian Kanoon; transportinfo.in	The Central Motor Vehicles Rules, 1989 — Rule 41	https://indiankanoon.org/doc/28072254/ ; https://www.transportinfo.in/simba/cmvr/cmvr.htm	—	Rule 41	India	In force	Summary
W2-S57	Rule 126 test agencies	PIB; TÜV	Rule 126 CMVR	https://www.pib.gov.in/newsite/erecontent.aspx?relid=279664 ; https://www.tuv.com/market-access-services/en/certification-filter/cmvr-(central-motor-vehicle-rules-notified-by-the-ministry-of-road-transport-and-highways)-document/930900675/Regulatory-Update-Nov-Dec-2022	—	Rule 126	India	In force	Summary
W2-S58	AIS-137 (Part 9) draft Amd 1 for CNG/LPG retrofit	Scribd (regulatory update)	Regulatory Updates: CMVR Nov-Dec 2022	https://www.scribd.com/document/930900675/Regulatory-Update-Nov-Dec-2022	12-2022	—	India	Draft	Fragment
W2-S59	CMVR-TSC minutes (ARAI hydrogen projects)	ACMA	MoM 62nd CMVR-TSC	https://www.acma.in/uploads/docmanager/MoM_62_CMVR-TSC.pdf	—	—	India	—	Blocked
W2-S60	Analyst: H2-ICE retrofit not yet legalised	S&P Global Mobility	Hydrogen trucks in India: green hydrogen potential	https://www.spglobal.com/automotive-insights/en/blogs/2025/03/hydrogen-trucks-in-india-green-hydrogen-potential	03-2025	—	India	—	Opinion
W2-S61	CV OEM hydrogen shift	Business Standard	India's vehicles industry quietly begins shift to hydrogen-powered vehicles	https://www.business-standard.com/industry/auto/indian-cv-majors-drive-hydrogen-transition-with-h2-vehicles-125061600725_1.html	16-06-2025	—	India	—	Press
W2-S62	Gazette 27-03-2026 doc citing AIS 157:2020	Gazette of India via Scribd	CG-DLE-27032026-...	https://www.scribd.com/document/1019065447/271279	27-03-2026	—	India	Unknown	Content unresolved
W2-S63	MNRE NGHM regulations directory	MNRE	National Green Hydrogen Mission Portal — Regulations	https://nghm.mnre.gov.in/regulations?language=en	—	—	India	Current	Blocked
W2-S64	State regulation of retrofit centres	Kerala MVD	Conversion of vehicles into CNG/LPG mode — Regulation and control of Retro fitment centres	https://mvd.kerala.gov.in/en/node/1386	—	—	Kerala	—	Blocked
W2-S65	NCR age restrictions after CNG conversion	Delhi Govt (Switch Delhi) / press	Switch Delhi — retro-fitment	https://ev.delhi.gov.in/retro-fitment	—	—	Delhi	—	Secondary
W2-S66	87 standards adopted / 59 under development	CEEW	How can India fix gaps for green hydrogen safety standards?	https://www.ceew.in/publications/how-can-india-adopt-international-standards-for-national-green-hydrogen-mission	2024	—	India	—	Think-tank
W2-S67	PESO hydrogen presentations at PNGRB	PESO via PNGRB	Statutory rules/regulations for CNG stations (PESO); Stakeholders interaction on hydrogen	https://www.pngrb.gov.in/pdf/press-note/PESO_15032024.pdf ; https://pngrb.gov.in/pdf/press-note/20250306_PESO.pdf	03-2024 / 03-2025	—	India	—	Blocked
W3-S1	72.10 lakh CNG vehicles registered	AIR News (newsonair.gov.in) reporting Lok Sabha written reply, MoS PNG Suresh Gopi	Total sales of CNG vehicles have increased over the past few years: Suresh Gopi	https://www.newsonair.gov.in/total-sales-of-cng-vehicles-have-increased-over-the-past-few-years-suresh-gopi	—	—	India	vehicles observed (Vahan)	page blocked; figure from search passage
W3-S2	58.61 lakh (Mar 2023) → 81.95 lakh (Mar 2025) CNG vehicles; +25% FY25; PNGRB VAT plea	ET Energyworld / UNI / The Hindu BusinessLine / Energywatch quoting PNGRB	PNGRB expands CGD network to 784 districts, sets 2034 target of 12.6 crore PNG connections; City Gas Distribution network now covers nearly 100 pc...; PNGRB in talks with states to rationalise VAT	https://energy.economictimes.indiatimes.com/news/oil-and-gas/pngrb-expands-cgd-network-to-784-districts-sets-2034-target-of-12-6-crore-png-connections/123553997 ; https://www.thehindubusinessline.com/markets/commodities/pngrb-in-talks-with-states-to-rationalise-vat-on-natural-gas-cng/article69991018.ece	—	—	India	vehicles observed (PNGRB citing Vahan)	secondary reporting of PNGRB statement
W3-S3	18,336 CNG stations & 12.63 crore PNG by 2034; ~1,206 stations added FY25	Business Standard; IBEF	PNGRB targets 12.6 crore PNG connections, 18,336 CNG stations by 2034	https://www.business-standard.com/economy/news/pngrb-targets-12-6-crore-png-connections-18336-cng-stations-by-2034-125082700329_1.html ; https://www.ibef.org/news/india-eyes-126-million-png-connections-18-000-cng-stations-by-2034	—	—	India	stations target / observed	secondary
W3-S4	738 (2014) → 8,150+ CNG stations; 18,000 by 2030	Hardeep Singh Puri (MoPNG) on X; The Hindu BusinessLine	One Nation, One Gas Grid (X post); India to double CNG stations to 18,000 by 2030 amid One Nation One Gas Grid plan: Puri	https://x.com/HardeepSPuri/status/1961324393344491893 ; https://www.thehindubusinessline.com/economy/logistics/india-to-double-cng-stations-to-18000-by-2030-amid-one-nation-one-gas-grid-plan-puri/article6998711.ece	—	—	India	stations observed (ministerial) / target	ministerial statement, not a PNGRB table
W3-S5	outdated "2,830 stations" figure	PRS Legislative Research	National Gas Grid Including PNG and CNG (Standing Committee report summary)	https://prsindia.org/policy/report-summaries/national-gas-grid-including-png-and-cng	—	—	India	stations	outdated used only to flag staleness
W3-S6	17,700 stations by 2030 (MWP, rounds to 11A); "17,500 by 2030"; "7,400 in 2025"	Autocar Professional; Construction World	India's CNG infra development in high gear; India Plans 17,500 CNG-PNG Stations... by 2030	https://www.autocarpro.in/feature/indias-auto-industry-set-for-a-cng-boom-as-gas-infra-inches-to-100-penetration-117791 ; https://www.constructionworld.in/policy-updates-and-economic-news/india-plans-17500-cng-png-stations-aims-for-120-million-connections-by-2030/52110	—	—	India	stations	commitment/ target industry press citing MoPNG/PNGRB
W3-S7	CNG ₹86.98/kg Delhi (IGL, +₹3.89); ₹88/kg Mumbai (MGL, +₹2); reasons	Goodreturns; Equentis; ETV Bharat; Infralog	CNG Price in New Delhi Today; CNG, PNG Price Hike: New Rates in Mumbai and Delhi-NCR; CNG Price Hiked By Rs 2 Per Kg In Mumbai; MGL hikes CNG... amid rising spot RLNG costs	https://www.goodreturns.in/cng-price-in-new-delhi.html ; https://www.equentis.com/blog/cng-png-price-hike-new-rates-in-mumbai-and-delhi-ncr/ ; https://www.etvbharat.com/en/business/cng-price-hiked-by-rs-2-per-kg-in-mumbai-enn26090104129	—	—	Delhi; Mumbai	₹/kg observed (retail)	aggregator pages; IGL/MGL notices not opened
W3-S8	Pune CNG ₹97.75/kg from 1 Sep 2026	MNGL (official rate card); Times of India	CNG Rate Card — Revision in Prices w.e.f. midnight of 1st-Sep-2026: Pune GA Rs 97.75/kg; MNGL hikes CNG price by Rs2 per kg in Pune	https://mngl.in/cng/rate-card ; https://timesofindia.indiatimes.com/city/pune/mngl-hikes-cng-price-by-rs2-per-kg-in-pune/articleshow/133662320.cms	—	—	Pune GA	₹/kg observed	primary text visible in passage
W3-S9	Ahmedabad CNG ₹94.02/kg	CarDekho / Goodreturns city price pages	CNG price in India Today	https://www.cardekho.com/cng-price ; https://www.goodreturns.in/cng-price.html	—	—	Ahmedabad	₹/kg observed (aggregator)	supplier notice not verified

W3-S10	Maruti FY25 CNG 6.12-6.2 lakh, 33%; 71.6% CNG-segment share (2024)	Autocar India; Autocar Professional	One in every three Maruti sold in FY2025 was a CNG; One-in-Three Maruti Vehicles Sold in FY25 Ran on CNG	https://www.autocarindia.com/industry/one-in-every-three-maruti-sold-in-fy2025-was-a-cng-434994 ; https://www.autocarpro.in/news/one-in-three-maruti-vehicles-sold-in-fy25-ran-on-cng-125695	India	units; %	observed (company/industry data)	secondary
W3-S11	Maruti FY26 sales 24.22 lakh, exports 4.47 lakh; biogas plants ₹561 crore	Maruti Suzuki (press releases, not accessible); Automotive World; scanx	Maruti Suzuki announces Financial Results for FY2025-26; Annual Integrated Report 2025-26	https://www.marutisuzuki.com/corporate/media/press-releases/2026/april/maruti-suzuki-announces-financial-results-for-fy2025-26 ; https://www.marutisuzuki.com/corporate/media/press-releases/2026/august/maruti-suzuki-annual-integrated-report-2025-26 ; https://www.automotiveworld.com/news/annual-integrated-report-maps-maruti-suzukis-growth-push/	India	units	observed	primary blocked; via secondary
W3-S12	Maruti FY26 CNG >7 lakh; 42% share "last month"; FY27 target 9 lakh	ET Auto; Business Standard; Moneycontrol	Maruti Suzuki targets nearly 30% jump in CNG sales to 9 lakh units in FY27; Maruti Suzuki targets 900,000 units of CNG vehicle sales this fiscal	https://auto.economictimes.indiatimes.com/news/passenger-vehicle/maruti-suzuki-aims-for-30-increase-in-cng-vehicle-sales/132637759 ; https://www.business-standard.com/companies/news/maruti-suzuki-targets-900-000-units-of-cng-vehicle-sales-this-fiscal-126072600153_1.html	India	units; %	observed / target 42% is a monthly figure	
W3-S13	PV fuel mix FY26: petrol 52.6% (24.61 of 46.83 lakh), CNG 21.7%; Jul 2026 CNG 24.67%	AutopunditZ (Vahan-based analyst)	Passenger Cars Fuel-wise Sales Analysis - India FY2026	https://www.autopunditZ.com/post/india-passenger-vehicle-fuelwise-sales-analysis-fy2026	India	% of retail registrations	observed (analyst compilation)	not SIAM; CNG % seen via social summary
W3-S14	SIAM quarterly highlights FY26 (not accessible)	SIAM	Key Highlights from Q1/Q2 FY2025-26	https://www.siam.in/pressrelease-details.aspx?mpgid=48&pgidtrail=50&pid=583;...pid=591	India	—	—	blocked; no CNG split retrieved
W3-S15	*4,000+ authorised installation centres*; aftermarket ₹2,500-3,500 crore/yr; brand shares	LetsGO CNG (vendor blog)	CNG Conversion Statistics India 2026: 50 Key Facts	https://lets gocng.in/cng-conversion-statistics-india-2026/	India	count; ₹ crore	vendor estimate	unofficial; low confidence
W3-S16	RFC registries (Vahan CNG Maker portal; IGL list; Delhi RFC application)	MoRTH Vahan; IGL; Delhi Transport Dept	Available Fitment Center (CNG Maker); RTO Approved CNG Kit Retrofit Centres; Application for CNG Retro Fitment	https://vahan.parivahan.gov.in/cngmaker/vahan/user/retrofitment/availableFitmentCenter.xhtml ; https://www.iglonline.net/cng-retrofitment-center ; https://transport.delhi.gov.in/transport/application-cng-retrofitment	India / Delhi	—	registry	no totals retrievable
W3-S17	Delhi aggregator scheme EV milestones; notified 21 Nov 2023	Legitquest (text reproduction); Hindustan Times; Delhi Transport Dept (X); Medianama	Delhi Motor Vehicle Aggregator and Delivery Service Provider Scheme, 2023	https://www.legitquest.com/act/delhi-motor-vehicle-aggregator-and-delivery-service-provider-scheme-2023/D483 ; https://www.hindustantimes.com/cities/delhi-news/delhi-govt-notifies-cab-aggregator-policy-fleets-to-switch-to-electric-101701281802617.html ; https://x.com/TransportDelhi/status/1730101116590059896	Delhi	% of new onboarding	rule	6-month/1-year milestones only via HT
W3-S18	~60,000 taxis registered in Delhi; CNG/clean-fuel taxi permits 15 years; SC CNG requirement	Outlook India	Delhi: All Taxis Running On CNG, Other Clean Fuels To Have Permit Validity Of 15 Years	https://www.outlookindia.com/national/delhi-all-taxis-running-on-cng-other-clean-fuels-to-have-permit-validity-of-15-years-news-296571	Delhi	vehicles	observed (dept data, undated)	secondary
W3-S19	CAQM: only CNG/EV new aggregator/delivery vehicles in NCR from 1 Jan 2026; Haryana rules	Deccan Herald; Free Press Journal; Swarajya	No new petrol or diesel vehicles for cab aggregators, delivery firms in NCR from January 2026: CAQM; Haryana Cabinet Mandates CNG, EV Or Cleaner Fuel For All New Aggregator Vehicles In NCR From 2026	https://www.deccanherald.com/amp/story/india%2Fno-new-petrol-or-diesel-vehicles-for-cab-aggregators-delivery-firms-in-ncr-from-january-2026-caqm-3571433 ; https://www.freepressjournal.in/amp/india/haryana-cabinet-mandates-cng-ev-or-cleaner-fuel-for-all-new-aggregator-vehicles-in-ncr-from-2026	NCR	—	rule	secondary
W3-S20	Delhi 8.76 m vehicles (19 Mar 2026); taxis limited growth; e-buses 4,338	The Wire (reporting Delhi Economic Survey 2025-26); Delhi Planning Dept PDF (blocked)	Service Sector, Transport, Water: Highlights of Delhi's Economic Survey 2025-26	https://m.thewire.in/article/economy/service-sector-transport-water-highlights-of-delhis-economic-survey-2025-26/amp ; https://delhiplanning.delhi.gov.in/sites/default/files/2026-03/economic_survey_english_0.pdf	Delhi	vehicles	observed	PDF blocked
W3-S21	60-driver Delhi study; longer hours	Springer (book chapter)	Are Ride-Sharing Platforms Good for Indian Drivers? An Investigation of Taxi and Auto-Rickshaw Drivers in Delhi	https://link.springer.com/chapter/10.1007/978-3-030-64697-4_10	Delhi	—	observed (survey)	small sample
W3-S22	ICCT ride-hailing TCO: BEV ₹8.8→5.8/km (2019→2020); ICCT 2019 not competitive		Updated analysis of incentives for electrifying India's four-wheeler ride-hailing fleet; Are electric four-wheelers becoming more cost attractive as ride-hailing vehicles in India?; (vendor rule-of-thumb: Zevpoint EV fleet charging guide)	https://theicct.org/publication/updated-analysis-of-incentives-for-electrifying-indias-four-wheeler-ride-hailing-fleet/ ; https://theicct.org/are-electric-four-wheelers-becoming-more-cost-attractive-as-ride-hailing-vehicles-in-india/ ; https://www.zevpoint.com/blogs/guides/ev-fleet-charging-solutions-taxi-logistics-india	Delhi; Telangana	₹/km	modelled	PDFs blocked; daily-km assumption not seen
W3-S23	BluSmart collapse: SEBI order 15 Apr 2025; ₹978 crore diversion; leasing model	Business Today; Down To Earth; Zentrum Law	SEBI turns heat on Gensol, BluSmart's house of cards wobbles; What BluSmart got wrong—and right	https://www.businesstoday.in/latest/corporate/story/sebi-turns-heat-on-gensol-blusmarts-house-of-cards-wobbles-472292-2025-04-16 ; https://www.downtoearth.org.in/air/what-blusmart-got-wrongand-right	India	₹ crore	observed	secondary
W3-S24	Everest (20,500 cars, ~2,000 EVs, 10,000 EV target); Reflex 2,000+ EVs; Evera 4,500 by Jun 2026; fleet constraints	Business Standard; EVreporter; EV Drive	Next stop for BluSmart's EVs could be Everest Fleet; India's Leading Electric 4-Wheeler Passenger Fleet Operators; Evera Fleet Expansion 4500 EVs by June 2026	https://www.business-standard.com/companies/start-ups/everest-fleet-in-talks-to-acquire-blusmart-evs-for-uber-green-125051201066_1.html ; https://evreporter.com/indias-leading-electric-4-wheeler-passenger-fleet-operators/ ; https://evdrive.in/evera-fleet-expansion-4500-electric-vehicles/	India	vehicles	observed / announced	company statements
W3-S25	Uber-Tata 25,000 Xpres-T EVs; Xpres-T price ₹13.04 lakh	Autocar India; Tata Motors Xpres-T site	Tata Motors to supply 25,000 XPRES-T EVs in largest ever fleet order; Tata XPRES-T EV Price	https://www.autocarindia.com/car-news/tata-motors-to-supply-25000-xpres-t-evs-in-largest-ever-fleet-order-427364 ; https://xprest.tatamotors.com/electric/price	India	units; ₹ lakh	announced / list price	delivery status unknown
W3-S26	used Dzire CNG from ₹3.71 lakh (Delhi); WagonR ₹1.6-6.0 lakh; new Dzire Tour S on-road ₹7.35-8.33 lakh	Cars24; Orange Book Value; CarDekho	Second Hand Maruti Dzire CNG Cars in New Delhi; Used Wagon R price and valuation; Dzire Tour S On Road Price in New Delhi	https://www.cars24.com/buy-used-maruti-dzire-cng-cars-new-delhi/ ; https://orangebookvalue.com/used-cars/maruti-suzuki-wagon-r-price ; https://www.cardekho.com/maruti/dzire-tour-s/price-in-new-delhi	Delhi	₹ lakh	listings (indicative)	not valuations
W3-S27	Tata EV prices/specs (Tiago, Punch, Nexon, Tigor)	Tata.ev; ZigWheels; Team-BHP; V3Cars	Tiago EV / Punch EV / Nexon EV price pages; Tigor EV 2026	https://ev.tata.cars/tiago/ev/price.html ; https://ev.tata.cars/punch/ev/price.html ; https://ev.tata.cars/nexon/ev/price.html ; https://www.v3cars.com/tata-cars/tigor-ev ; https://www.zigwheels.com/tata-cars/tiago-ev/ ; https://www.mgmotor.co.in/vehicles/comet-ev-electric-car-in-india ; https://www.zigwheels.com/mg-motor-cars/windsor-ev/	India	₹ lakh; kWh; km	list price / ARAI	ex-showroom, variant-dependent
W3-S28	MG Comet, Windsor prices/specs	JSW MG Motor India; ZigWheels	MG Comet EV page; MG Windsor EV 2026	https://www.mgmotor.co.in/vehicles/comet-ev-electric-car-in-india ; https://www.zigwheels.com/mg-motor-cars/windsor-ev/	India	₹; kWh; km	list price	—
W3-S29	e Vitara, Creta Electric, eC3X prices/specs	Autocar India; CarWale/Team-BHP	Price and specs of Maruti e Vitara, Hyundai Creta Electric, MG Windsor EV... compared; 2026 Citroen eC3X launched at Rs 11.99 lakh; Hyundai Creta Electric	https://www.autocarindia.com/auto-features/maruti-e-vitara-vs-rivals-prices-compared-440249 ; https://www.autocarindia.com/car-news/2026-citroen-ec3x-launched-at-rs-1025-lakh-439987 ; https://www.carwale.com/hyundai-cars/creta-electric/	India	₹ lakh; kWh; km	list price	—
W3-S30	real-world range/efficiency tests	Autocar India; ZigWheels	Real world EV range tested of top EV cars in India; Tata EVs and their real-world range	https://www.autocarindia.com/auto-features/every-mass-market-ev-real-world-range-tested-434105 ; https://www.zigwheels.com/news-features/ev-news/heres-a-look-at-all-the-tata-evs-and-their-real-world-range/54737/	India	km; km/kWh	observed (media test)	single-test conditions

W3-S31	Tata battery warranty (8 yr/1.6 lakh km; lifetime for Curvv.ev/Nexon.ev 45)	Tata Motors; Tata.ev; Spinny	Tata Motors introduces Lifetime HV Battery Warranty...; Tata EV warranty page; Tata EV Warranty Explained	https://www.tatamotors.com/press-releases/reimagining-ev-ownership-tata-motors-introduces-lifetime-hv-battery-warranty-for-curvev-ev-and-nexon-ev-45-kwh/ ; https://ev.tata.cars/service/warranty.html ; https://www.spinny.com/blog/tata-ev-warranty/	India	years/km	company terms	fleet terms differ
W3-S32	Delhi EV tariff ₹4.50/kWh LT, ₹4.00/kVAh HT; DERC PM E-DRIVE order	Mercom India; thediscombill; EMobility+	Delhi Slashes Tariffs for EV Charging; Delhi Electricity Tariff 2025-26; DERC directs EV charging infra costs... borne by operators	https://www.mercomindia.com/delhi-tariffs-ev-charging-electric-mobility ; https://thediscombill.com/tariffs/delhi/ ; https://emobilityplus.com/2026/07/03/derc-directs-ev-charging-infrastructure-costs-under-pm-e-drive-scheme-to-be-borne-by-operators-not-consumers-in-delhi/	Delhi	₹/kWh	tariff order (secondary)	DERC order not opened
W3-S33	public DC ₹18-25/kWh; AC ₹8-18; jio-bp ₹15.99	Pulse Energy; Team-BHP	EV Charging Rates in India 2026; Pricing of just Rs 15.99 per kWh — Jio BP	https://pulseenergy.io/blog/ev-charging-rates ; https://www.team-bhp.com/forum/electric-cars/273196-pricing-just-rs-15-99-per-kwh-will-jio-bp-disrupt-ev-charging-market.html	India	₹/kWh	observed (operator prices)	vendor blog
W3-S34	GST: EV 5%; small cars 18%; large 40% (GST 2.0, Sep 2025)	Bajaj Finserv; The Munim	GST on Vehicles in 2026; GST on Electric Vehicles in India 2026	https://www.bajajfinserv.in/gst-on-vehicles ; https://themunim.com/blog/gst-on-electric-vehicles/	India	%	rule	secondary explainer
W3-S35	Delhi EV Policy 2026 (w.e.f. 1 Jul 2026) provisions	Business Standard; YourStory; EV Infrastructure News	Delhi govt notifies EV Policy 2026, rolls out subsidies, tax exemptions; Delhi government notifies Delhi EV Policy 2026	https://www.business-standard.com/india-news/delhi-govt-notifies-ev-policy-2026-rolls-out-subsidies-tax-exemptions-126070100488_1.html ; https://yourstory.com/2026/07/delhi-government-notifies-delhi-ev-policy-2026 ; https://www.evinfrastructurereads.com/ev-policies/delhi-notifies-ev-policy-2026-with-tax-exemptions-for-ev	Delhi	₹; dates	rule (notified)	gazette not opened
W3-S36	5 pilot projects: 37 vehicles, 9 HRS, 10 routes, ₹208 crore; awardees; ARAI SIA	PIB (PRID 2107795, blocked); Down To Earth; DD News; Energetica India; Swarajya	As told to Parliament (March 25, 2025): 37 cross-country vehicles, 9 fuelling stations...; India rolls out pilot projects for hydrogen-powered buses and trucks; MNRE Initiates 5 Pilot Projects...	https://www.pib.gov.in/PressReleasePage.aspx?PRID=2107795 ; https://www.downtoearth.org.in/environment/as-told-to-parliament-march-25-2025-37-cross-country-vehicles-9-fuelling-stations-planned-under-green-hydrogen-mission ; https://ddnews.gov.in/en/india-rolls-out-pilot-projects-for-hydrogen-powered-buses-and-trucks/ ; https://www.energetica-india.net/news/mnre-initiates-5-pilot-projects-for-using-hydrogen-in-buses-and-trucks	India	vehicles; stations; ₹ crore	sanctioned	PIB blocked
W3-S37	12 pilot projects: 70 vehicles (27 buses, 43 trucks), 16 HRS, 21 routes, ₹410.77 crore, ₹41.582 crore released	PIB / MNRE (PRID 2291541, blocked); finews; EQ Mag; MNRE minister social posts	Centre Launches 12 Pilot Projects Under National Green Hydrogen Mission for Deployment of 70 Hydrogen-Powered Vehicles Across 21 Routes; Project 12 NGHM are under implementation	https://www.pib.gov.in/PressReleasePage.aspx?PRID=2291541&reg=48&lang=1 ; https://www.finews.com/2026/08/01/project-12-nghm-are-under-implementation ; https://www.eqmagpro.com/centre-launches-12-pilot-projects-under-national-green-hydrogen-mission-for-deployment-of-70-hydrogen-powered-vehicles-across-21-routes-eq/	India	vehicles; stations; ₹ crore	sanctioned / under implementation	PIB blocked; 21-route list not retrieved
W3-S38	SIGHT 8.62 lakh tpa awarded (19 cos); 3,000 MW electrolyser awards; ~8,000 tpa commissioned (Feb 2026); ₹292 crore spent; 30 KTPA to 4 refineries; Tranche II 450,000 tpa; NGHM outlay/target	Business Today (column); Business Standard; ANI; Mercom India	India's green hydrogen ambitions face a reality check; Govt spends ₹292 crore under NGHM so far: MNRE; Centre awards 30 KTPA...; MNRE to Incentivize 450,000 MTPA...	https://www.businesstoday.in/opinion/columns/story/indias-green-hydrogen-ambitions-face-a-reality-check-536276-2026-06-11 ; https://www.business-standard.com/amp/industry/news/govt-spends-292-crore-under-national-green-hydrogen-mission-so-far-mnre-126081201853_1.html ; https://www.aninews.in/news/business/centre-awards-30-ktpa-green-hydrogen-capacity-to-four-oil-refineries-under-national-green-hydrogen-mission-20260811180606/ ; https://www.mercomindia.com/mnre-450000-mtpa-green-hydrogen	India	tpa; ₹ crore	awarded vs operating	secondary reporting of MNRE/SECI data
W3-S39	Numaligarh ₹279/kg; SECI green ammonia ₹55.75/kg; SIGHT auction Mar 2025; cost estimates	S&P Global; PIB (PRID 2153006, blocked); Renewable Watch; Reslink (vendor)	India's Numaligarh hydrogen tender ends at record low Rupees 279/kg; SECI conducts first-ever auction for procurement of Green Ammonia; SECI announces results of green hydrogen auction under SIGHT; India Green Hydrogen Price Per Kg 2026	https://www.spglobal.com/energy/en/news-research/latest-news/energy-transition/021226-indias-numaligarh-hydrogen-tender-ends-at-record-low-rupees-279kg ; https://www.pib.gov.in/PressReleasePage.aspx?PRID=2153006 ; https://renewablewatch.in/2025/03/20/seci-announces-results-of-green-hydrogen-auction-under-sight-programme/ ; https://www.reslink.org/blogs/india-s-green-hydrogen-mission-is-moving-from-paper-to-projects/	India	₹/kg	observed (tender)	₹397/kg and \$5.3-6.7/kg are vendor figures
W3-S40	California retail H2 \$32-36/kg; avg \$34.55 (Sep 2024); True Zero \$36	S&P Global; Stillwater Associates; hydrogenfuelfinder	California hydrogen pump prices for light-duty vehicles reach new highs; 2025 Cost Showdown for Drivers; Why Hydrogen Fuel Costs \$36/kg in 2025	https://www.spglobal.com/commodity-insights/en/news-research/latest-news/energy-transition/100124-california-hydrogen-pump-prices-for-light-duty-vehicles-reach-new-highs ; https://stillwaterassociates.com/2025-cost-showdown-for-drivers-is-hydrogen-fuel-cheaper-than-gasoline/ ; https://hydrogenfuelfinder.com/blog/hydrogen-fuel-cost-2025-reality-check.html	California	\$/kg	observed (retail)	—
W3-S41	Germany €12.95/kg avg; €12.85; Hylane-H2 Mobility €8/kg (2026, trucks)	Atawey; Energy News; Driving Hydrogen; glpautogas	Understanding the Cost and Price of Hydrogen (2026); Hydrogen price goes to €12.85/kg; €8/kg hydrogen enables cost parity with diesel: Hylane & H2 Mobility	https://atawey.com/en/hydrogen-prices-in-2026-cost-per-kg-refuelling-costs-comparisons-and-2030-outlook/ ; https://energynews.biz/hydrogen-price-goes-up-to-e12-85-kg/ ; https://drivinghydrogen.com/2025/11/20/e8-kg-hydrogen-enables-cost-parity-with-diesel-hylane-h2-mobility-milestone/	Germany	€/kg	observed / contracted	dates of averages not exact
W3-S42	Japan ¥700/kg subsidy; Tokyo auction ~\$23/kg; METI priority regions	Driving Hydrogen; METI; Hydrogen Insight	Japan to introduce ¥700/kg hydrogen fuel subsidy...; First Selection of Priority Regions for FC Commercial Vehicles; Hydrogen prices of \$23/kg — Tokyo... auction	https://drivinghydrogen.com/2025/05/30/japan-to-introduce-%C2%A5700-kg-hydrogen-fuel-subsidy-to-kickstart-commercial-uptake/ ; https://www.meti.go.jp/english/press/2025/0519_002.html ; https://www.hydrogeninsight.com/production/hydrogen-prices-of-23-kg-tokyo-city-government-announces-results-for-first-double-sided-green-hydrogen-auction/2-1-1830862	Japan	¥/kg; \$/kg	policy / auction	pump price NOT retrieved
W3-S43	Standards: 34 standards in 14 months; PESO/OISD/PNGRB/MoRTH/BIS; Gas Cylinder Rules 2016; ISO 19880 harmonisation; CMVR 4th Amdt 2025 H2 plates; MoRTH 18 Aug 2026 draft	PIB (static backgrounder); PNGRB/PESO; autopdp; Business Standard	Green Hydrogen Standards and Approval Systems in India: Stakeholders interaction on pathways for transmission of hydrogen; Automotive Regulation Updates India: Govt proposes 5-yr age extension for EV, CNG, hydrogen commercial vehicles	https://static.pib.gov.in/WriteReadData/specificdocs/india%20Stakeholders%20interaction%20on%20pathways%20for%20transmission%20of%20hydrogen%20documents/202404/may/doc/2024510336201.pdf ; https://www.pngrb.gov.in/pdf/press-note/PESO_15032024.pdf ; https://autopdp.com/regulatory-updates ; https://www.business-standard.com/industry/auto/govt-proposes-5-yr-age-extension-for-ev-cng-hydrogen-commercial-vehicles-126081800231_1.html	India	—	rule / draft	PDFs blocked; passages only
W3-S44	NTPC Leh: 5 FC buses; 80 kg/day; 350 bar; 1.7 MW solar; Nov 2024 flag-off; Jun 2025 handover	NTPC; UT Ladakh; pv magazine India; AIR News	NTPC Hands Over Five Hydrogen Fuel Cell Buses to Leh Administration; Union Minister Manohar Lal flags off the fleet of green hydrogen buses of NTPC in Ladakh	https://ntpc.co.in/media/press-releases/ntpc-hands-over-five-hydrogen-fuel-cell-buses-leh-administration ; https://ladakh.gov.in/ntpc-hands-over-five-hydrogen-fuel-cell-buses-to-sidco-at-green-hydrogen-mobility-station-leh/ ; https://www.pv-magazine-india.com/2025/06/19/the-hydrogen-stream-ntpc-hands-over-five-hydrogen-fuel-cell-buses-to-leh-administration/ ; https://www.newsonair.gov.in/union-minister-manohar-lal-flags-off-the-fleet-of-green-hydrogen-buses-of-ntpc-in-ladakh	Leh	buses; kg/day; operating bar		costs/H2 price not disclosed

W3-S45	Tata 16-truck trials (4 Mar 2025); 40 H2-ICE trucks (Feb 2026); IOCL HRS at 4 sites	Tata Motors; AIR News; electricve	Tata Motors drives India's green future with country's first hydrogen truck trials; First trials of hydrogen powered heavy duty trucks flagged off; Tata Motors to deploy 40 H2 ICE trucks in India	https://www.tatamotors.com/press-releases/tata-motors-drives-indias-green-future-with-countrys-first-hydrogen-truck-trials/ ; https://www.newsonair.gov.in/first-trials-of-hydrogen-powered-heavy-duty-trucks-flagged-off/ ; https://www.electrive.com/2026/02/27/tata-motors-to-deploy-40-h2-ice-trucks-in-india/	India	trucks	trial / announced primary blocked		
W3-S46	IOCL Faridabad & Vadodara stations; 15-bus trials; Panipat 10 kTPA by Dec 2027; Air Products "first station"	IOCL; Indian Chemical News; Air Products; VahanBazaar (unofficial)	Green Drive: IOC plans hydrogen production units; Panipat green hydrogen unit on track for December 2027 completion; India's First Hydrogen Fuelling Station Inaugurated	https://iocl.com/MediaDetails/59096 ; https://www.indianchemicalnews.com/interviews/panipat-green-hydrogen-unit-on-track-for-december-2027-completion-dr-alok-sharma-director-rd-iocl-551 ; https://investors.airproducts.com/news-releases/news-release-details/indias-first-hydrogen-fuelling-station-inaugurated	Haryana; Gujarat	—	operating (captive) / under construction	dates & pressures not verified	
W3-S47	HCNG 50-bus Delhi trial (20 Oct 2020); 70% CO / 25% HC reduction; NOx same	fuelcellsworld; Busworld; IOCL media; pv magazine India	Shri Pradhan inaugurates H-CNG plant and launches trials in Delhi of 50 hydrogen-powered buses; Trial run of H-CNG powered city buses to begin in Delhi Oct 20; Hydrogen CNG, biogas: the many faces of new auto fuel	https://fuelcellsworld.com/news/india-shri-pradhan-inaugurates-h-cng-plant-and-launches-trials-in-delhi-of-50-hydrogen-powered-buses/ ; https://news.busworld.org/article/11296/trial-run-of-hcng-powered-city-buses-to-begin-in-delhi-oct-20 ; https://www.iocl.com/MediaDetails/56688 ; https://www.pv-magazine-india.com/2020/10/21/pollution-hit-delhi-starts-hydrogen-enhanced-cng-trial-for-buses/	Delhi	% emissions	pilot (IOCL-reported)	formal report not found	
W3-S48	Ashok Leyland-Reliance H2-ICE truck; launch in 18-24 months (Oct 2024)	Business Standard; Rushlane	Ashok Leyland to launch first H2-powered truck in the next 2 years; Ashok Leyland, Reliance Unveil India's First Hydrogen-ICE Powered Heavy Duty Truck	https://www.business-standard.com/companies/news/ashok-leyland-to-launch-first-h2-powered-truck-in-the-next-2-years-124100700992_1.html ; https://www.rushlane.com/first-hydrogen-ice-powered-heavy-duty-truck-12460051.html	India	—	prototype / announced	launch not verified	
W3-S49	Toyota Mirai-iCAT pilot, Mar 2022	Autocar Professional; India TV	Toyota Kirloskar, iCAT begin pilot project with Toyota Mirai; Nitin Gadkari drives India's first green hydrogen powered car to Parliament	https://www.autocarpro.in/news-national/toyota-kirloskar-icat-begin-pilot-project-with-toyota-mirai-81384 ; https://www.indiatvnews.com/news/india/nitin-gadkari-drives-green-hydrogen-powered-car-to-parliament-toyota-mirai-pics-2022-03-30-766664	India	—	pilot	—	
W3-S50	TVS hydrogen iQube; Bajaj exploring H2 2W/3W	Autocar India	TVS working on hydrogen powered scooter; Bajaj exploring hydrogen-powered two-wheelers under Chetak brand	https://www.autocarindia.com/bike-news/hydrogen-powered-tvs-scooter-in-the-works-425268 ; https://www.autocarindia.com/bike-news/bajaj-exploring-hydrogen-powered-two-wheelers-under-chetak-brand-431355	India	—	concept	—	
W3-S51	Mirai sales US 499 (2024) / 210 (2025); global 1,702 (Jan-Nov 2024); cumulative 21,475 (Nov 2022); 2026 fuel credit	GoodCarBadCar; CarFigures; CarBuzz; Wikipedia; Toyota.com	Toyota Mirai Sales Figures; Toyota's Biggest Flop Of 2025...; Toyota Mirai (Wikipedia, "Markets and sales"); 2026 Toyota Mirai	https://www.goodcarbadcar.net/toyota-mirai-sales-figures-usa-canada-monthly-yearly/ ; https://carfigures.com/us-market-brand/toyota/mirai ; https://carbuzz.com/toyotas-biggest-flop-2025-sold-fewer-than-1000/ ; https://en.wikipedia.org/wiki/Toyota_Mirai ; https://www.toyota.com/mirai/	US; Japan; global	units	observed	2025 global total not found	
W3-S52	Nexo gen-2 (Seoul 2025; Korea sales Jun 2025; 720 km); CR-V e:FCEV (~300/yr; 2,000 plan)	electricve; TopElectricSUV; Honda News; Green Car Reports	Next-gen Hyundai Nexo to debut in Q2 2025; First Look Review: Next-gen Hyundai Nexo; Honda Reveals 2025 Honda CR-V e:FCEV; First Drive: 2025 Honda CR-V e:FCEV	https://www.electrive.com/2024/05/10/next-gen-hyundai-nexo-2025/ ; https://topelectricusuv.com/first-look-review/hyundai-nexo-next-gen/ ; https://hondanews.com/en-US/honda-automobiles/releases/honda-reveals-2025-honda-cr-v-efcev-americas-first-production-plug-in-hydrogen-fuel-cell-electric-vehicle ; https://www.greencarreports.com/news/1142702_2025-honda-cr-v-efcev-test-drive-review	Korea; US	units; km	production	sales figures partial	
W3-S53	Global FCEV sales +24.4% (2025); H1 2026 4,643 units electricve; SNE Research (+13%); Europe 14		Global H ₂ vehicle sales rise by nearly 25%; Global FCEV Sales Reach 4,643 Units in H1 2026, Up 13.0% YoY	https://www.electrive.com/2026/02/23/global-h2-e2%82%82-vehicle-sales-rise-by-nearly-25/ ; https://www.sneresearch.com/en/insight/release_view/707/page/0	global	units	observed (SNE)	absolute 2025 total not seen	
W3-S54	IEA GHR 2025: heavy trucks only fast-growing FCEV market; China ~95% FC CV stock; OMV closure; low-emissions H2 <1%; ~1,160 HRS end-2024; country counts	IEA; TÜV SÜD; Interact Analysis; IEA AFC TCP	Global Hydrogen Review 2025 (Executive summary; PDF); Over 1,000 hydrogen refuelling stations in operation worldwide in 2024; 80% of global HRS in five countries	https://www.iea.org/reports/global-hydrogen-review-2025/executive-summary ; https://iea.blob.core.windows.net/assets/12d92ecc-e960-40f3-aff5-b2de6690abb6/GlobalHydrogenReview2025.pdf ; https://www.tuvsud.com/en/newsroom/press-releases/2025/february/over-1000-hydrogen-refuelling-stations-in-operation-worldwide-in-2024 ; https://interactanalysis.com/insight/80-of-global-hydrogen-refueling-stations-are-located-in-just-five-countries/	global	stations; %	observed	IEA site blocked; FCEV stock number not retrieved; "72,193 FCEVs / 1,022 HRS" seen without year	
W3-S55	Hydrogen HiAce pilot (CPB, 2024) results; 2025 hybrid trial	Toyota Australia; CPB Contractors; Roads Australia; just-auto	Toyota Unveils Hydrogen HiAce; CPB Contractors completes Australian-first Toyota hydrogen-fuelled vehicle pilot; Toyota to test hydrogen-electric hybrid vans in 2025	https://www.toyota.com.au/discover/news/toyota-unveils-hydrogen-powertrain-technology-for-pilot-program-in-australia ; https://www.cpbcon.com.au/news-and-media/2024/cpb-contractors-completes-australian-first-toyota-hydrogen-fuelled-vehicle-pilot ; https://roads.org.au/cpb-contractors-completes-historic-hydrogen-vehicle-trial/ ; https://www.just-auto.com/news/toyota-to-begin-testing-hydrogen-electric-hybrid-vans-in-2025/	Australia	—	pilot	2025 results not found	
W3-S56	GR Corolla H2 (LH2) 2025 racing status	Toyota Global Newsroom; Driving Hydrogen	Pushing Hydrogen Engine Technology to New Heights in the Super Taikyu Series Final Race; Hydrogen-Powered GR Corolla... Fuji 24 Hours; Toyota's superconducting pump and liquid H2 take to the grid	https://global.toyota/en/newsroom/corporate/43550115.html ; https://global.toyota/en/newsroom/corporate/44458612.html ; https://drivinghydrogen.com/2025/11/21/toyotas-superconducting-pump-and-liquid-h2-take-to-the-grid/	Japan	—	development	—	
W3-S57	HySE members; Dakar 2024 4th, 2025 2nd in class	Global Suzuki; Toyota Times	HySE to participate in the Dakar 2025 with the HySE-X2; HySE Buggy 2nd in Class at Dakar 2025	https://www.globalsuzuki.com/globalnews/2024/1112.html ; https://toyotatimes.jp/en/spotlights/1082.html	Japan/Saudi	—	research	—	
W3-S58	Burgman Hydrogen at JMS 2025; "not launching anytime soon"; Access CNG concept	Global Suzuki; BikeWale; Team-BHP	Hydrogen Engine Burgman — Japan Mobility Show 2025; Suzuki Burgman Hydrogen Showcased at JMS 2025; Suzuki Access CNG & hydrogen Burgman showcased	https://www.globalsuzuki.com/mobilityshow2025/burgman/ ; https://www.bikewale.com/news/suzuki-burgman-hydrogen-showcased-at-japan-mobility-show-2025/ ; https://www.team-bhp.com/forum/motorbikes/301240-suzuki-access-cng-hydrogen-powered-burgman-400-showcased-japan-mobility-show-2025-a.html	Japan	—	concept	70 MPa spec not restated	
W3-S59	Cummins B6.7H production Mar 2025 (India, Tata); X15H 2027	The Buzz EV News; Cummins; Truckinginfo; autonocion	Cummins starts H2-ICE production at new India facility; Cummins X15 Off-Highway... Agritechnica; Cummins Debuts 15L Hydrogen Engine	https://www.thebuzzevnews.com/cummins-h2-ice/ ; https://www.cummins.com/news/releases/2025/11/10/cummins-x15-highway-receives-powertrain-magazines-2026-alternative-engine ; https://www.truckinginfo.com/10170361/cummins-debuts-15l-hydrogen-engine	India; US	—	production (limited)	X15H date secondary	

W3-S60	JCB H2 engine EU type-approval (May 2025); 11 authorities (Jan 2025); 130+ engines	JCB; Power Progress	JCB secures full EU type-approval for hydrogen engine; Landmark start to 2025...; JCB receives type approval from 11 licensing authorities	https://www.jcb.com/en-GB/explore/insight/news/2025/05/jcb-secures-full-eu-type-approval-for-pioneering-hydrogen-engine/ ; https://www.jcb.com/en-GB/explore/insight/news/2025/01/landmark-start-to-2025-as-jcbs-hydrogen-engine-approved-for-use/ ; https://www.powerprogress.com/news/jcb-receives-type-approval-from-11-licensing-authorities-for-h2-ice/8049767.article	EU/UK	engines	approved / evaluation	off-highway only
W3-S61	KEYOU conversions; Daimler H2-ICE 2027	Daimler Truck; gasworld; Allison Transmission	Daimler Truck Partners with KEYOU on Hydrogen Combustion Technology; Daimler hydrogen ICE truck to launch in 2027; KEYOU hands over first hydrogen 18-ton truck... to EP-Trans	https://www.daimlertruck.com/en/newsroom/pressrelease/daimler-truck-partners-with-keyou-on-hydrogen-combustion-technology-53511488 ; https://www.gasworld.com/story/daimler-hydrogen-ice-truck-to-launch-in-2027-under-keyou-partnership/2252665.article/ ; https://www.allisontransmission.com/newsroom/article/2024/12/12/zero-emission-and-efficient---keyou-hands-over-first-hydrogen-18-ton-truck-with-allison-fully-automatic-transmission-to-ep-trans	Germany	trucks	pioneer fleet / announced	—
W3-S62	MAN hTXG 200 units; Deutz TCG 7.8 H2; Westport H2 HPDI	Hydrogen Insight; MAN; Deutz; IAA/trans.info	MAN to produce limited series of 200 vehicles with H2 combustion engines; DEUTZ TCG 7.8 H2; Westport unveils H2 HPDI; Westport successfully demonstrates H2 HPDI	https://www.hydrogeninsight.com/transport/hydrogen-sceptical-truck-maker-man-to-produce-limited-series-of-200-vehicles-with-h2-combustion-engines/2-1-1622908 ; https://press.mantruckandbus.com/corporate/man-expands-its-zero-emission-portfolio/ ; https://www.deutz.com/en/products/product-highlights/deutz-tcg-78-h2 ; https://trans.info/en/westport-successfully-demonstrates-h2-hpdi-fuel-system-374302	EU	units; kW	limited series / prototype	—
W3-S63	ULEMCo conversions (Aberdeen 35 vehicles; ambulance; H2/HVO)	Van Fleet World; ULEMCo; Autocar UK	ULEMCo wins deal for largest fleet-wide hydrogen conversion to date; ULEMCo completes first hydrogen/HVO dual-fuel conversion; Ulemco reveals first hydrogen-fuelled ambulance	https://vanfleetworld.co.uk/ulemco-wins-deal-for-largest-fleet-wide-hydrogen-conversion-to-date/ ; https://ulemco.com/ulemco-completes-first-hydrogen-hvo-dual-fuel-conversion/ ; https://www.autocar.co.uk/car-news/new-cars/ulemco-reveals-first-hydrogen-fuelled-ambulance	UK	vehicles	commercial (small)	prices NOT FOUND
W3-S64	Stellantis ends FC programme 16 Jul 2025; Symbio fallout	Stellantis (blocked); Driving Hydrogen; US News/Reuters; Global Banking & Finance	Stellantis Discontinues Hydrogen Fuel Cell Technology Development Program; Stellantis halts hydrogen van plans...; Symbio cuts 70% of staff...; Stellantis to exit Symbio JV, faces €235 million cost	https://www.stellantis.com/en/news/press-releases/2025/july/stellantis-discontinues-hydrogen-fuel-cell-technology-development-program ; https://drivinghydrogen.com/2025/07/16/stellantis-halts-hydrogen-van-plans-ends-fuel-cell-development-programme/ ; https://money.usnews.com/investing/news/articles/2025-07-16/stellantis-discontinues-hydrogen-fuel-cell-programme-and-van-production ; https://drivinghydrogen.com/2025/12/08/symbio-cuts-70-of-staff-as-stellantis-walks-away/	EU	—	decision (observed)	primary blocked
W3-S65	Shell closes 7 California LD stations (6 Feb 2024)	S&P Global; InsideEVs; Automotive News	Shell permanently closes light-duty hydrogen fueling stations; Shell is Immediately Closing All Of Its California Hydrogen Stations	https://www.spglobal.com/energy/en/news-research/latest-news/electric-power/021224-shell-permanently-closes-light-duty-hydrogen-fueling-stations ; https://insideevs.com/news/708156/shell-closes-california-hydrogen-stations/ ; https://www.autonews.com/mobility-report/shell-closes-california-light-duty-hydrogen-fueling-stations/	California	stations	observed	—
W3-S66	H2 Mobility: 22 closures by mid-2025; 14 more 31 Dec 2025	electrive; MobilityEnergy; H2.LIVE; CleanTechnica	H2 Mobility to shut down 22 hydrogen fuel stations in Germany; Fourteen hydrogen filling stations in Germany to close this year; Network consolidation 700 bar	https://www.electrive.com/2025/03/03/h2-mobility-to-shut-down-22-hydrogen-fuel-stations-in-germany/ ; https://www.mobilityenergy.com/en/industry/2025/11/17/fourteen-hydrogen-filling-stations-in-germany-to-close-this-year-focusing-on-forward-looking-locations/ ; https://h2.live/en/news/3815/	Germany	stations	observed	—
W3-S67	Honda Clarity ended Aug 2021; GM-Honda FCSM JV to end before end-2026	InsideEVs; Automotive Manufacturing Solutions; Headlight.news	Official: Honda Discontinues Clarity PHEV And Clarity Fuel Cell; Honda-GM fuel cell venture ends as EV priorities reshape	https://insideevs.com/news/514412/honda-discontinues-clarity-phev-fcv/ ; https://www.automotivemanufacturingsolutions.com/electrification/hondagm-fuel-cell-venture-ends-as-ev-priorities-reshape/2591276 ; https://headlight.news/2025/02/24/honda-general-motors-continue-to-drift-apart-ending-fuel-cell-partnership/	US/Japan	—	observed	—
W3-S68	Suzuki Growth Strategy FY2030 (CNG/biogas/ethanol; ~70% CNG share); multi-pathway statements 2025	Suzuki Motor Corp (globalsuzuki.com blocked; mirror suzuki.hu; Automotive World); Autocar Professional; S&P Global Mobility	Suzuki Announces Growth Strategy for FY2030; Suzuki backs multi-pathway route to carbon neutrality, says only BEVs not practical; India's multifuel path to automotive sustainability	https://www.globalsuzuki.com/globalnews/2023/0126.html ; https://www.suzuki.hu/corporate/en/company/growth-strategy-2030/growth-strategy ; https://www.autocarpro.in/news/suzuki-backs-multi-pathway-route-to-carbon-neutrality-says-only-bevs-not-practical-129454 ; https://www.spglobal.com/automotive-insights/en/blogs/2025/11/indias-multifuel-path-to-automotive-sustainability	Japan/India	—	company strategy	context only
W3-S69	SRDI established 28 Aug 2022; role; Banas Suzuki biogas plant 6 Dec 2025; 11 biogas plants	Global Suzuki; SRDI site: MarketScreener; BioEnergy Times	Suzuki establishes R&D center in India; About Suzuki R&D India; Suzuki holds the opening ceremony of Banas Suzuki Biogas Plant in India; Suzuki, Maruti Suzuki to add 11 biogas plants	https://www.globalsuzuki.com/globalnews/2022/0828b.html ; https://www.suzukiindia.com/about-us ; https://uk.marketscreener.com/news/suzuki-motor-december-9suzuki-holds-the-opening-ceremony-of-banas-suzuki-biogas-plant-in-india-ce7d51d2db8ff22d ; https://bioenergytimes.com/suzuki-maruti-suzuki-to-add-11-biogas-plants-using-cow-dung-farm-waste/	India	—	company	context only
W3-S70	Delhi draft EV Policy 2.0 proposed phasing out CNG autos; previous policy extended to 31 Mar 2025	Deccan Herald	Draft EV policy 2.0 recommends phasing out of CNG-driven autorickshaws in Delhi; Delhi electric vehicle policy extended till March 31, 2025	https://www.deccanherald.com/amp/story/india%2Fdelhi-%2Fdraft-ev-policy-20-recommends-phasing-out-of-cng-driven-autorickshaws-in-delhi-3482789 ; https://www.deccanherald.com/amp/story/india%2Fdelhi-%2Fdellhi-electric-vehicle-policy-extended-till-march-31-2025-3295696	Delhi	—	draft / rule	superseded by 2026 policy
W3-S71	Context: India's alternative-fuel shift "becoming a CNG Down To Earth story"	Down To Earth	India's CNG Dominance: A Step Towards Cleaner Transport or a Roadblock to Zero Emissions?	https://www.downtoearth.org.in/energy/indias-transport-sectors-alternative-fuel-shift-is-becoming-a-cng-story-but-what-comes-next	India	—	commentary	—
W3-S72	CNG kit prices ₹40-70k (sequential)	Cars24; Spinny	CNG Kit Price in India; CNG Kit in India: Price, Function, Conversion	https://www.cars24.com/article/cng-kit-price-in-india/ ; https://www.spinny.com/blog/cng-kit-in-india-price-function-conversion/	India	₹	market guide	vendor content
W3-S73	Creta Electric BaaS ₹10.99 lakh + ₹3.90/km	Team-BHP / CarDekho	Hyundai Creta Electric 2026	https://www.team-bhp.com/new-cars/hyundai/creta-electric/	India	₹	list price	—
W3-S74	~85,000 taxis in Delhi (2022)	Statista (summary only)	Number of taxis in Delhi	(URL not captured — search summary only)	Delhi	vehicles	observed (secondary) analysis	unverified; low confidence
W3-S76	expert scepticism on 2030 gas targets	IEEFA	Is India's 2030 gas consumption target feasible?	https://ieefa.org/resources/indias-2030-gas-consumption-target-feasible	India	—		title only seen

W3-S77	PNGRB 2030/2040 projections (CGD 87 mmscmd 2030 base case) and critique	PNGRB press note; PSU Watch	Natural Gas Projections 2030–40 (press note 7 Apr 2025); PNGRB projects a natural gas boom, but experts urge a sobering reality check	https://pngrb.gov.in/pdf/press-note/20250407_NG_Projections2030_40.pdf ; https://psuwatch.com/newsupdates/pngrb-projects-a-natural-gas-boom-but-experts-urge-a-sobering-reality-check		India	mmscmd	modelled	PDF blocked	
W4-S1	700-bar Type IV system \$12.1–13.3/kWh (median 12.7, US DOE Hydrogen Program 2020s) at 100k/yr; carbon fibre dominant		Record #24006 "Onboard Type IV Compressed Hydrogen Storage System—Cost and Performance Status"	https://www.hydrogen.energy.gov/docs/hydrogenprogramlibraries/pdfs/24006-onboard-storage-cost-performance-status.pdf	28 Apr 2025 / 2024 analysis	Record summary (page not verified)	USA (ANL/USA analysis)	\$/kWh (2020s) modelled	Host blocked; values from search summary; 10k & 500k rows unread	
W4-S2	\$14.2 / 15.7 / 21.3 per kWh at 500k / 100k / 10k systems/yr (2016s)	US DOE	Record #19008 onboard storage cost & performance status	https://www.hydrogen.energy.gov/pdfs/19008_onboard_storage_cost_performance_status.pdf	25 Nov 2019	Record p.1 summary	USA	\$/kWh	modelled	search summary
W4-S3	2013: 350-bar \$15/kWh, 700-bar \$19/kWh single tank (500k/yr); Record 13010 \$16–20/kWh	US DOE / Strategic Analysis	Record #13010: "Hydrogen Storage Cost Analysis Final Public Report" (OSTI 1082754)	https://www.hydrogen.energy.gov/docs/hydrogenprogramlibraries/pdfs/13010_onboard_storage_performance_cost.pdf ; https://digital.library.unt.edu/ark:/67531/metad827980/m2/1/high_res_d/1082754.pdf	11 Jun 2013 / 2013	abstract/summary	USA	\$/kWh	modelled	old dollar-year
W4-S4	DOE targets \$10/9/8 per kWh = \$333/300/266 per kg	US DOE	Technical Targets for Onboard Hydrogen Storage for LDVs	https://www.energy.gov/cmei/fuels/doe-technical-targets-onboard-hydrogen-storage-light-duty-vehicles	current page	Storage system cost row	USA	\$/kWh, \$/kg	target	targets not costs
W4-S5	12% cost reduction 2013–2015; volumes 10k–500k/yr	US DOE FCTO	Webinar: Onboard Type IV compressed H2 storage system cost	https://www.energy.gov/sites/prod/files/2016/03/f30/fcto_webinarslides_compressed_h2_storage_system_cost_022516.pdf	25 Feb 2016	slide 8	USA	%	modelled	snippet
W4-S6	Compressed-gas storage USD 400–700 per kg H2 (Cerri et al.)	MDPI Energies 16(13):5233	A Review on the Cost Analysis of Hydrogen Gas Storage Tanks for Fuel Cell Vehicles	https://www.mdpi.com/1996-1073/16/13/5233	2023	Table citing Cerri et al. [33]	global	USD/kg H2	literature	secondary citation
W4-S7	Type IV 700-bar tanks ~\$700–1,200 per kg capacity	Dataintelto (market report)	Cryo-Compressed Hydrogen Storage Market	https://dataintelto.com/report/cryo-compressed-hydrogen-storage-market	2025	Opportunities & threats	global	\$/kg	estimate	low-grade commercial source
W4-S8	Time Technoplast PESO approvals (Mar/Jun 2024), 250 L, BV cert, ₹2.30 crore trial order, ₹555 crore 9MFY26	Business Standard; Autocar Professional; Time Technoplast press release; Scanx	multiple	https://www.business-standard.com/markets/capital-market-news/time-technoplast-gets-final-approval-from-peso-for-manufacturing-composite-cylinders-124062100121_1.html ; https://www.autocarpro.in/news/time-technoplast-secures-international-certification-for-hydrogen-composite-cylinder-126444 ; https://www.timetechnoplast.com/pdf/investors-calendar/press-release/approval-for-manufacturing-of-type-iv-composite-cylinders-for-hydrogen.pdf	2024–2026	news	India	—	observed	no unit prices
W4-S10	Bosch H2 PFI "suitable for retrofitting existing fleets"	Bosch Mobility	Port fuel injection for hydrogen engines	https://www.bosch-mobility.com/en/solutions/powertrain/gas/port-fuel-injection-for-hydrogen-engines/	current	product page	global	—	observed	no price
W4-S11	CNG kit prices ₹45–65k (B56 OBD-II incl. install), Lovato ₹31,999–58,000, Landi Renzo ₹32,000, install ₹3–7k, RTO ₹2–3k	mycarhelpline; Cars24; IndiaMART; Autonexa; IOAGPL	Best CNG kit brands 2026; CNG kit price in India; listings	https://www.mycarhelpline.com/index.php?option=com_easyblog&view=entry&id=557&Itemid=91 ; https://www.cars24.com/article/cng-kit-price-in-india/ ; https://www.indiamart.com/proddetail/landirenzo-cng-kit-23070488673.html ; https://autonexa.com/cng-fitting-cost-in-india-complete-guide-price-break-down-kits-and-savings ; https://ioagpl.com/cng-kit-price/	2025–2026	retail/trade listings	India	₹	observed (retail)	listings vary by city
W4-S12	ARAI retrofit approval under Rule 115; "full approval in 45* days"; fee schedule not public; Scribd cost-breakdown doc exists; ₹15–25 lakh homologation anecdote	ARAI; Scribd; Quora	Retrofitment certification; Vehicle type approval; "Retrofitment Kit Approval"	https://www.araiindia.com/certification/certification/retrofitment ; https://www.araiindia.com/certification/certification/vehicle ; https://www.scribd.com/document/975757543/Retrofitment-Kit-Approval-1	current	pages	India	₹	observed / anecdote	fees NOT FOUND
W4-S13	PFI gasoline conversion needs injectors + plugs; DI needs ignition/injection; custom pistons	Ultimate Cell UK; OSTI 1008335 (NETL small H2 engine)	trade guides	https://ultimatecelldistributors.co.uk/insights/hydrogen-retrofit-for-diesel-petrol-engines/ ; https://www.osti.gov/servlets/purl/1008335	2020s / 2011	—	UK/US	—	qualitative	low grade
W4-S14	HORIBA H2-ICE test bed Pune, ₹28–30 crore, ≤380 kW, Dec 2024	HORIBA; Autocar Professional	HORIBA inaugurates first H2-ICE test bed facility in India	https://www.horiba.com/int/mobility/news/detail/news/12/2024/20241210/ ; https://www.autocarpro.in/news/horiba-opens-hydrogen-internal-combustion-engine-test-bed-facility-in-pune-123903	10 Dec 2024	press release	India	₹ crore	observed	—
W4-S15	ARAI ₹500 crore H2-ICE test infra (from 2025); Takwe cylinder test facility; ₹100 crore labs	Autocar Professional; Pune.news; Telematics Wire	ARAI to invest Rs 500 crore...; ARAI to establish hydrogen cylinder testing facility	https://www.autocarpro.in/news/arai-to-invest-rs-500-crore-to-enhance-infra-h2ice-testing-likely-from-2025-120865 ; https://www.autocarpro.in/news/arai-to-establish-a-hydrogen-cylinder-testing-facility-at-greenfield-takwe-near-pune-115086	2023–2024	news	India	₹ crore	announced	plans, not completions
W4-S16	CNG Delhi ₹86.98/kg	Goodreturns; CarDekho	CNG price in New Delhi today	https://www.goodreturns.in/cng-price-in-new-delhi.html ; https://www.cardekho.com/cng-price-in-delhi-state	14 Sep 2026	price page	Delhi	₹/kg	observed (aggregator)	IGL site blocked
W4-S17	MGL CNG ₹88/kg from 1 Sep 2026 (+₹2), PNG ₹53; spot RLNG cited	BusinessToday; IndianPSU; Goodreturns	MGL hikes CNG, PNG prices in Mumbai from September 1	https://www.businesstoday.in/latest/story/mgl-hikes-cng-png-prices-in-mumbai-from-september-1-check-revised-rates-552419-2026-08-31 ; https://indiainpsu.com/mgl-cng-price-hike-mumbai-88-september-2026/ ; https://www.goodreturns.in/cng-price-in-mumbai.html	31 Aug–14 Sep 2026	news	Mumbai	₹/kg	observed	—
W4-S18	Ahmedabad CNG ₹94.02 (7 Sep 2026); ₹86.02 (16 May 2026); ₹83.77 (3 Apr 2026)	Goodreturns; DeshGujarat; Vibes of India	CNG price in Ahmedabad; Adani CNG hikes	https://www.goodreturns.in/cng-price-in-ahmedabad.html ; https://deshgujarat.com/2026/05/16/adani-gas-imposes-another-price-hike-cng-rate-in-ahmedabad-now-at-rs-86-02-per-kg/ ; https://deshgujarat.com/2026/04/03/adani-cng-price-hiked-by-%E2%82%B91-50-rate-reaches-%E2%82%B983-77-amid-global-gas-crisis/	2026	news/price page	Ahmedabad	₹/kg	observed	—
W4-S19	Petrol Delhi ₹102.12/L (25 May 2026), crude basket \$113–114; aggregator ₹111.21 (14 Sep 2026); Ahmedabad ₹102.15 vs ₹94.59 conflict	Univest; PingTV; CarDekho; Goodreturns; DriveSpark; Business Standard	Petrol price hike to Rs 100 in Delhi...; petrol price pages	https://univest.in/blogs/petrol-price-hike-crosses-100-delhi-fourth-hike-10-days-may-2026 ; https://www.pingtvindia.com/fuel-price-hike-india-petrol-price-delhi-crossed-100/ ; https://www.cardekho.com/petrol-price ; https://www.goodreturns.in/petrol-price-in-ahmedabad.html ; https://www.drivespark.com/petrol-price-in-ahmedabad/	May–Sep 2026	news/price pages	Delhi/ Ahmedabad	₹/L	observed	Sep figures unverified vs IOCL
W4-S20	DERC EV tariff ₹4.50/kWh LT, ₹4.00/kVAh HT, zero fixed; domestic ₹3–8; DERC 1 Jul 2026 upstream-cost order	thediscombill; bridgewaypower; easyEVplug; Energetica India; MeraEV	Delhi electricity tariff 2025–26; DERC directs...	https://thediscombill.com/tariffs/delhi/ ; https://www.easyevplug.com/ev-charger-installation-delhi ; https://energetica-india.net/news/derc-directs-full-upstream-ev-charging-infra-costs-to-be-included-in-demand-notes-under-pm-e-drive	2025–2026	tariff summaries	Delhi	₹/kWh	observed (secondary)	DERC order not opened
W4-S21	Public DC fast charging ₹16–24/kWh; Tata Power ₹17; ChargeZone ₹20–25; Statiq ₹15–22	Pulse Energy; searchev.co; MeraEV; GreenTax	EV charging rates in India 2026	https://pulseenergy.io/blog/ev-charging-rates ; https://searchev.co/cost-for-charging-electric-cars/ ; https://www.meraev.com/blog/ev-charging-stations-india-types-costs-guide	2026	guides	India	₹/kWh	observed (secondary)	operator tariffs vary by state
W4-S22	CEEW TCO: charging tariffs; H2 production/dispensing/delivered cost 2030–2050; taxi/4W findings; H2 bus competitiveness; maintenance Table 5; 3W/2W ₹/km	CEEW	What Drives Vehicle Ownership Costs in India? / Cost of Ownership for India's Road Transport Sector	https://www.ceew.in/publications/cost-of-ownership-for-road-transport-sector-for-different-vehicle-segments-fuels-and-powertrains ; https://www.ceew.in/sites/default/files/cost-of-ownership-for-different-vehicle-segments-fuels-and-powertrains.pdf	Jun–Jul 2025 / 2024 base	Exec summary; Fuel costs; Green hydrogen; Maintenance; Results \$4 (verbatim extracts)	India, 14 states	USD/kg, ₹/kWh, ₹/km	modelled	Tables 3 & 5 values not readable

W4-S23	ADAC charging losses AC 5–10% (wallbox), 24–30% (socket), DC 5–15%	Zecar; InsideEVs; TheNextWeb; vision-mobility (reporting ADAC)	EV charging losses up to 24%...; How efficient is each type of EV charger	https://zecar.com/reviews/ev-charging-losses-adac-study ; https://insideevs.com/features/711659/ev-charger-efficiency-losses/ ; https://vision-mobility.de/en/news/adac-analysis-less-energy-is-lost-at-fast-chargers-384109.html	2022–2024	test summaries	Germany	%	measured	ADAC report not opened
W4-S24	GST: CNG small cars & parts 28%~18% (Sep 2025); CNG-tank classification dispute 18% vs 28%	EY; Taxo; SAG Infotech	GST Council announces major rate rationalization (4 Sep 2025); GST notices to CNG kit providers	https://www.ey.com/content/dam/ey-unified-site/ey-com/en-in/alerts-hub/2025/09/gst-council-announces-major-rate-rationalization-and-trade-facilitation-measures.pdf ; https://taxo.online/latest-news/28-12-2024-gst-notices-sent-to-over-a-dozen-cng-kit-providers/	Sep 2025; Dec 2024	alert; news	India	%	observed	kit rate unresolved
W4-S25	Delhi EV Policy 2026: notified 30 Jun 2026, in force 1 Jul 2026–31 Mar 2030; 100% road tax & registration waiver BEV ≤₹30 lakh; hybrids excluded	EV Infrastructure News; Akashvani News; Coverfox; MeraEV	Delhi notifies EV Policy 2026...	https://www.evinfrastructurereviews.com/ev-policies/delhi-notifies-ev-policy-2026-with-tax-exemptions-for-ev ; https://newsonair.gov.in/electric-cars-priced-up-to-30-lakh-rupees-and-registered-in-delhi-to-be-granted-100-exemption-on-road-tax/	Jun–Jul 2026	news	Delhi	—	observed	gazette not opened
W4-S26	Tata 8 yr/160,000 km (Tiago.ev, Punch.ev, Curvv.ev, Nexon.ev 4s), lifetime first owner; MG ZS EV 8 yr/150,000 km; Windsor/Comet lifetime	Tata.ev; Cars24 news; ev.care	Tata EV warranty; MG ZS EV warranty explained	https://ev.tata.cars/service/warranty.html ; https://www.cars24.com/news/autotata-nexon-ev-lifetime-battery-warranty-details-revealed/ ; https://ev.care/blog/mg-zs-ev-warranty-guide	2025–2026	warranty pages	India	years/km	observed	—
W4-S27	Geotab degradation 1.8%/yr (2024) → 2.3%/yr (2025 update); >90% after 160,000 km	Geotab	EV battery health study; 2024 battery degradation update	https://www.geotab.com/blog/ev-battery-health/ ; https://www.geotab.com/uk/press-release/2024-battery-degradation/	2024–2025	blog/press	global fleets	%/yr	measured	fleet mix not Indian
W4-S28	Used Dzire CNG 2022 ₹6.09 lakh (56k km); True Value 2022 ₹6.40 lakh / 2024 ₹7.40 lakh; new Tour S CNG ₹7.09 lakh; on-road ₹6.24–8.51 lakh	Cars24; Maruti True Value; Cars24 new cars; CarDekho	listings	https://www.cars24.com/buy-used-maruti-dzire-cars-new-delhi/ ; https://www.marutisuzukitruevalue.com/used-dzire ; https://www.cars24.com/new-cars/maruti-suzuki/dzire-tour/ ; https://www.cardekho.com/maruti/dzire-tour-s/price-in-new-delhi	Sep 2026	listings	Delhi	₹ lakh	observed (asking not taxi-registered cars prices)	
W4-S29	NBFC rates: Shriram from 10%; taxi 9–15%; Chola 9.35–17.35% used car, 10–21% CV; NBFC 13–18%	Shriram Finance; upniva (Cholamandalam); Asymmetrical/calciwise	rate pages	https://www.shriramfinance.in/articles/used-car-loan/2025/ ; https://upniva.com/loans/cholamandalam-used-car-loan/ ; https://asymmetrical.in/calculators/car-loan-rates/	2025–2026	rate pages	India	% p.a.	observed	indicative
W4-S30	Taxi 70,000 km/yr; car 11,500; bus 75,000 (Table A.3)	NITI Aayog	Scenarios Towards Viksit Bharat and Net Zero — Sectoral Insights: Transport	https://www.niti.gov.in/sites/default/files/2026-02/Scenarios-Towards-Viksit-Bharat-and-Net-Zero-Sectoral-Insights-Transport.pdf	Feb 2026	Annex Table A.3 (verbatim)	India	km/yr	modelling assumption	—
W4-S31	ICCT ride-hailing TCO ₹7.6/km; –34% Delhi / –20% Hyderabad with incentives; ₹60,000 bulk discount	ICCT	Updated analysis of incentives for electrifying India's four-wheeler ride-hailing fleet	https://theicct.org/wp-content/uploads/2021/06/update-electrifying-india-ride-hailing-fleet-apr2021.pdf	Apr 2021	working paper (snippets)	Delhi, Hyderabad, Bengaluru	₹/km	modelled	daily-km assumption not extracted
W4-S32	Driver anecdotes –300 km/day incl. empty running	Quora	—	https://www.quora.com/How-many-kilometers-on-average-do-Ola-or-Uber-cab-drivers-drive-in-a-single-day	n.d.	—	India	km/day	anecdotal	low grade
W4-S33	Green H2 USD 3.5–5/kg (industry); breakeven <USD 2; USD 1.8/kg policy levers	CEEW	How can India Boost Investment for Domestic Green Hydrogen?	https://www.ceew.in/publications/how-can-indian-policy-makers-boost-investments-for-domestic-green-hydrogen-financing	2024–2025	Challenge 1 (verbatim)	India	USD/kg	estimate	—
W4-S34	₹397–560/kg (\$4.6–6.7/kg), 2–3× grey	GH2 India	India's Green Molecules Moment: ACME's new deals...	https://www.gh2.org.in/indias-green-molecules-moment-acmes-new-deals-show-the-market-is-finally-moving-from-announcements-to-offtake	2025–2026	article (verbatim)	India	₹/kg	estimate	advocacy body
W4-S36	Green H2 \$4–7/kg today; \$1.60 (2030), \$0.70 (2050) best case; \$1.4/kg most aggressive; \$1/kg goal	NITI Aayog & RMI	Harnessing Green Hydrogen	https://www.niti.gov.in/sites/default/files/2022-06/Harnessing_Green_Hydrogen_V21_DIGITAL_29062022.pdf ; https://www.pib.gov.in/PressReleasePage.aspx?PRID=1838104	29 Jun 2022	cost chapter (summary)	India	USD/kg	modelled/target	search summary
W4-S37	At-pump green H2 –\$5/kg (₹350) mid, \$3.5 (₹245) optimistic (2050)	ICCT; EurekAlert	Hydrogen fuel for transport in India	https://theicct.org/publication/india-fuel-hydrogen-cost-jan22/ ; https://www.eurekalert.org/news-releases/941197	Jan 2022	summary	India	USD/kg	modelled	year attribution uncertain
W4-S38	Green ammonia auctions ₹55.75 (first), ₹49.75 lowest, ₹53.27 weighted avg, 724,000 t/yr; ₹54.73 8th; ₹52.50 Onix	PIB; pv magazine India; S&P Global; ACME; Fuel Cells Onix	SECI first green ammonia auction...; Green ammonia only 10% costlier...; Low-priced Indian renewable ammonia auctions	https://www.pib.gov.in/PressReleasePage.aspx?PRID=2153006 Jul–Sep 2025 ; https://www.pv-magazine-india.com/2025/08/28/green-ammonia-is-only-10-costlier-than-grey-in-the-latest-sight-mode-2-auctions/ ; https://www.spglobal.com/commodity-insights/en/news-research/latest-news/energy-transition/090425-low-priced-indian-renewable-ammonia-auctions-a-meaningful-signal-h2global		news	India	₹/kg NH3	observed (auction)	conversion to H2 is a calculation
W4-S39	SIGHT Mode-1 Tranche-II 450,000 t/yr, ₹2,239 crore; ₹50/40/30 per kg	Renewable Watch; JMK; SolarQuarter	SECI announces results of green hydrogen auction under SIGHT	https://renewablewatch.in/2025/03/20/seci-announces-results-of-green-hydrogen-auction-under-sight-programme/ ; https://jmkresearch.com/seci-awards-450000-mt-annual-capacity-under-sight-tranche-ii-for-green-hydrogen-production/	Mar 2025	news	India	t/yr, ₹	observed	—
W4-S40	Electrolyser Tranche-II 1.5 GW, 11 winners, ₹4,440 crore FY26–30	Mercom; pv magazine India; MNRE	MNRE allocates ₹44.4 bn...; SECI awards LoAs...	https://www.mercomindia.com/mnre-electrolyzer-manufacturing-sight-ii-program ; https://www.pv-magazine-india.com/2025/01/21/the-hydrogen-stream-seci-awards-loas-for-electrolyser-production-linked-incentives/	Jan 2025	news	India	MW, ₹	observed	—
W4-S41	Electrolyser ₹30,000/kW alkaline, ₹42,000/kW PEM (1 MW, 300 MW/yr)	JMK Research / World Bank technical report	India's \$2.1bn leap towards its green hydrogen vision; Technical and Economic Characteristics...	https://jmkresearch.com/wp-content/uploads/2025/02/Indias-2.1bn-leap-towards-its-Green-Hydrogen-vision.pdf ; https://documents1.worldbank.org/curated/en/099022326123011297/pdf/P506220-b45b0865-8706-4862-8792-d8cc71f84569.pdf	Feb 2025 / 2026	summary	India	₹/kW	modelled	attribution between the two docs unclear
W4-S42	Grey H2 \$1.5–1.6/kg stable, \$3–7.5 under LNG shock; gap \$1.6–3.2; SMR ₹181–229/kg (LinkedIn)	RMI; LinkedIn post	Green Hydrogen Production Pathways for India	https://rmi.org/green-hydrogen-production-pathways-for-india/ ; https://www.linkedin.com/posts/sanjeeshharmagh2_refineryhydrogen-greenhydrogen-iocl-activity-7454393902108889088-B4dk	2024–2026	exec summary (verbatim)	India	USD/kg, ₹/kg	estimate	LinkedIn low grade
W4-S43	Station cost \$1,200–3,000 per kg/day (770–1,620 kg/day, 700 bar)	US DOE	Record 21002 Hydrogen Fueling Station Cost	https://www.hydrogen.energy.gov/docs/hydrogenprogramlibraries/pdfs/21002-hydrogen-fueling-station-cost.pdf	2 Nov 2020	summary	California	\$(/kg/day)	observed (grants)	blocked; snippet
W4-S44	Delivery+dispensing –\$11/kg @450 kg/d, –\$8/kg @1,000 kg/d (liquid; HDSAM 3.1)	US DOE	Record 20007 Hydrogen Delivery and Dispensing Cost	https://www.hydrogen.energy.gov/docs/hydrogenprogramlibraries/pdfs/20007-hydrogen-delivery-dispensing-cost.pdf	25 Aug 2020	summary	USA	\$/kg	modelled	blocked; snippet
W4-S45	350-bar delivery+dispensing \$2.70–3.70/gge (HDSAM 2.3, mature)	US DOE	Record 16011 H2 delivery cost 2015	https://www.hydrogen.energy.gov/pdfs/16011_h2_delivery_cost_2015.pdf	2015	summary	USA	\$/gge	modelled	mature-market case
W4-S46	200 kg/day station \$6–8/kg (gaseous), \$8–9 (liquid); utilisation 50% (2030), 70% (2050)	ICCT	Cost of renewable hydrogen produced onsite at HRS in Europe	https://theicct.org/publication/fuels-eu-on-site-hydro-cost-feb22/	Feb 2022	white paper summary	EU	\$/kg	modelled	snippet
W4-S47	NREL 88818 scope (700 bar, 2022\$); ATB 2024 H2 price ranges 6.2–12.0 / 8.2–18.0; future 6.2–7.0; 4.0 target	NREL	Levelized Cost of Dispensed Hydrogen for Heavy-Duty Vehicles (TP-5400-88818); ATB 2024 Hydrogen	https://docs.nrel.gov/docs/fy24osti/88818.pdf ; https://atb.nlr.gov/transportation/2024/Hydrogen	Mar 2024	ATB table (verbatim)	USA	2022\$/kg	modelled	88818 tables unread
W4-S48	LCOH \$11.51 (2025) → \$4.26/kg (2050); UR 0.3/0.5/1.0; breakeven \$4.90–5.85	Elsevier (journal article, S2590116826000421)	Breakeven levelized cost of hydrogen for the zero-emission freight transition	https://www.sciencedirect.com/science/article/pii/S2590116826000421	2026	Highlights; \$4.3 (verbatim)	USA	\$/kg	modelled	freight context
W4-S49	350-bar compressor/storage cost inputs used by CEEW Wiley (Chem. Ing. Tech. 96)		Detailed Cost Analysis of Hydrogen Refueling Costs for Fleets (Eiðler et al.)	https://onlinelibrary.wiley.com/doi/full/10.1002/cite.202300123 2024		—	Germany	—	modelled	not opened

W4-S50	California \$32–36/kg; True Zero \$36	S&P Global; hydrogenfuelfinder; Stillwater	California hydrogen pump prices reach new highs; hydrogen fuel costs \$36/kg in 2025	Why https://www.spglobal.com/energy/en/news-research/latest-news/energy-transition/100124-california-hydrogen-pump-prices-for-light-duty-vehicles-reach-new-highs ; https://hydrogenfuelfinder.com/blog/hydrogen-fuel-cost-2025-reality-check.html	2024-2025	news	California	\$/kg	observed	—
W4-S51	Germany €10–15/kg retail; €8/kg hylane deal from 1 Jan 2026	H2.LIVE / H2 MOBILITY: Driving Hydrogen; glpautogas	Hydrogen for 8 euros per kilogram now possible	https://h2.live/en/press/hydrogen-for-8-euros-per-kilogram-now-possible-hylane-and-h2-mobility-enable-cost-effective-operation-of-hydrogen-trucks/ ; https://www.glpautogas.info/en/hydrogen-sale-price-germany.html	Nov 2025	press release	Germany	€/kg	observed	fleet deal
W4-S52	Japan Iwatani ¥1,210 → ¥1,650/kg (Jun 2025); ¥700/kg subsidy	Hydrogen Insight; Driving Hydrogen	Price hike...; Japan to introduce ¥700/kg subsidy	https://www.hydrogeninsight.com/transport/price-hike-hydrogen-vehicle-drivers-to-be-charged-33-more-for-fuel-at-majority-of-h2-filling-stations-in-japan/2-1-1628765 ; https://drivinghydrogen.com/2025/05/30/japan-to-introduce-¥700-kg-hydrogen-fuel-subsidy-to-kickstart-commercial-uptake/	2025	news	Japan	¥/kg	observed	—
W4-S53	NTPC Leh: 1.7 MW solar, 80 kg/day 350-bar station, 5 FC buses (25 kg, 300 km)	NTPC	NTPC hands over five hydrogen fuel cell buses to Leh Administration	https://ntpc.co.in/media/press-releases/ntpc-hands-over-five-hydrogen-fuel-cell-buses-leh-administration	Jun 2025	press release	Leh, India	—	observed	no cost data
W4-S54	MNRE Phase-1 pilots: 5 projects, 37 vehicles (15 FCEV + 22 H2-ICE), 9 HRS, ₹208 crore, 10 routes, 18–24 months	PIB; Down To Earth	Pilot Projects on Hydrogen Fuelled Buses and Trucks Launched under NGHM	https://www.pib.gov.in/PressReleasePage.aspx?PRID=2107795 ; https://www.downtoearth.org.in/environment/as-told-to-parliament-march-25-2025-37-cross-country-vehicles-9-fuelling-stations-planned-under-green-hydrogen-mission	Mar 2025	press release	India	₹ crore	observed (sanction)	—
W4-S55	Phase-2: 12 projects, 70 vehicles (27 buses/43 trucks), 16 HRS, 21 routes, ₹410.77 crore (₹41.58 released); ₹292 crore NGHM spend to date	pv magazine India; Fuel Cells Works; EQ; Business Standard; Renewable Watch	India awards 12 green hydrogen mobility pilots...; Govt spends ₹292 crore under NGHM so far	https://www.pv-magazine-india.com/2026/08/03/the-hydrogen-stream-india-awards-12-green-hydrogen-mobility-pilots-with-inr-410-77-crore-support/ ; https://fuelcellsworld.com/2026/08/02/fuel-cells-india-s-hydrogen-vehicle-pilot-nearly-doubles-70-vehicles-16-refuelling-stations-410-77-crore ; https://www.business-standard.com/amp/industry/news/govt-spends-292-crore-under-national-green-hydrogen-mission-so-far-mnre-126081201853_1.html	Jul-Aug 2026	news	India	₹ crore	observed (sanction)	MNRE order not opened
W4-S56	HRS ₹14–20 crore; 200–500 kg/day break-even; "₹4/km" hydrogen	VahanBazaar; Tata AIG blog	Hydrogen refuelling stations India 2030	https://vahanbazaar.in/tips/hydrogen-refuelling-stations-india-2030-vahanbazaar ; https://www.tataaig.com/knowledge-center/car-insurance/hydrogen-cars-in-india	2026	blog	India	₹ crore	estimate	low grade
W4-S57	IEA GHR 2026: India/Middle East/Chile "just above USD 3/kg" by 2030; China <USD 2	Green Hydrogen Organisation (reporting IEA GHR 2026)	IEA Global Hydrogen Review 2026...	https://gh2.org/iea-global-hydrogen-review-2026-green-hydrogen-can-help-prevent-next-food-and-energy-crisis	18–19 Jun 2026	article (verbatim)	India	USD/kg	modelled	IEA site blocked
W4-S58	IEA GHR 2025: project pipeline shrank; electrolysis >80% of drop; assumptions annex	IEA	Global Hydrogen Review 2025 (+ Assumptions Annex)	https://www.iea.org/reports/global-hydrogen-review-2025 ; https://iea.blob.core.windows.net/assets/15673ab3-a86a-4434-bf44-490bb42d3563/GlobalHydrogenReview2025AssumptionsAnnex.pdf	Sep 2025	exec summary	global	—	modelled	blocked; India LCOH NOT FOUND
W4-S59	CEA WAEF FY2024-25 0.710 (prov.), FY2023-24 0.727; BM 0.512; CM definition	CEA (user guide v21.0); batchwise.ai; reclaimatize; LinkedIn (R. Goenka)	CO2 Baseline Database for the Indian Power Sector, User Guide v21.0	https://cea.nic.in/wp-content/uploads/baseline/2025/12/User_Guide_V_21.0.pdf ; https://batchwise.ai/methodology/cea-grid-emission-factors/all-india/	Dec 2025 / FY2024-25	CM definition (verbatim); India WAEF (secondary)	India	tCO2/MWh	observed (provisional)	primary blocked; T&D note unread
W4-S60	NEP 2022-32: EF 0.548 (2026-27), 0.430 (2031-32) kg/kWh net; non-fossil 57.4% / 68.4%	CEA; PIB	National Electricity Plan 2022-32 (Vol. I Generation)	https://cea.nic.in/wp-content/uploads/rfp/2023/05/NEP_2022_32_FINAL_GAZETTE-1.pdf ; https://www.pib.gov.in/PressReleaseframePage.aspx?PRID=1928750®=48&lang=2	May 2023	summary	India	kg CO2/kWh	projected	snippet
W4-S61	India default fuel EFs exist (GLEC-aligned)	Smart Freight Centre India	India Default GHG Emission Values V1.0	https://smart-freight-centre-media.s3.amazonaws.com/documents/India_Default_GHG_Emission_Values_V1_SFC_INDIA_latest.pdf	2023–2024	—	India	kg CO2e	published	not opened
W4-S62	IPCC default EFs 69.300 (gasoline), 56.100 (natural gas) kg CO2/TJ; NCV 44.3 / 48.0 TJ/Gg	IPCC	2006 IPCC Guidelines Vol. 2 Energy, Ch. 1 Tables 1.2 & 1.4	https://ghgprotocol.org/Third-Party-Databases/IPCC-Emissions-Factor-Database	2006	Tables 1.2, 1.4	global	kg/TJ	default	from memory — verify before use
W4-S63	ICCT 2025 LCA values (235/234/203/188/163/63/52/50/175 g/km; H2 13.6 & 1.2 kg CO2e/kg; battery 72.8 kg/kWh; +40% production; 17,000 km; TTW non-CO2)	ICCT (Negri & Bieker)	Life-cycle GHG emissions from passenger cars in the EU: a 2025 update	https://theicct.org/wp-content/uploads/2025/07/ID-392-%E2%80%93Life-cycle-GHG_report_final.pdf ; https://theicct.org/publication/electric-cars-life-cycle-analysis-emissions-europe-jul25/	8 Jul 2025	Key findings; Fuel-cycle \$; Figure 4 (verbatim extracts)	EU	g CO2e/km; kg CO2e/kg; kg CO2e/kWh	kg modelled	battery chemistry attribution unverified
W4-S64	Unabated NG 10–14 kg CO2e/kg (upstream 1–5); solar PV electrolysis 0.9–2.5; downstream 0.4–2.7	IEA; CRS R48196	Towards hydrogen definitions based on their emissions intensity; Hydrogen Production: Overview and Issues for Congress	https://www.iea.org/reports/towards-hydrogen-definitions-based-on-their-emissions-intensity ; https://www.congress.gov/crs-product/R48196	Apr 2023; Oct 2024	exec summary	global	kg CO2e/kg	modelled	snippet
W4-S65	India green-H2 standard ≤2 kg CO2e/kg well-to-gate	MNRE (via CSEP/RMI summaries)	Green Hydrogen Standard for India	https://csep.org/key-insights/green-hydrogen-pathways-in-india-from-a-grid-perspective/	Aug 2023	—	India	kg CO2e/kg	regulation	primary not opened
W4-S66	Medium petrol car 54.1 t, PHEV 36.9 t, BEV 25.0 t (15 yr, 42 km/day, STEPS); BEV lower after 2 yr bZ4X 26 t (25–31); Corolla Cross hybrid 53 t (44–70)	IEA	EV Life Cycle Assessment Calculator	https://www.iea.org/data-and-statistics/data-tools/ev-life-cycle-assessment-calculator	2024–2025	tool defaults (verbatim)	global	t CO2e	modelled	global assumptions
W4-S67	bZ4X 26 t (25–31); Corolla Cross hybrid 53 t (44–70)	Green NCAP	European LCA results 2024	https://www.greencap.com/european-lca-results/	2024	results table (verbatim)	EU	t CO2e	modelled	EU mix
W4-S68	DI H2 engine λ 2.4–3.0, BTE 43%, WLTC tailpipe NOx <35 mg/km with urea-SCR	ACS Omega 11(7):12788 (PMC12947045)	NOx/N2O Formation and Coordinated Urea-SCR Control in an Ultra-Lean Hydrogen-Fueled Automotive SI Engine	https://pmc.ncbi.nlm.nih.gov/articles/PMC12947045/	2026	abstract	China (China 6b)	mg/km	measured/ modelled	full text not indexed
W4-S69	55 kW 4-cyl PFI H2ICE NRRM tailpipe NOx variants	Environ. Sci.: Processes & Impacts (doi 10.1039/d4em00448e)	Evaluating low NOx hydrogen engines designed for off-road and construction applications	https://doi.org/10.1039/d4em00448e	Feb 2025	abstract	UK	g/kWh	measured	values inaccessible
W4-S70	Base H2 car cannot stay ultra-lean at high load; hybrid EMS —90% engine-out NOx, ~16% H2; Euro 7 60 mg/km fuel-neutral	arXiv (ETH Zürich / Bosch)	Energy Management of Hydrogen Hybrid Electric Vehicles — A Potential Study	https://arxiv.org/abs/2309.09804	Sep 2023	\$1, \$3.1–3.2, Table 4 (verbatim)	Europe	mg/km (relative)	simulated	simulation only
W4-S71	BS-VI M1 NOx 60 mg/km petrol; 80 diesel	ICCT; DieselNet; ARAI	India Bharat Stage VI emission standards (policy update); Indian Emission Regulation Booklet	https://theicct.org/wp-content/uploads/2021/06/India-BS-VI-Policy-Update-vF.pdf ; https://dieselnet.com/standards/in/d.php ; https://www.araiindia.com/pdf/Indian_Emission_Regulation_Booklet.pdf	2016/2021	limit tables	India	mg/km	regulation	CNG row not verbatim
W4-S72	H2 indirect GWP100 11 ± 4; GWP20 37 ± 18; OH-bias 20% overestimate; leakage 1–10%	Nature 2025 (PMC12711571); Ocko & Hamburg 2022 (ACP); arXiv 2408.05127; PMC10976891	The global hydrogen budget; Climate consequences of hydrogen leakage; Model underestimates of OH reactivity...	https://pmc.ncbi.nlm.nih.gov/articles/PMC12711571/ ; https://acp.copernicus.org/preprints/acp-2022-91/acp-2022-91.pdf ; https://arxiv.org/abs/2408.05127	2022–2025	abstracts	global	kg CO2e/kg H2	modelled	O&H exact values not verified
W4-S73	India VCM ₹1,350–2,800/t (2026); MSCI index \$3.50/t (2025), \$4.30 (2024); CCTS offset mechanism	Grow Billion Trees; Senken; Regreener; CEEW	Carbon credit rate in India 2026; Carbon credit prices 2026; India's voluntary offset scheme	https://growbilliontrees.com/pages/carbon-credit-rate-in-india-updates-monthly-prices ; https://www.senken.io/academy/understanding-carbon-markets/carbon-credit-price ; https://www.ceew.in/publications/voluntary-carbon-offset-mechanism-and-challenges-in-carbon-credit-trading-scheme-market-for-india	2025–2026	price trackers	India/global	₹/t, \$/t	observed	Indian tracker low grade

W4-S74	Transport pilot guidelines 14 Feb 2024: ₹496 crore to FY26; buses/trucks/cars <3.5 t; steering committee	JSA; Lexology; Business Standard; Deloitte	MNRE issues Scheme Guidelines for pilot projects on use of green hydrogen in the transport sector	https://www.jsalaw.com/newsletters-and-updates/mnre-issues-scheme-guidelines-for-pilot-projects-on-use-of-green-hydrogen-in-the-transport-sector/ ; https://www.business-standard.com/industry/news/govt-issues-guidelines-for-transport-pilot-projects-on-green-hydrogen-s-use-124021401729_1.html	Feb 2024	summaries	India	₹ crore	regulation	guideline PDF blocked; funding share per vehicle not extracted
W4-S75	R&D scheme ₹400 crore; guidelines 15 Mar 2024; 1st call 23 projects; 2nd call 14 Jul-21 Aug 2025; TRL ≥4; 100%/80% funding; ₹200 crore testing-infra call	MNRE; Mercom; SolarQuarter; CSIR; pv magazine India	Second Call for Proposals under R&D Scheme of NGHM	https://mnre.gov.in/en/notice/second-call-for-proposals-under-research-and-development-scheme-of-national-green-hydrogen-mission-nghm/ ; https://www.mercomindia.com/mnre-invites-second-round-of-rd-proposals-under-green-hydrogen-mission/ ; https://www.csir.res.in/en/news-calls-and-events/second-call-proposals-under-research-and-development-scheme-national-green/ ; https://www.pv-magazine-india.com/2025/08/12/mnre-launches-second-call-for-proposals-to-build-and-upgrade-green-hydrogen-testing-infrastructure/	Mar 2024-Aug 2025	notices/news	India	% of cost	regulation	remaining budget NOT FOUND
W4-S76	Gujarat GH2 Policy 2025 incentives (electrolyser, HRS, buses, MSME)	Mercom India; iamrenew; JMK (policy PDF)	Gujarat's new Green Hydrogen Policy targets 3 MMTPA by 2035; Gujarat Green Hydrogen Policy 2025	https://www.mercomindia.com/gujarats-new-green-hydrogen-policy-targets-3-mmtpa-by-2035/ ; https://jmkresearch.com/wp-content/uploads/2025/12/Gujarat-Green-Hydrogen-Project-Policy-2025-.pdf	late 2025	summaries	Gujarat	₹	regulation	gazette not opened
W4-S77	UP GH2 Policy 2024: 30% (40% first 5) capex, cap ₹225 crore; land ₹15,000/acre/yr	Invest UP; Business Standard	Uttar Pradesh Green Hydrogen Policy 2024	https://invest.up.gov.in/wp-content/themes/investup/sector-assets/policies_schemes/Uttar_Pradesh_Green_Hydrogen_Policy_2024.pdf ; https://www.business-standard.com/economy/uttar-pradesh/uttar-pradesh-green-hydrogen-policy-2024-key-features-incentives-eligibility-and-implementation-126090700729_1.html	2024 / Sep 2026 explainer	summaries	UP	₹	regulation	no mobility incentive found
W4-S78	Maharashtra GH2 Policy 2023: ₹50/kg blending; 500 FCEV cars 30% ≤₹60 lakh; 20 HRS 30% ≤₹4.5 crore; ₹8,500 crore	MAHAURJA (FAQ); JMK; Swarajya; Global Trade Alert	Maharashtra Green Hydrogen Policy 2023 FAQs	https://www.mahaurja.maharashtra.gov.in/Site/Upload/Pdf/GHfaq.pdf ; https://jmkresearch.com/wp-content/uploads/2023/11/Maharashtra-Green-Hydrogen-Policy-2023.pdf	Oct 2023	FAQ	Maharashtra	₹	regulation	fuel-cell wording
W4-S79	AIS 195:2023 for hydrogen ICE M/N vehicles (GSR 746(E)); AIS 195A:2024 CEV	MoRTH	GSR 746(E) dated 16 Oct 2023 — Hydrogen ICE	https://morth.nic.in/sites/default/files/notifications_document/8th%20GSR%20746(E)%20dated%2016th%20october%202023%20Hydrogen%20ICE.pdf	16 Oct 2023	notification	India	—	regulation	not opened
W4-S80	AIS-137 durability × DF; hydrogen reference fuel; CMVR 115/116/126	ARAI (AIS-137 Parts 3/4/6)	AIS-137 Automotive Industry Standard	https://hmr.araiindia.com/Control/AIS/45201991403AMAIS_137_Part_3_F.pdf ; https://hmr.araiindia.com/Control/AIS/53201963840PMAIS_137_Part_4_F.pdf	2019 + amendments	procedure sections	India	—	regulation	snippet
W4-S81	IOCL Faridabad: H2 dispensing since 2005; solar-electrolysis refuelling; PESO permission; Fluiftron dispenser Jan 2023	IndianOil; BusinessWire	R&D Centre pages; Fluiftron installs first domestically built hydrogen dispenser	https://iocl.com/pages/r-and-d-centre ; https://www.businesswire.com/news/home/20230116005549/en/Fluiftron-Installs-First-Domestically-Built-Hydrogen-Dispenser-in-Faridabad-India	2023-2025	pages	India	—	observed	—
W4-S82	Tata Prima H.55S H2 specs; trials 4 Mar 2025; 40 trucks VOC Port by 2028; Cummins Jamshedpur production Mar 2024; \$425 m JV	Tata Motors; electric; Cummins; Hydrogen Insight; Truck Junction	Tata Motors drives India's green future...; Tata Motors to deploy 40 H2 ICE trucks; Cummins produces H2-ICE for Tata Motors	https://www.tatamotors.com/press-releases/tata-motors-drives-indias-green-future-with-countrys-first-hydrogen-truck-trials/ ; https://www.electrive.com/2026/02/27/tata-motors-to-deploy-40-h2-ice-trucks-in-india/ ; https://www.cummins.com/en-na/news/releases/2024/05/01/cummins-produces-hydrogen-internal-combustion-engines-tata-motors-new	2024-2026	press	India	—	observed	—
W4-S83	Ashok Leyland-Reliance H2-ICE truck; AL fleet 2026	Team-BHP; Business Standard	Ashok Leyland & Reliance unveil India's first hydrogen-ICE truck; India's vehicles industry quietly begins shift...	https://www.team-bhp.com/news/ashok-leyland-reliance-unveil-india%E2%80%99s-first-hydrogen-ice-truck ; https://www.business-standard.com/industry/auto/indian-cv-majors-drive-hydrogen-transition-with-h2-vehicles-125061600725_1.html	Feb 2023; Jun 2025 news		India	—	observed	—
W4-S84	Suzuki hydrogen Burgman (JMS 2025); hydrogen Swift 1.4 L DI 100 kW/220 Nm; Toshihiro Suzuki BMGE 2025 quote; Maruti H-CNG	Suzuki (globalsuzuki); DriveSpark; Automotive World; evo India; V3Cars	Hydrogen Engine Burgman; Hydrogen-powered Suzuki Swift revealed; Suzuki unveils three new motorcycle models...	https://www.globalsuzuki.com/mobilityshow2025/burgman/ ; https://www.drivespark.com/four-wheelers/2026/suzuki-hydrogen-powered-swift-revealed-084763.html ; https://www.automotiveworld.com/news-releases/suzuki-unveils-three-new-motorcycle-models-including-an-ev-scooter-at-the-bharat-mobility-global-expo-2025/ ; https://www.v3cars.com/news/maruti-suzuki-hydrogen-cng-hcng-engine-for-future-vehicles	2025-2026	press/news	Japan/India	—	observed	no retrofit statement
W4-S86	IIT Madras-Hyundai HTWO centre operational by 2026	Careers360	IIT Madras unveils Hyundai HTWO Centre	https://news.careers360.com/iit-madras-unveils-hyundai-htwo-2025-centre-on-campus-operational-by-2026-iitm-hydrogen-india-research/amp	2025	news	India	—	announced	—